PAULA'S REVENGE V0.63

USER MANUAL

AN 8-TRACK AMIGA-STYLE SAMPLE TRACKER

FOR JUNGLE, DRUM & BASS AND BREAKS



THE MAIN SCREEN, LIGHT SKIN

Paula's Revenge is an original program inspired by the classic Amiga trackers of the early 90s. Paula and Akai sound, a mixer, plug-ins, MIDI out, automation, bouncing and stems.

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GETTING STARTED

Paula's Revenge is an 8-track sample tracker for jungle, drum & bass and anything else that wants the crunchy sound of the Amiga's Paula chip and the old Akai samplers. You write music as lines of notes and commands that play from top to bottom, one block at a time, in the order the song lists them.

INSTALL

1. Open **Releases** on the right of the GitHub repo page. Each release has both versions: `PaulasRevenge-macOS.zip` for a Mac and `PaulasRevenge-Windows.zip` for a PC (Windows 10 or 11, 64-bit).
2. **Mac:** follow `INSTALL.txt` inside: copy the app into Applications and clear the macOS quarantine.
   PC: unzip it anywhere and run `Paula's Revenge.exe`. The first time, Windows says *Windows protected your PC* because the app isn't code-signed yet: click **More info**, then **Run anyway**.
3. Open it. Audio and MIDI devices are under **Audio/MIDI Settings…** (`CMD` `,`) in the **File** menu (on a Mac also in the app menu). The File menu has New, Load…, Save, Save As…, **Save MOD…**, **Render WAV…** and **Export Samples…**; the View menu has Capture, Sample Audio, **Record Note-Offs**, **Full Velocity**, **Follow Instrument** and full screen.
4. The sound follows the computer's output: plug in headphones or connect AirPods and it moves there, with the status line saying where it went. **Follow System Output (Headphones)** in the menu turns this off; picking a device by hand in Audio/MIDI Settings turns it off too, so your choice stays.

> **PC** Every `CMD` shortcut in this manual is `CTRL` on a PC, and the app shows CTRL where the Mac shows CMD. Full screen is `F11`. Plug-ins are VST3 (Audio Units are Mac-only).

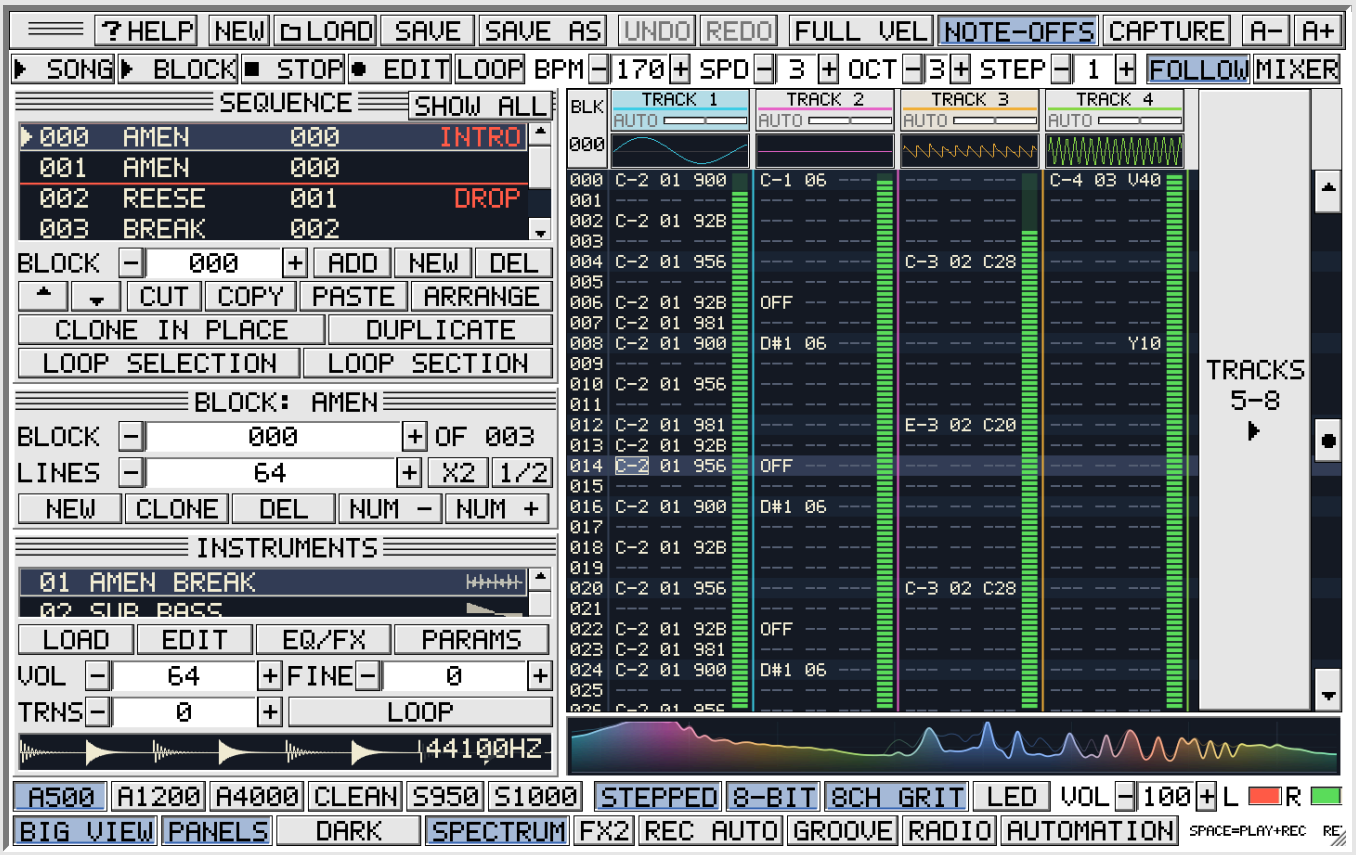
YOUR FIRST SOUND IN A MINUTE

1. A new song already has a few chip sounds in it, so the keys make sound straight away.

2. Press ESC to switch • EDIT on. The cursor line turns grey. (SPACE plays *and* switches EDIT on, to record as it plays; Space again stops and switches it off.)
3. Type notes with the bottom two rows of the keyboard (Z is C, S is C#, X is D…). Each note goes in with the current instrument and the cursor moves down by STEP.
4. Click LOAD in the instrument panel to pick a break from your drive. Every sample you highlight plays straight away.
5. Press RETURN to hear the block (and Return again to stop). SHIFT RETURN plays the whole song.

TIP For a break: put C-3 on line 0 and use 9xx offsets to jump around inside it, or slice it in the sample editor and play the slices from the keys.

THE SCREEN



THE MAIN SCREEN, LIGHT SKIN, PLAYING

AREA	WHAT IT DOES
Title bar	NEW, LOAD, SAVE, SAVE AS, UNDO, REDO, FULL VEL, NOTE-OFFS, CAPTURE (see Recording), and text size A- A+. Save MOD, Render WAV and Export Samples are in the **File** menu.
Transport	Play the song or the block, stop, EDIT mode, LOOP (the marked lines), BPM, SPD (ticks per line), octave and step, FOLLOW (CMD F) and MIXER (CMD 2).
Sequence	The song order. Click a position to go there, drag to highlight positions, and set which block plays where.
Block	Pick, add, clone or delete blocks; set the number of lines (1–256); X2 doubles a block, 1/2 halves it.
Instruments	63 slots. LOAD, EDIT, EQ/FX, PARAMS; DELETE clears the instrument picked (CLEAR and RENAME are on its right-click). The instrument's VOL, finetune, transpose, loop, and a little waveform.

AREA	WHAT IT DOES
Pattern	8 tracks. Click a track name to mute it, right-click to solo; drag in the bar under the name to pan the track. Under that, each track's scope (click it to mute, right-click to solo).
Output strip	The machine you hear (A500, A1200, A4000, CLEAN, S950, S1000), its options, the master VOL and an L/R meter. Stereo separation (SEP): right-click a machine button.
Bottom row	BIG VIEW, PANELS, DARK, SPECTRUM, FX2, REC AUTO, and the status line.

The app opens with the dark skin. DARK switches to the black-on-white WORKBENCH skin, as in this manual (more skins below). Every song has 8 tracks (a 4-channel MOD fills tracks 1–4). BIG VIEW shows 4 at a time with large text, with a tall TRACKS 5-8 ▶ tab beside them; click it for tracks 5–8, and ◀ TRACKS 1-4 brings you back (TAB past track 4 flips too). The song's name (in the window's title bar, and at the top when there's room) is its file's name: save it as `amen roller.prv` and it's AMEN ROLLER.

The status line at the bottom right says what just happened, and what the command under the cursor does.

RIGHT-CLICK MENUS

Right-click almost anything for the things you can do to it. Everything they do can be undone.

RIGHT-CLICK	MENU
A note in the pattern	Copy / cut / paste the track, clear the line, insert or delete a line, effect commands, VU meter colour, spectrum colour
An effect cell	The command list; on a `#xy`, the chord list
A highlight in the pattern	Bounce it to a new sample
An instrument	Copy, paste, duplicate to the next free slot, clear, rename, load a sample, sample (record audio), play `C-3`
A song position (sequence or ARRANGE)	Copy, cut, paste, repeat, delete, insert the block again, section label

RIGHT-CLICK	MENU
The block panel	Copy / cut / paste / clear the block, duplicate, new, delete, double, halve
The sample's waveform	The sample editor's edits on the highlight
A mixer strip's name	Copy / paste / reset the strip, open the channel
A mixer effect slot	Copy / paste the effect, clear the slot, bypass, open
A CHORDS slot	Copy, paste, clear (all the picked slots), colour, play, record chords, note
Any level meter: the bar at the right of each track, the L / R meter, the mixer's meters	Their colour, one for all of them: green, amber, cyan, red, blue, pink, cream / black, or each track's colour (kept for next time)
The spectrum strip	Its colour: the same choices, or rainbow (the track colours, as it starts)
EDIT	Recording: note-offs in step entry, detect chords, capture, NOTE-OFFS, FULL VEL, FOLLOW INSTRUMENT
DARK	The skins, and CRT
A HELP page	Copy the chapter, or all of HELP

HELP AND SEARCHING IT

HELP has this manual's text. Just type to search it: every match is highlighted and the chapter list shows only the chapters with matches. ENTER (or CMD G) goes to the next match, SHIFT ENTER the one before. ESC clears the search, then closes HELP.

In the dark skin, the plug-in pages (an effect opened from the mixer, a plug-in instrument's page) and the parameter pages are dark too.

SKINS AND CRT

Twenty-one skins: **WORKBENCH** (black on light grey), **DARK** (cream on blue-black), **KICKSTART 1.3** (white on the old blue, orange when pressed), **HIGH CONTRAST** (white on black, yellow when pressed), **GLASS**, **MODERN**, **STUDIO**, **NEON**, **HARDWARE**, **PRISM**, **BLUEWAVE**, **PAPER**, **VIOLET**, **EMBER**, **FOREST**, **PRISM II**, **AURORA**, **BOOMBOX**, **DARK II**, **GLASS II**

and **DARK III**. Click DARK to swap between WORKBENCH and DARK; right-click it, or use **View** → **Skin**, for all twenty-one. The button shows the skin's name (**KS 1.3**, **HI CON**, **GLASS**, **MODERN**, **STUDIO**, **NEON**, **HARDWARE**, **PRISM**, **BLUEWAVE**, **PAPER**, **VIOLET**, **EMBER**, **FOREST**, **PRISM II**, **AURORA**, **BOOMBOX**, **DARK II**, **GLASS II**, **DARK III**) when one of those is on. The skin is kept for next time.

GLASS is deep blue glass: every button and pane has a gloss over its top half and a bright rim, every field and list a soft shadow inside its top edge, and what's chosen glows cyan. Panel titles sit on their own at the left over a thin accent line. **MODERN** is light and flat: white and soft grey, no bevels – just thin borders and a faint drop under raised things – and what's chosen is solid blue with white text.

STUDIO is flat greys like a modern DAW, and the one skin with smooth type instead of the pixel font: numbers and the pattern in **JetBrains Mono**, labels in **Inter**. Every letter takes the same room as its pixel letter, so nothing moves or shrinks. Buttons are flat with slightly rounded corners; one that's on is filled with what it means – orange for chosen, red for recording, blue for modulating – with dark text on it.

Three more skins use smooth type the same way, each with its own label typeface (the pattern and numbers stay in JetBrains Mono, so columns line up):

- **NEON** – deep navy and rounded, buttons lit from above, and what's on glows cyan. Labels in **Space Grotesk**.
- **HARDWARE** – matte black with rubber-grey keys, like a drum machine. A key that's on keeps its face and lights an LED along its top (red-orange; red for recording). The pattern is an amber display. Labels in **Barlow Condensed**.
- **PRISM** – slate grey, soft and rounded, in bright track colours. In the pattern every note is a **bar** in its track's colour that lasts until the next note or OFF: the note and instrument in dark text where it starts, the bar fainter while it's held. You see at a glance how long each note rings. Labels in **Nunito**.

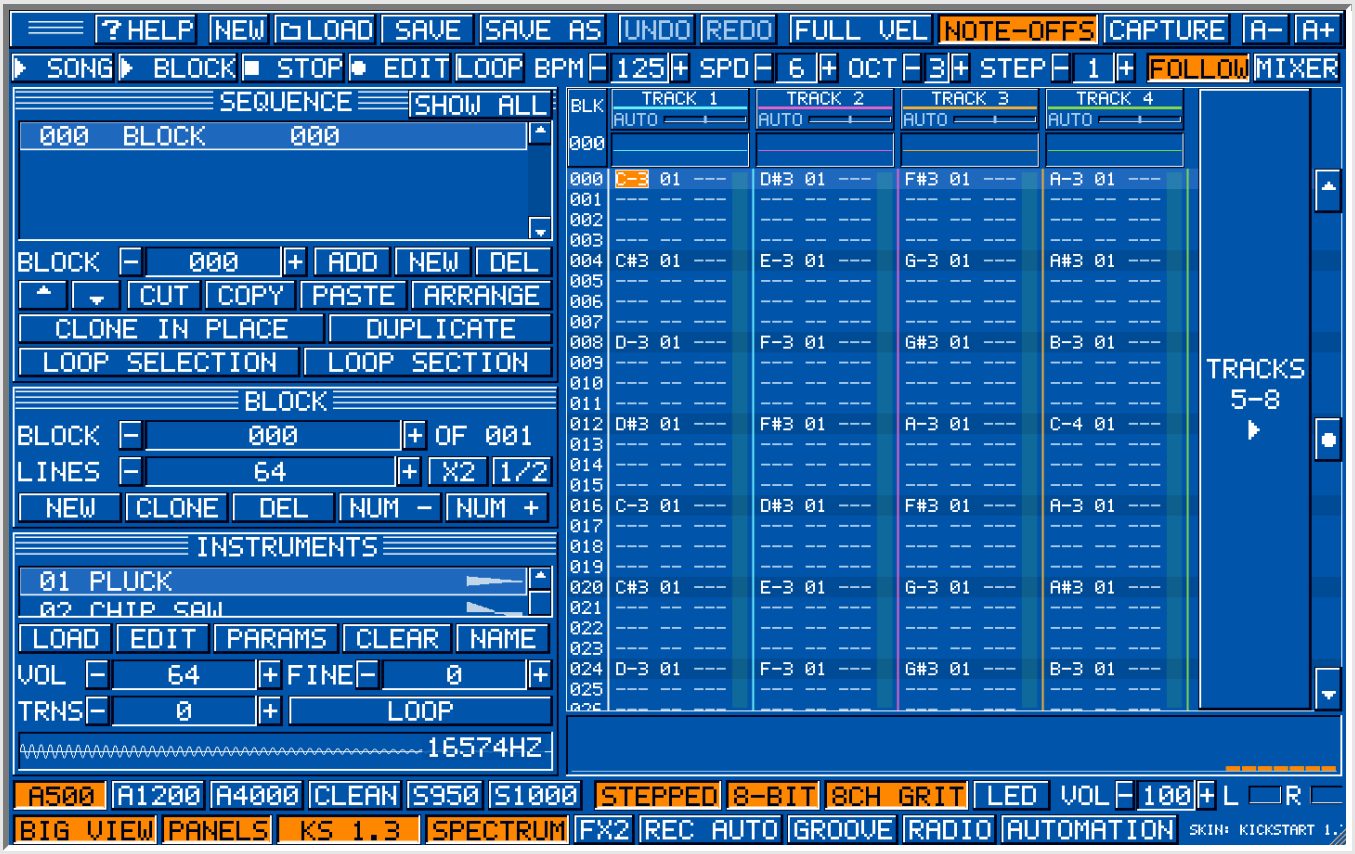
And five more, all with smooth type:

- **BLUEWAVE** – a wavetable synth's charcoal and slate blue-grey. What's on is bright blue, modulation is orange, and buttons are lit from above.
- **PAPER** – light and clean: white panes on soft grey; what's on is blue with white text on it.
- **VIOLET** – dark purple and rounded; what's on glows lilac, modulation is teal. Labels in Nunito.
- **EMBER** – warm charcoal with orange for what's on and yellow for modulation. Labels in Space Grotesk.
- **FOREST** – deep green with lime for what's on. Labels in Barlow Condensed.
- **PRISM II** – PRISM without the bars: every track's notes and instruments are in its colour on the plain slate background, and its effects a lighter shade of it.
- **AURORA** – near black with a soft purple and navy glow behind smoky grey panes, pill-shaped buttons, pastel mint for what's on and yellow for modulation, and every track a pastel colour. Labels in Inter.
- **BOOMBOX** – an 80s boombox: deep violet panels with a silver rim and a soft purple glow behind them, hot pink for what's on, orange for modulation, and every track's notes in an equaliser colour – orange, pink, magenta, purple, lavender. Labels in Space Grotesk.
- **DARK II** – DARK made modern: the same cream on blue-black, with smooth type, rounded keys lit from above and a soft blue glow behind. Labels in Inter.
- **GLASS II** – GLASS made modern: deep blue glass with smooth type, glossy rounded keys that glow when they're on, and a sheen across the panes. Labels in Inter.
- **DARK III** – DARK as it is, made modern: the same light grey panels and keys round the cream-on-blue-black pattern, with smooth type, rounded keys lit from above and soft edges instead of the Amiga bevels. Labels in Inter.

(All the typefaces are free under the SIL Open Font License.)

CRT (in the same menu) lays scanlines and darker corners over everything, like an old monitor. It is only a look: clicks go through.

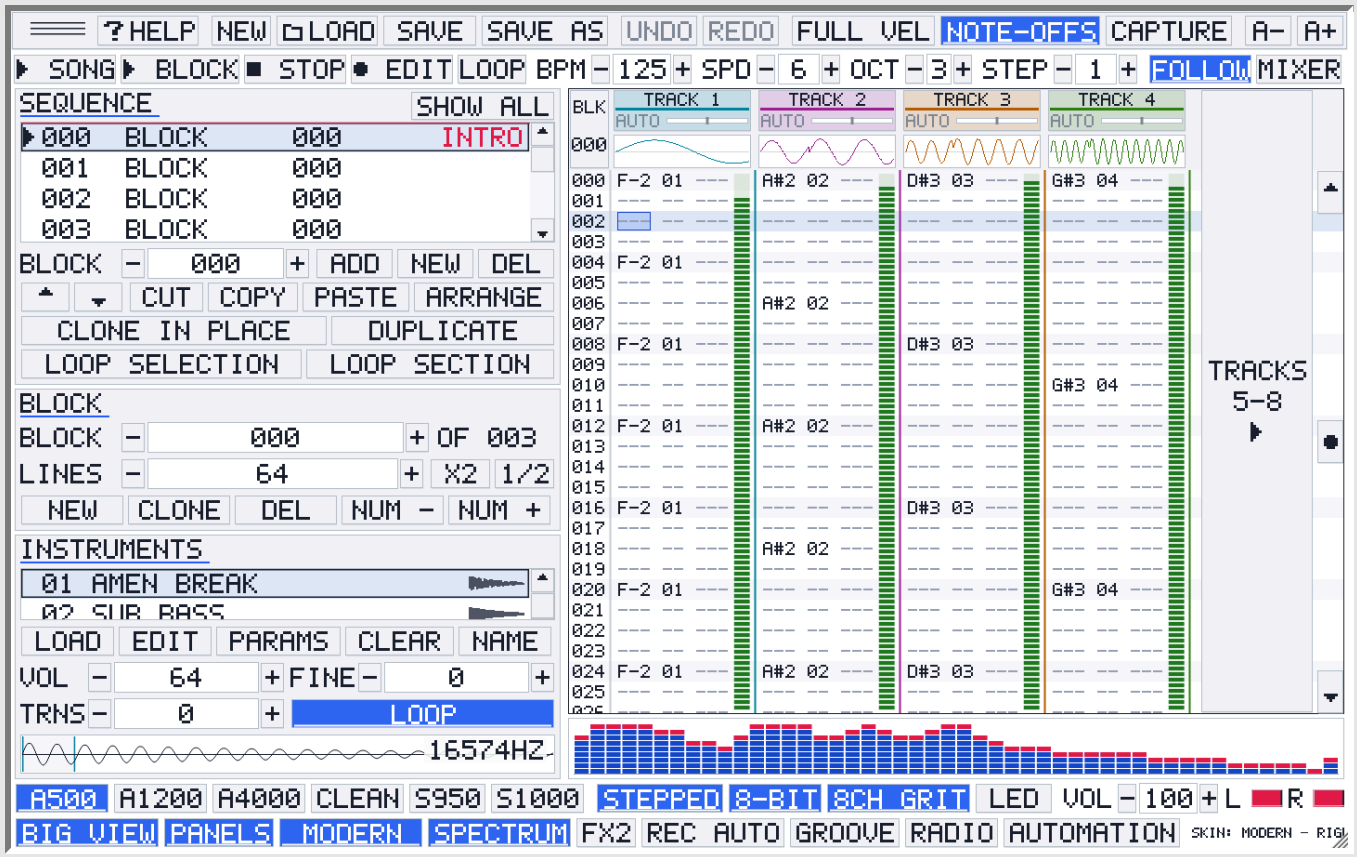
Colour means the same thing everywhere: a button that's on has a stripe along its foot - red for recording (EDIT, REC AUTO, RECORD), orange for learning and modulating, blue (in KICKSTART, white) for the rest. Orange also marks a modulated control. Text and bevels are drawn on whole screen pixels, so they stay sharp at any size and on Retina screens.



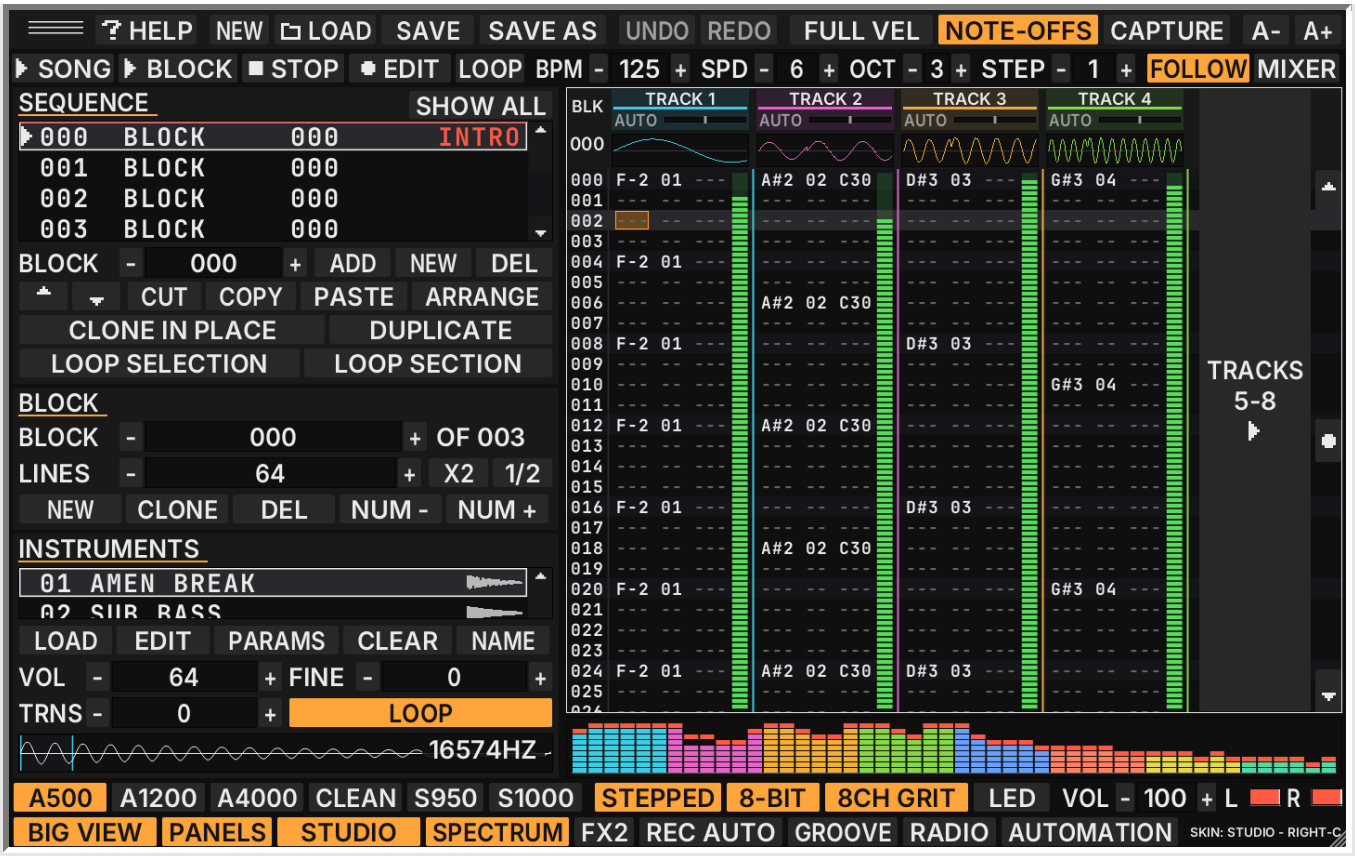
THE KICKSTART 1.3 SKIN



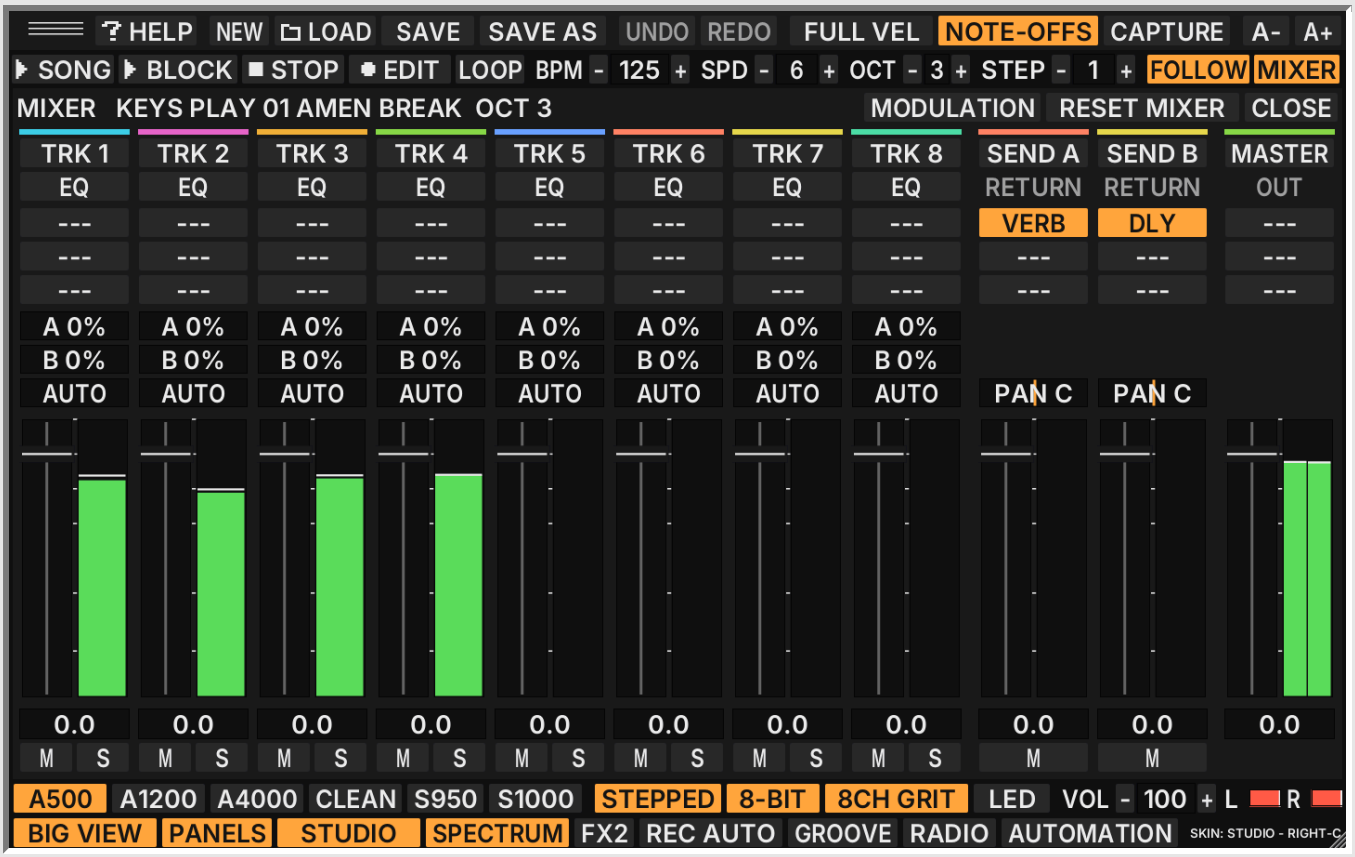
THE GLASS SKIN



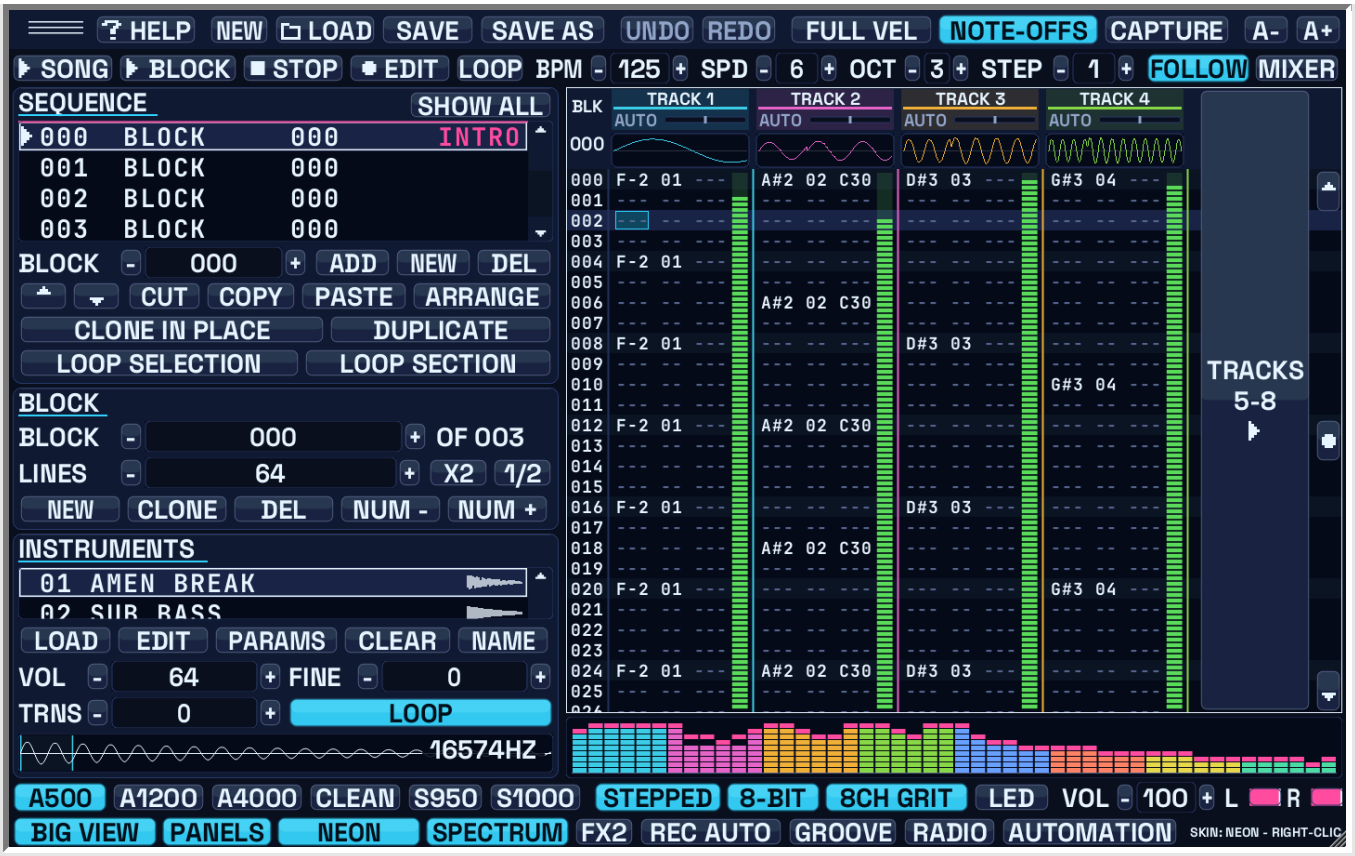
THE MODERN SKIN



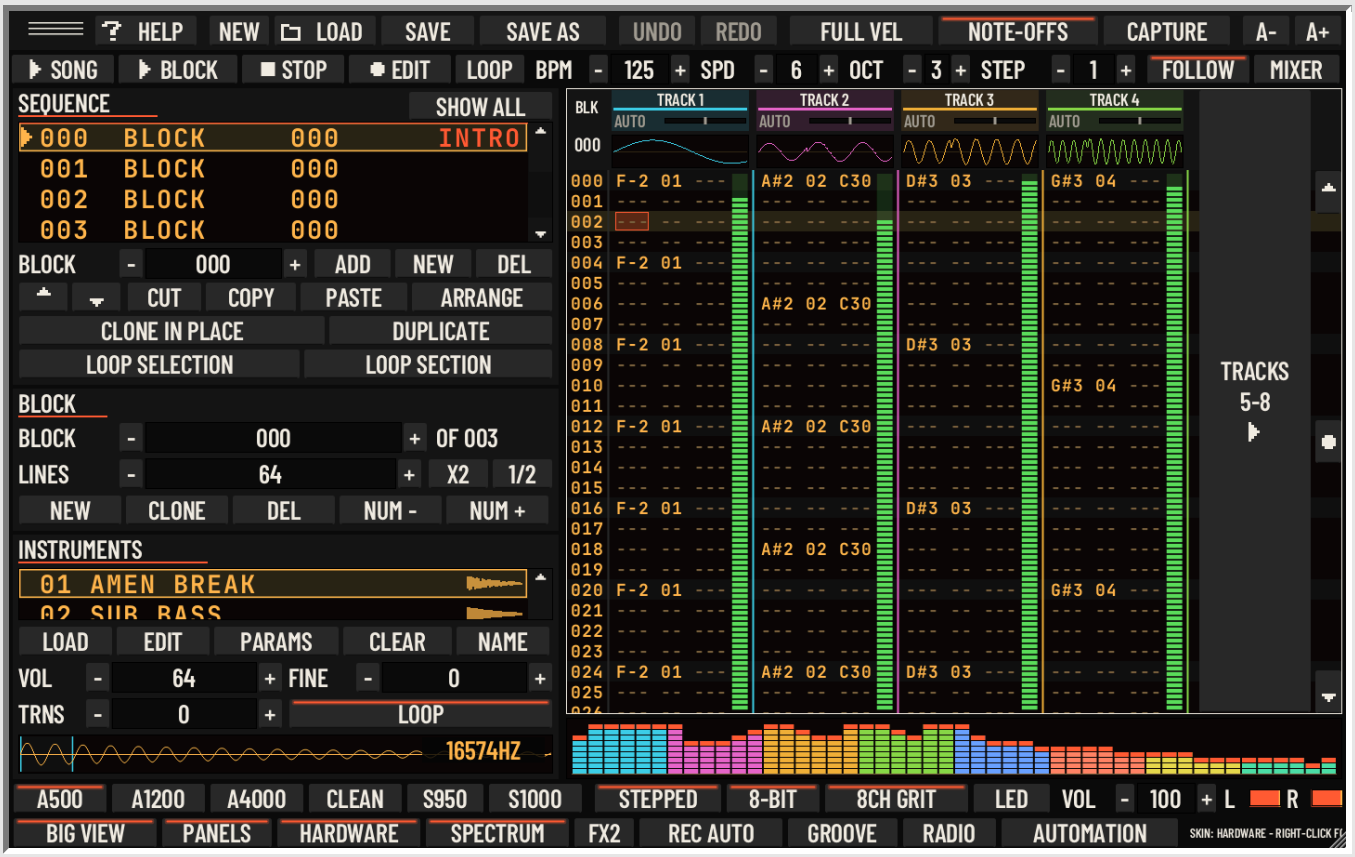
THE STUDIO SKIN



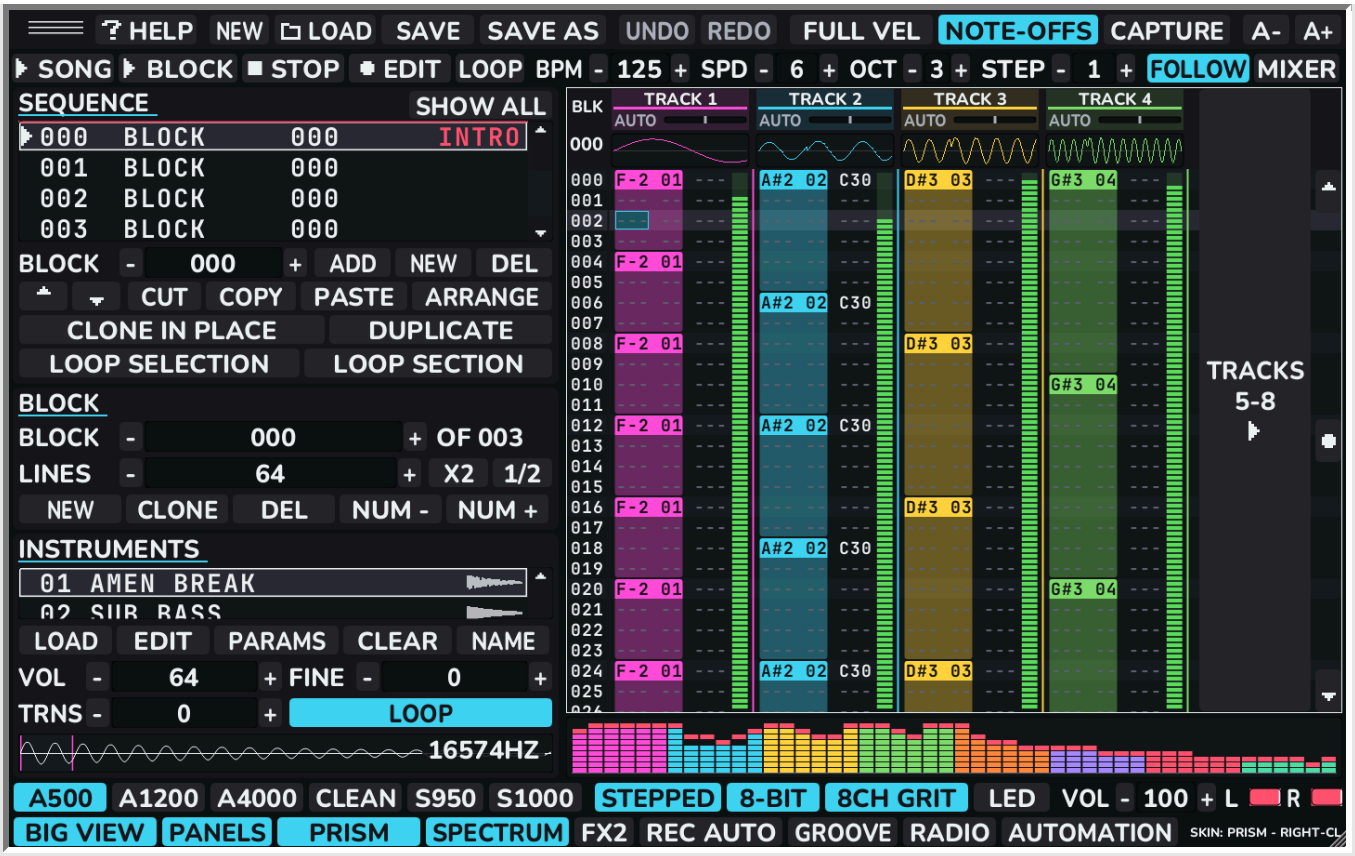
THE MIXER IN STUDIO



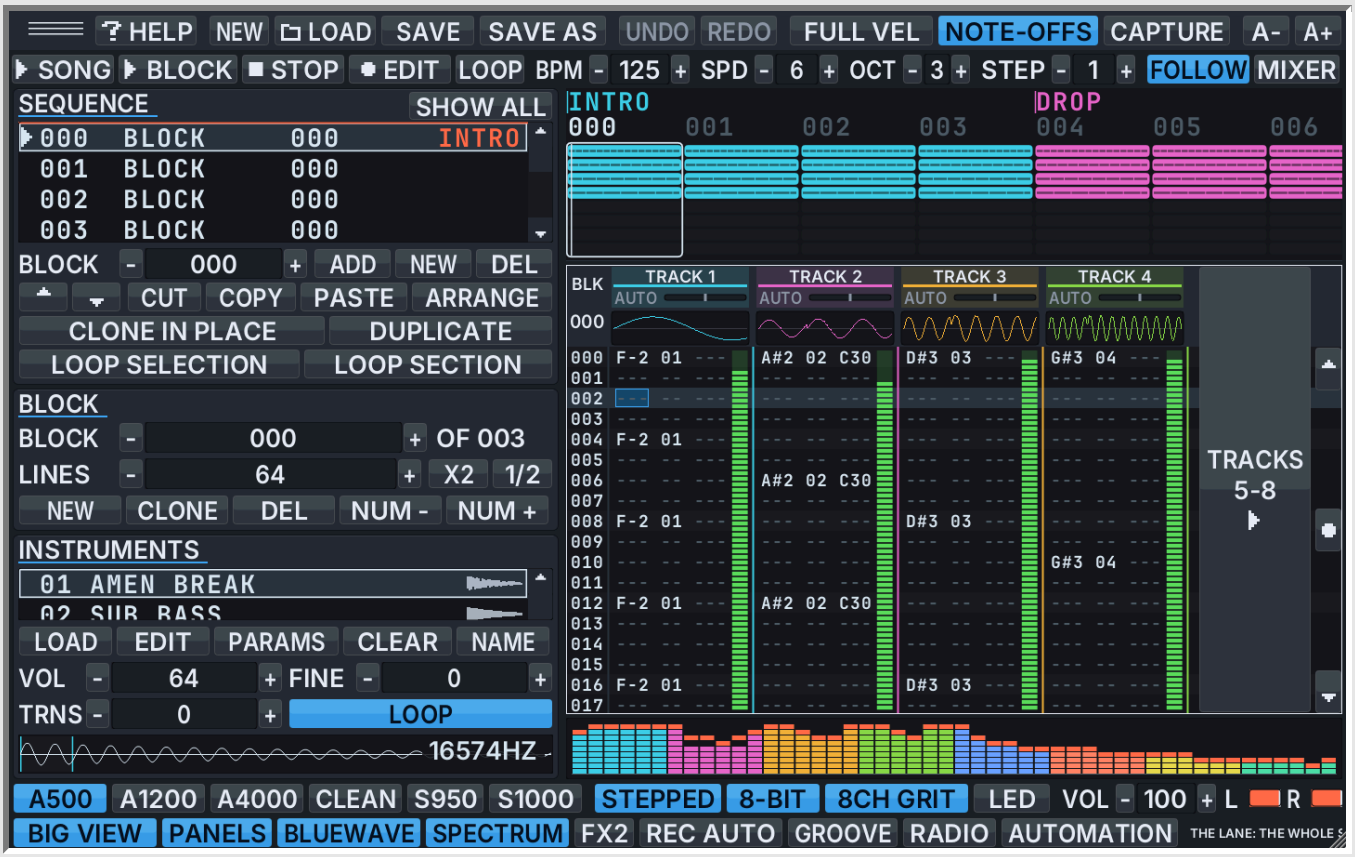
THE NEON SKIN



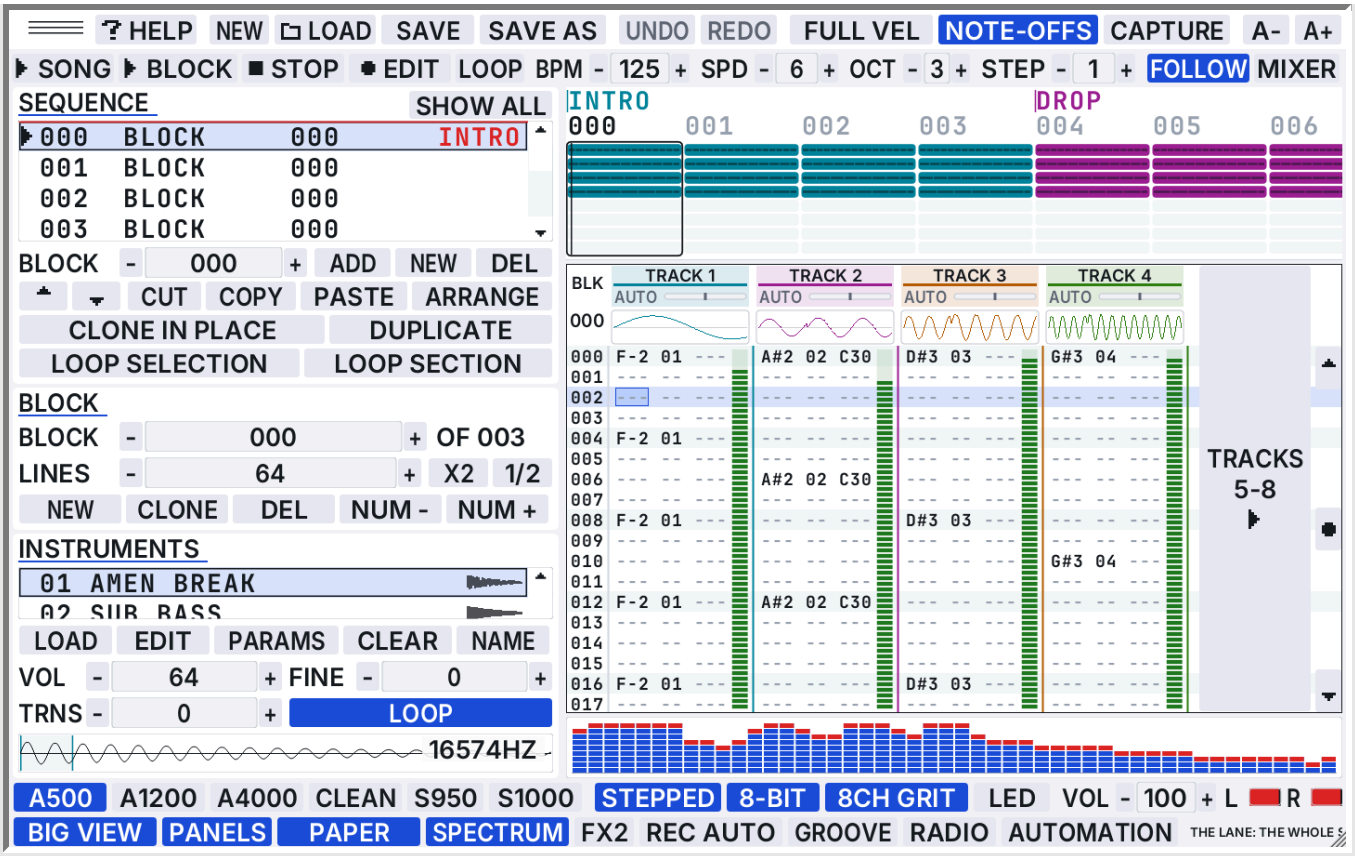
THE HARDWARE SKIN



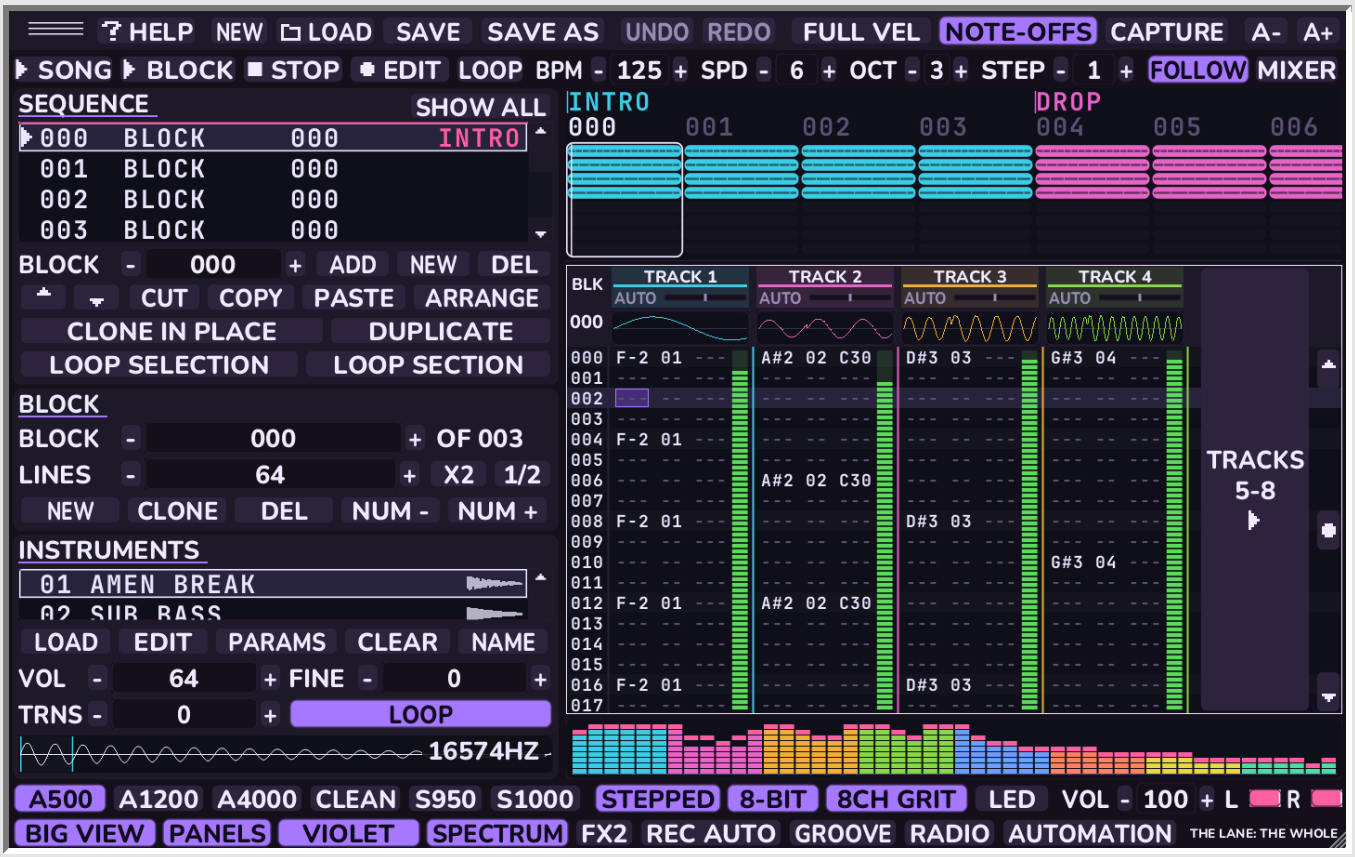
THE PRISM SKIN, WITH ITS NOTE BARS



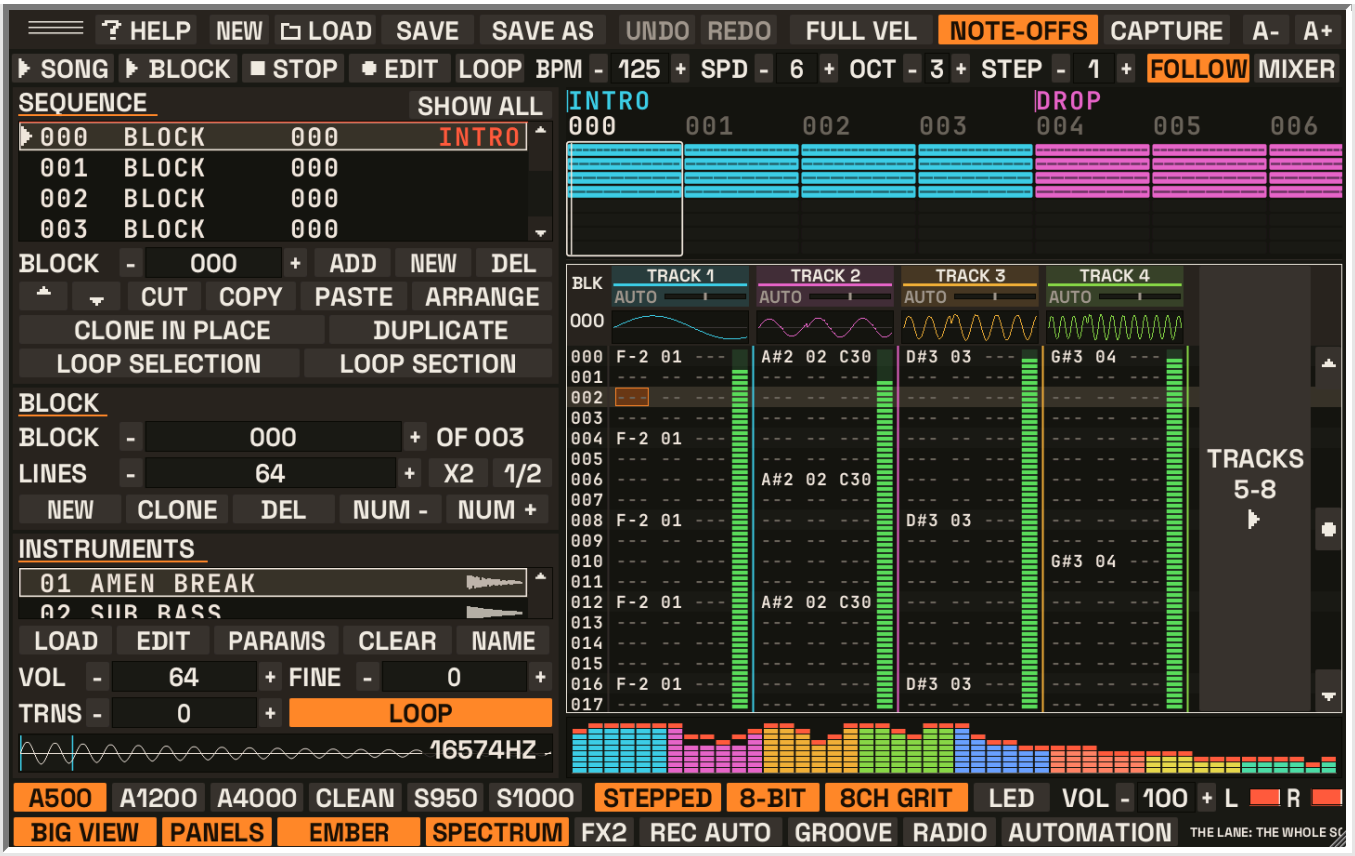
BLUEWAVE



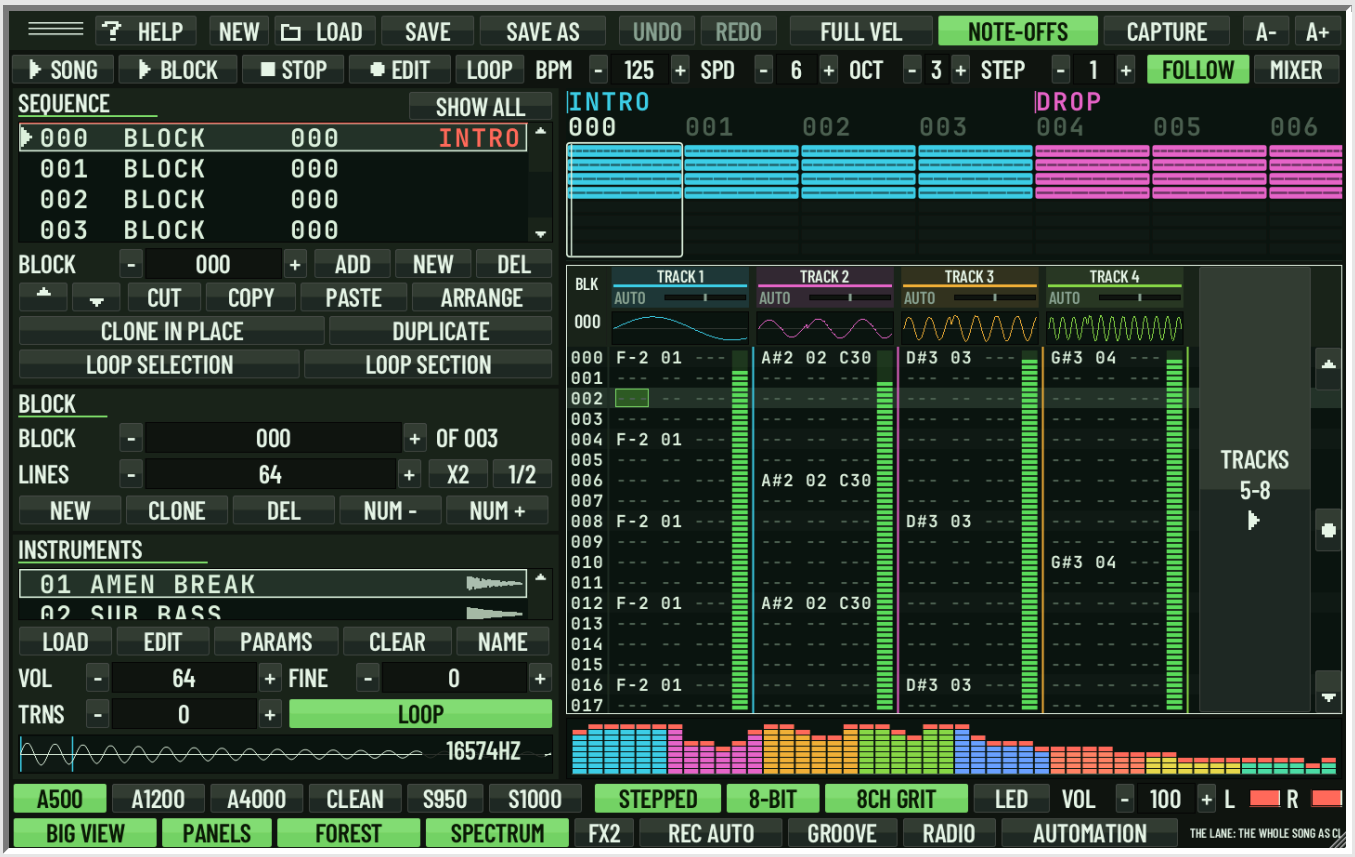
PAPER



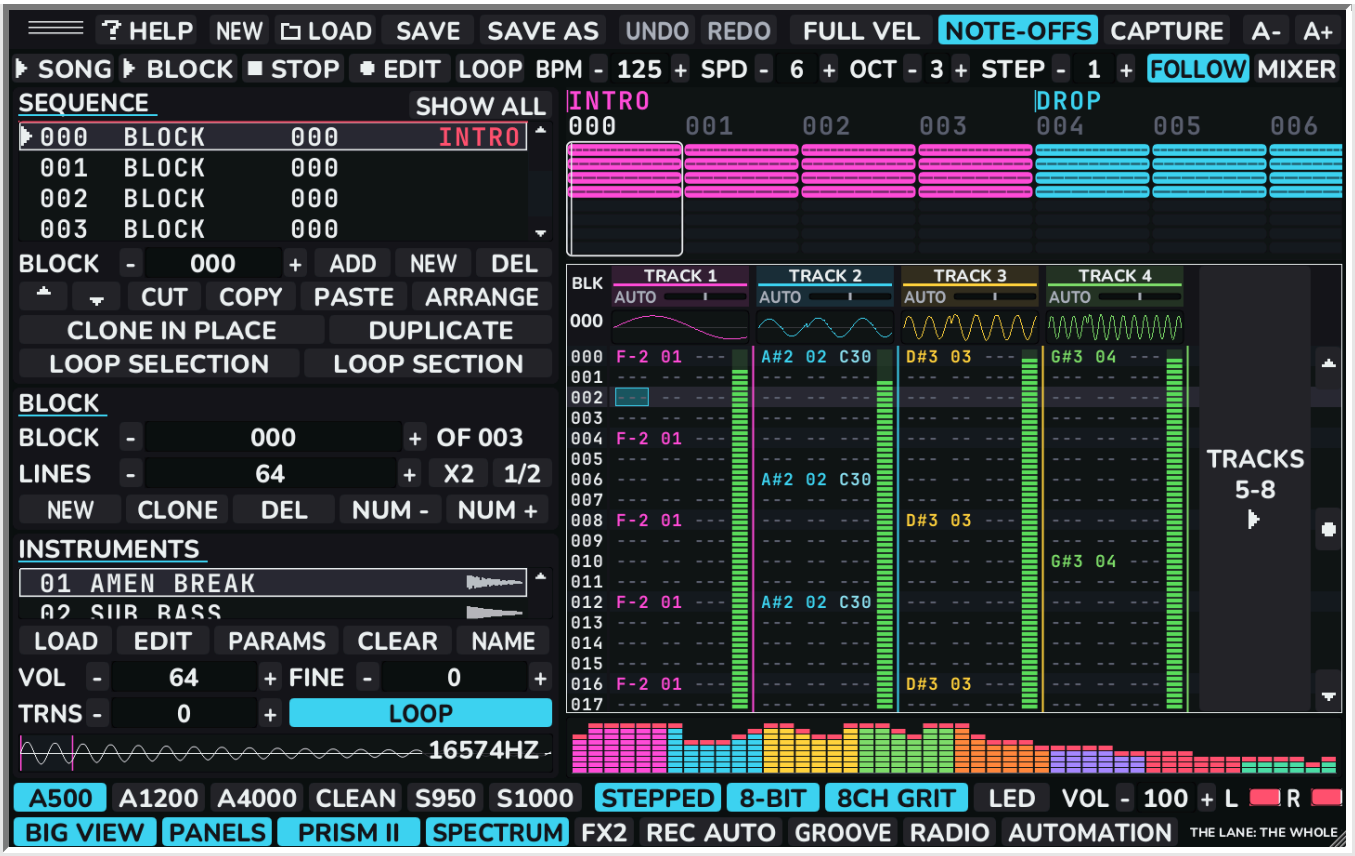
VIOLET



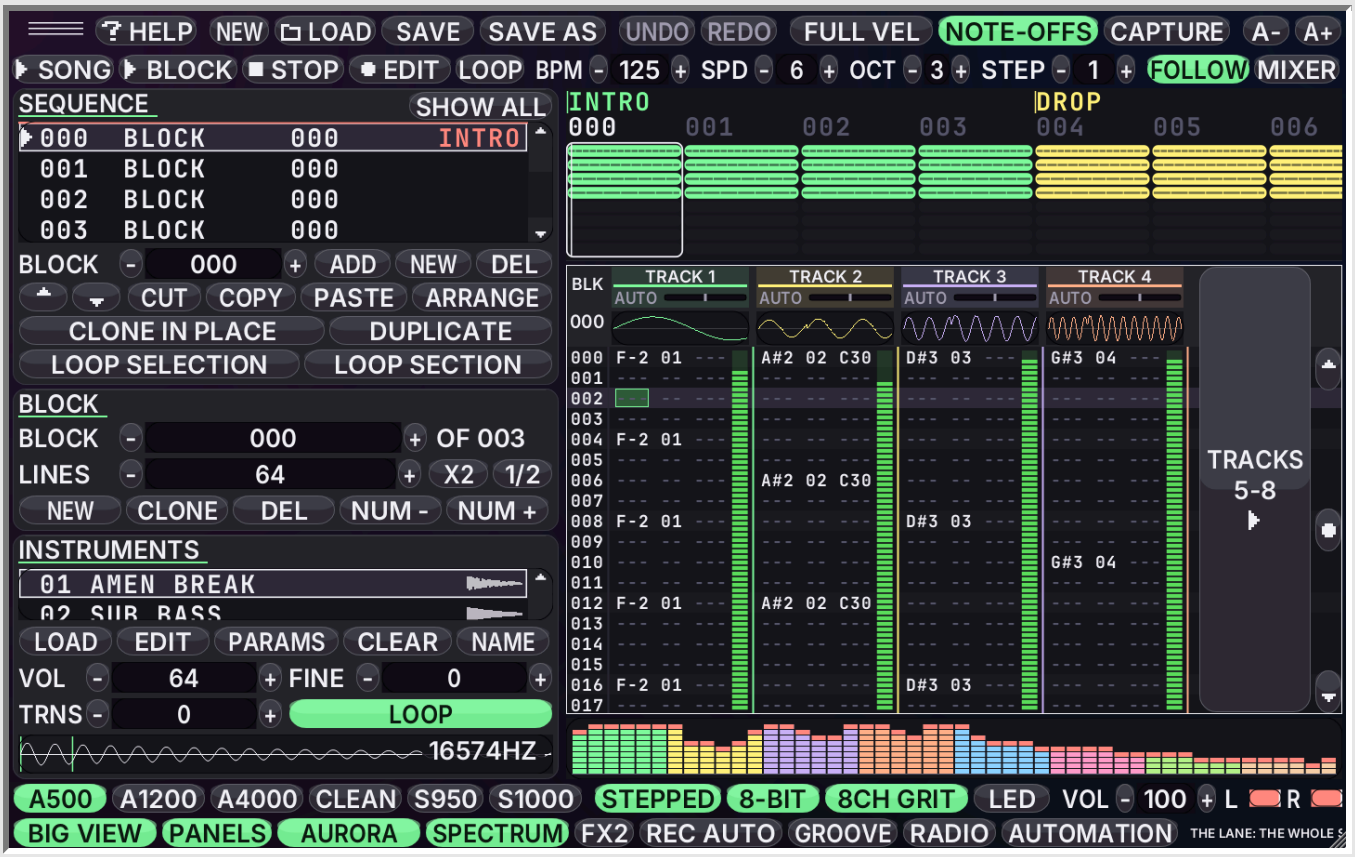
EMBER



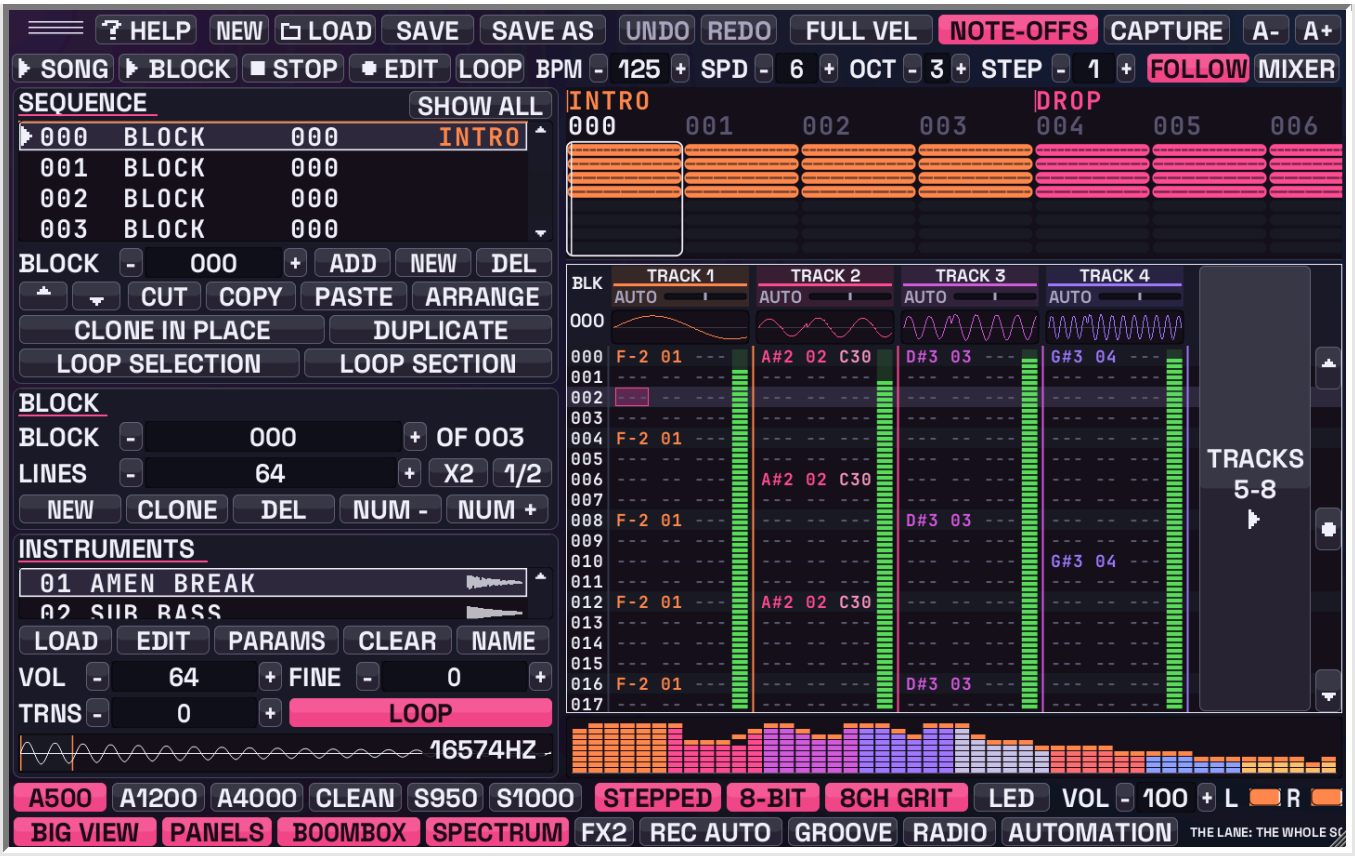
FOREST



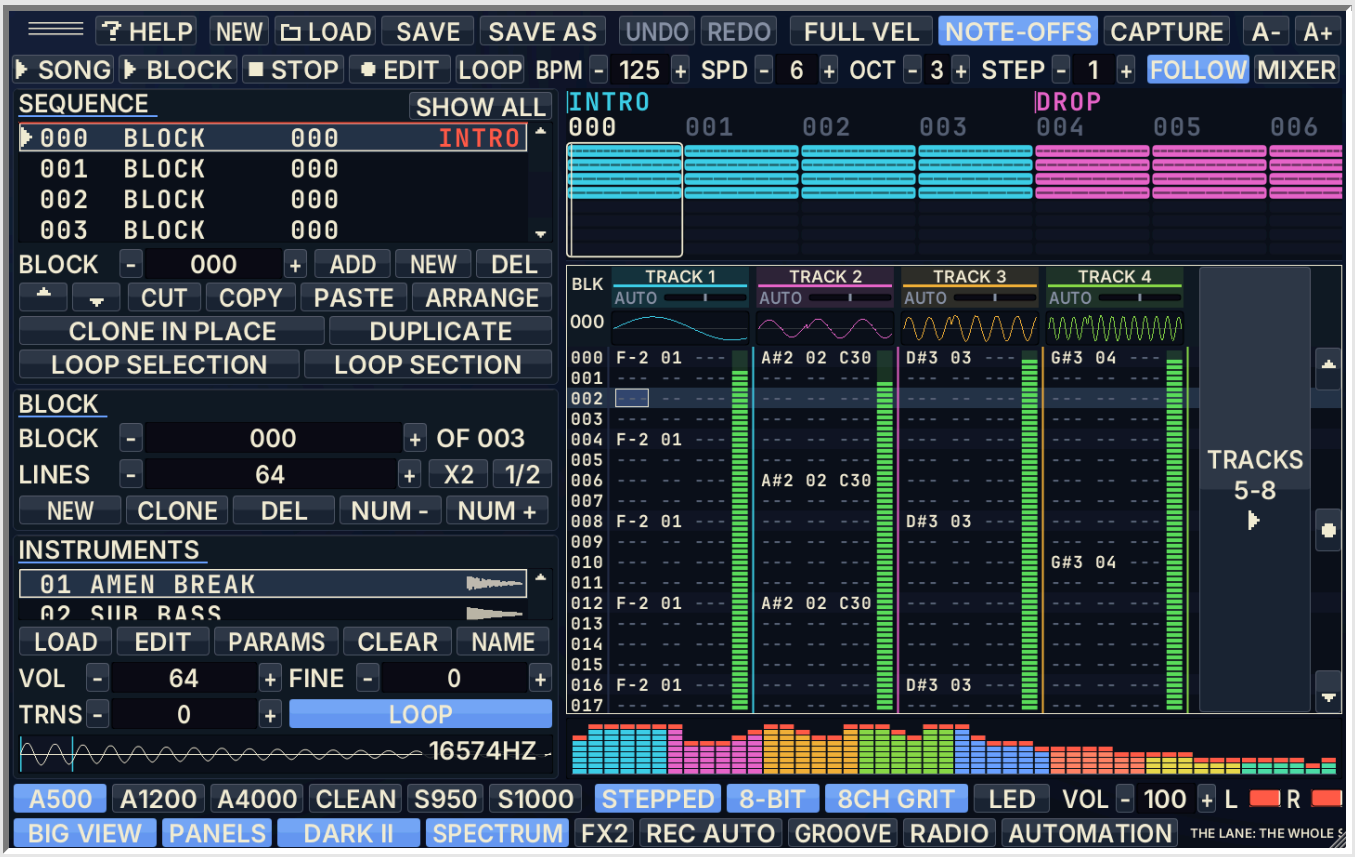
PRISM II: COLOURED NOTES AND EFFECTS, NO BARS



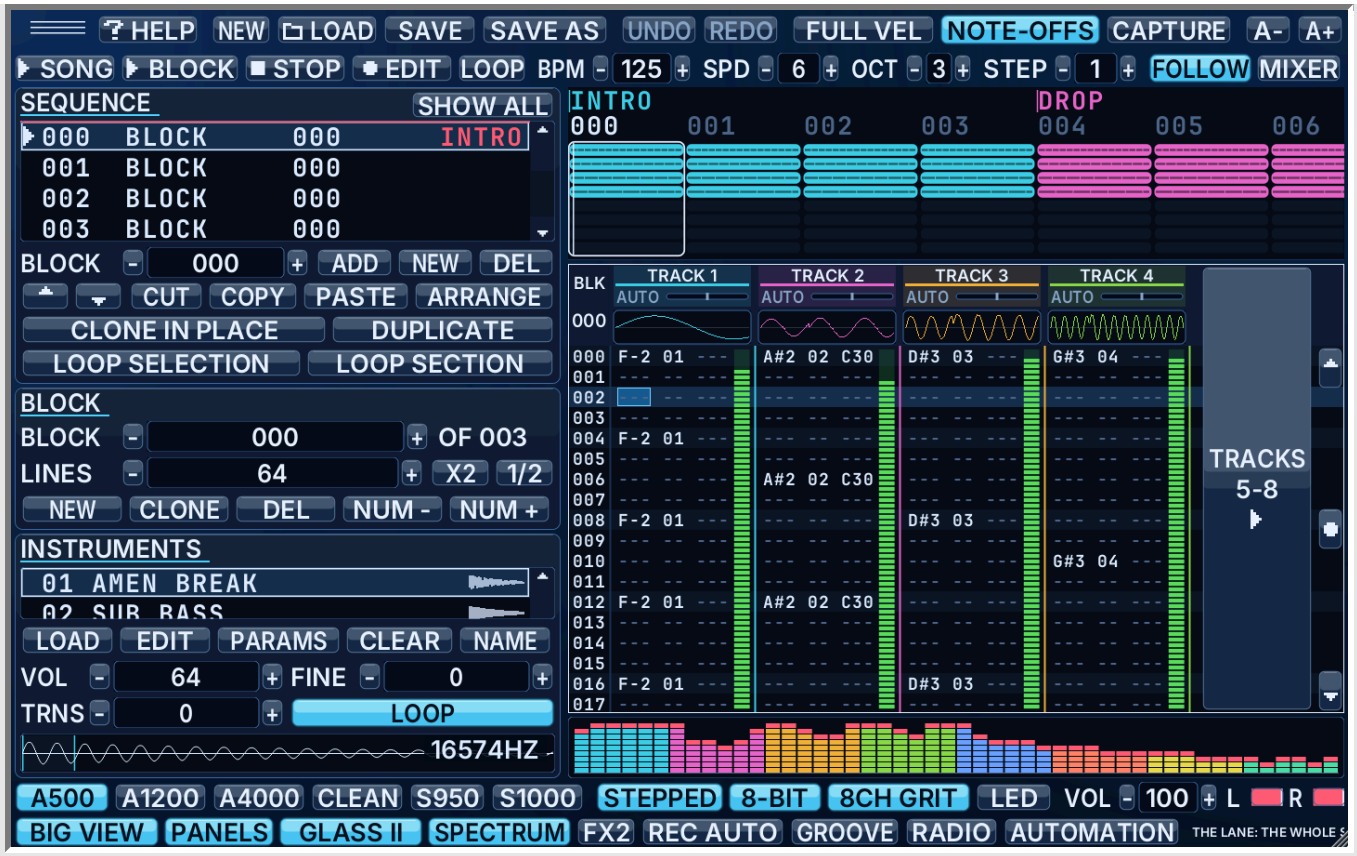
AURORA: PASTEL ON NEAR BLACK, PILL BUTTONS



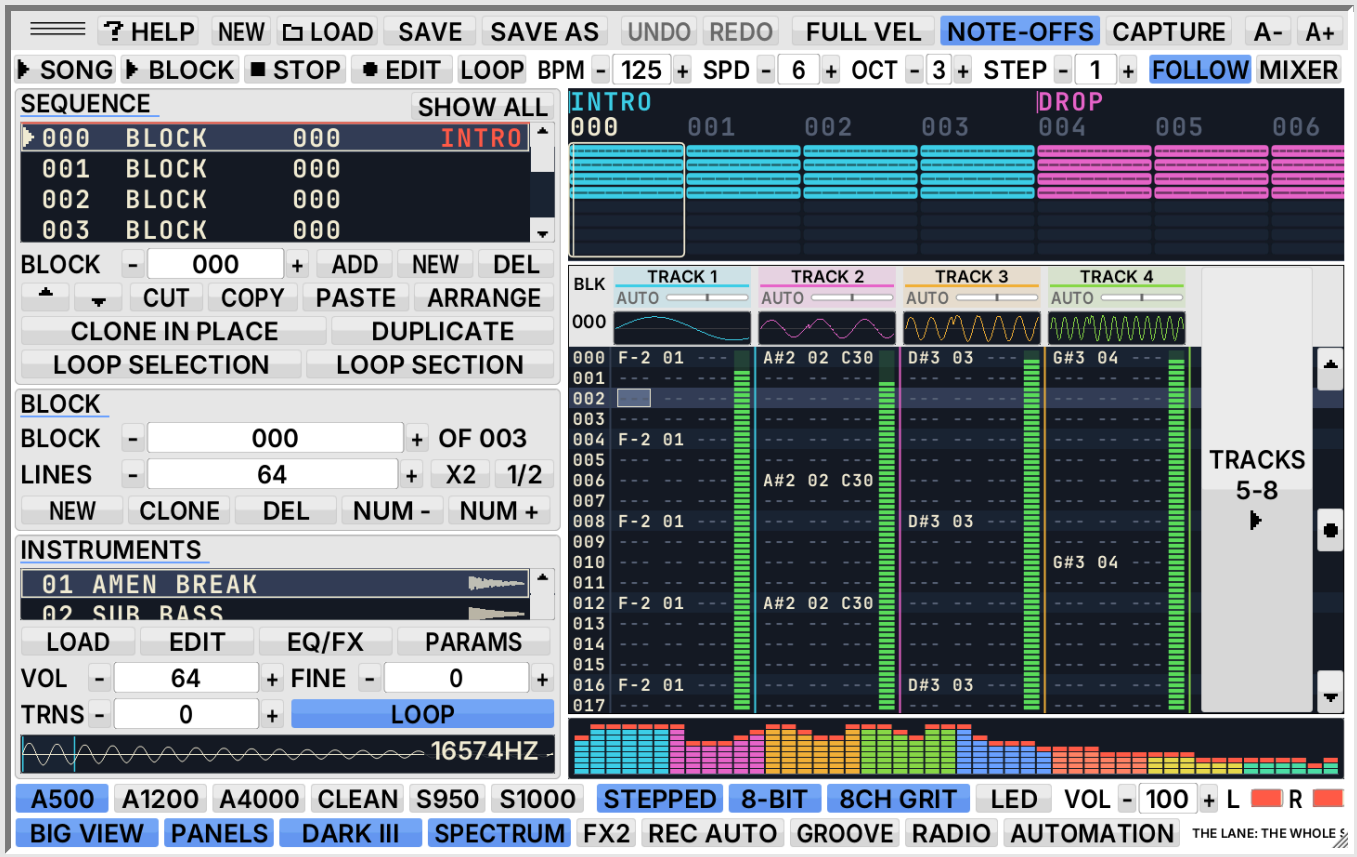
BOOMBOX: DEEP VIOLET, HOT PINK, EQUALISER COLOURS



DARK II: DARK, MADE MODERN



GLASS II: GLOSSY BLUE GLASS



DARK III: DARK, MADE MODERN

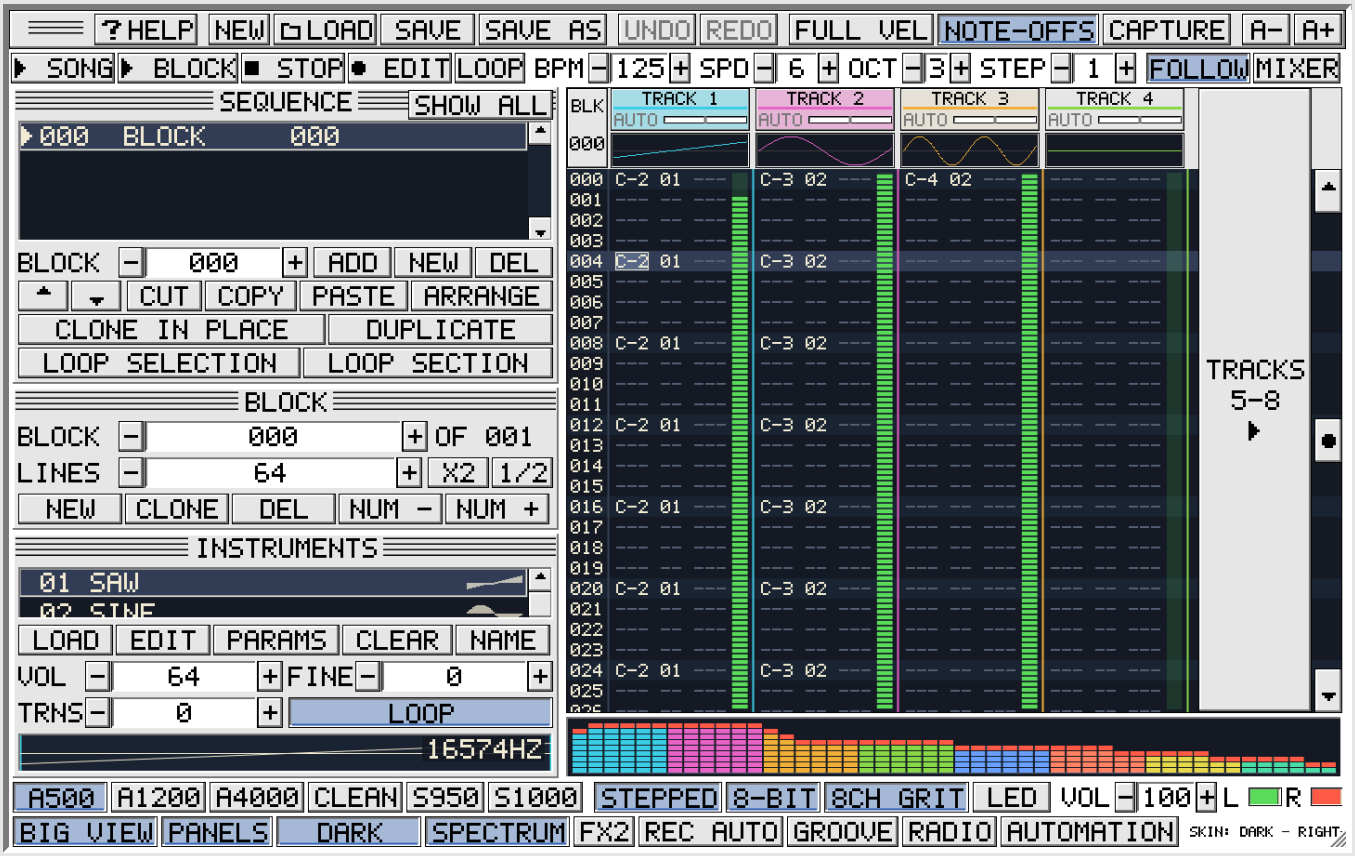
LIVE FEEDBACK

Under each track's name and pan bar is its **scope**: the track's sound as a moving line in the track's colour, held still on each cycle the

way ProTracker's scopes were. A silent track is a flat line; a muted one shows only the zero line. Click a scope to mute its track, right-click (or Alt-click) to solo it - the same as its name. **View → Track Scopes** hides them for more rows; the choice is kept.

Each new note lights the track's header up in its colour, fading out in a moment, so you can see which tracks are playing at a glance. The meters, scopes and play line move 60 times a second.

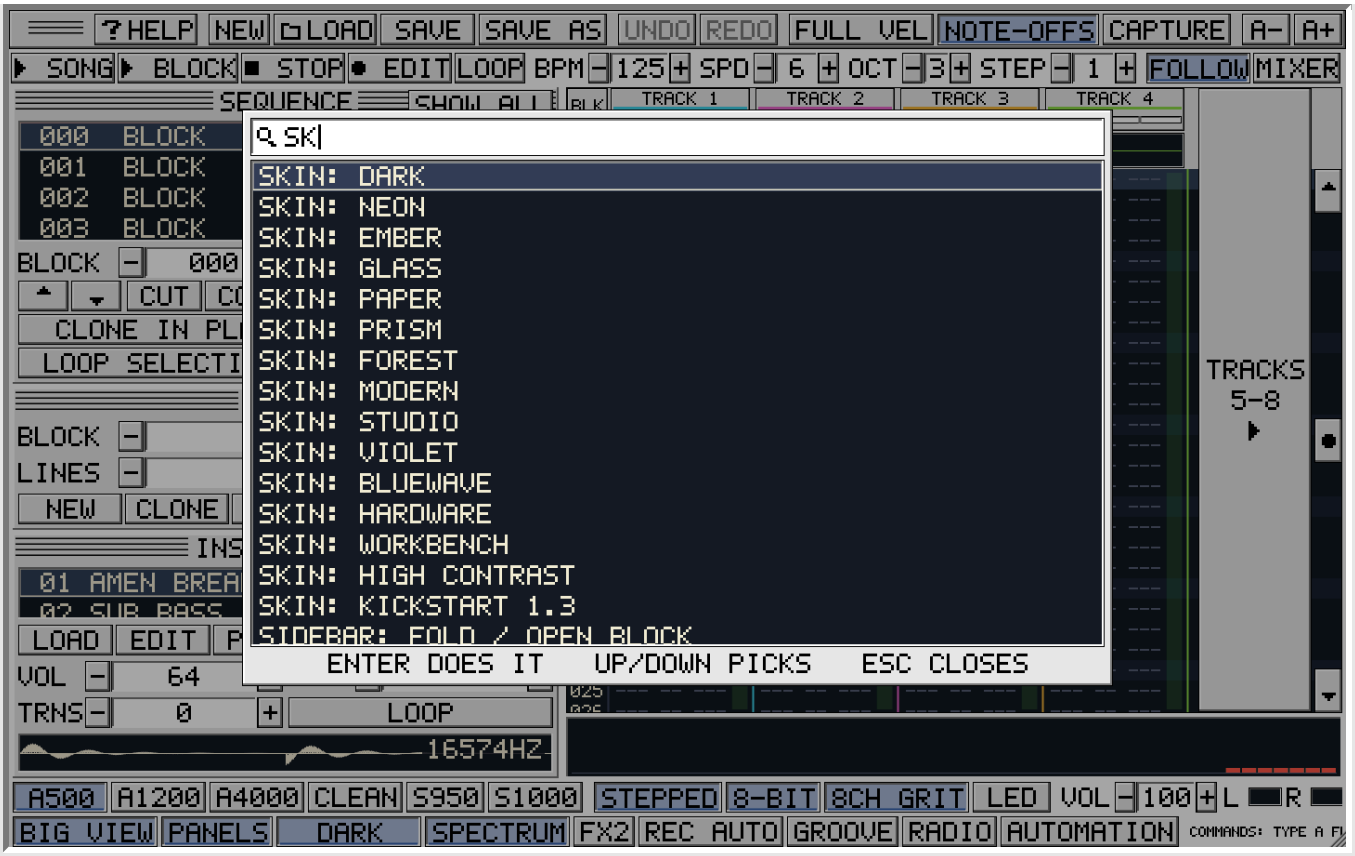
While you drag a slider, a mixer fader or one of the strips' small bars, its value shows big in a box beside it (**VOL -6.0**, **WIDTH 150%**), and goes when you let go. Buttons light up under the mouse and sink when you press them.



THE TRACK SCOPES WHILE PLAYING

THE COMMAND PALETTE

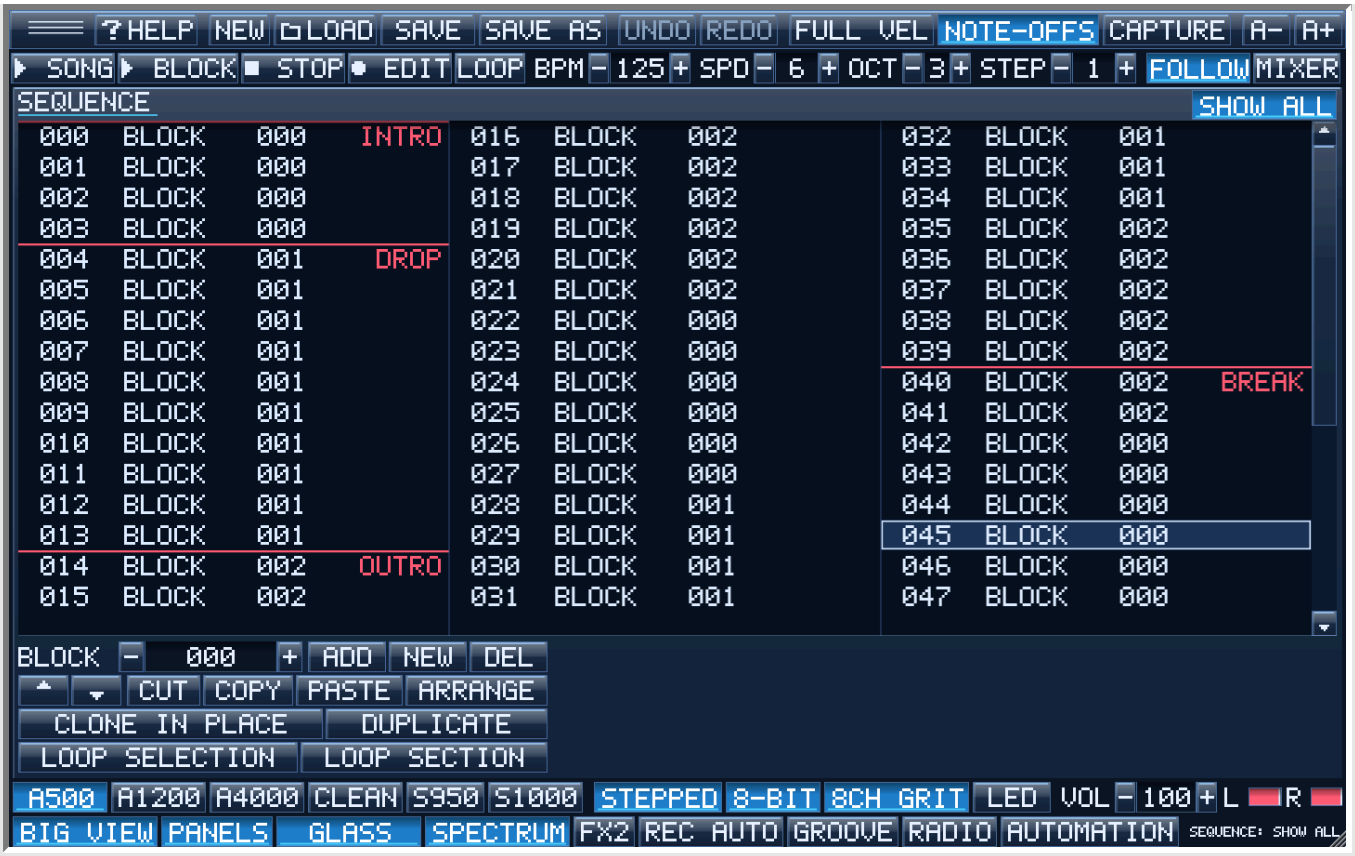
CMD K (or **View → Command Palette**) opens a box for typing a command. Type a few letters of anything the app can do - a button on screen, a page (**PAGE: MIXER**), a skin, a menu command - and the list narrows as you type. Letters in order are enough: **SVAS** finds SAVE AS. UP / DOWN pick, ENTER does it, ESC closes it. A menu command shows its shortcut beside it in orange, so the palette teaches the keys as you go.



CMD+K, WITH SK TYPED

SHOWING THE WHOLE SEQUENCE

SHOW ALL, at the right of the SEQUENCE title, gives the sequence the whole side: BLOCK and INSTRUMENTS fold down to their titles. When the song is longer than the side can show, the sequence spreads over the main area as well, in columns, so the whole song is in view at once - about 48 positions at 1440 X 900 (A- fits more; a longer song still scrolls). The buttons stay at the left, the size they always are. Clicking, dragging to move, highlighting and the section labels work in every column. Click SHOW ALL again to get the pattern and the other panels back as they were.



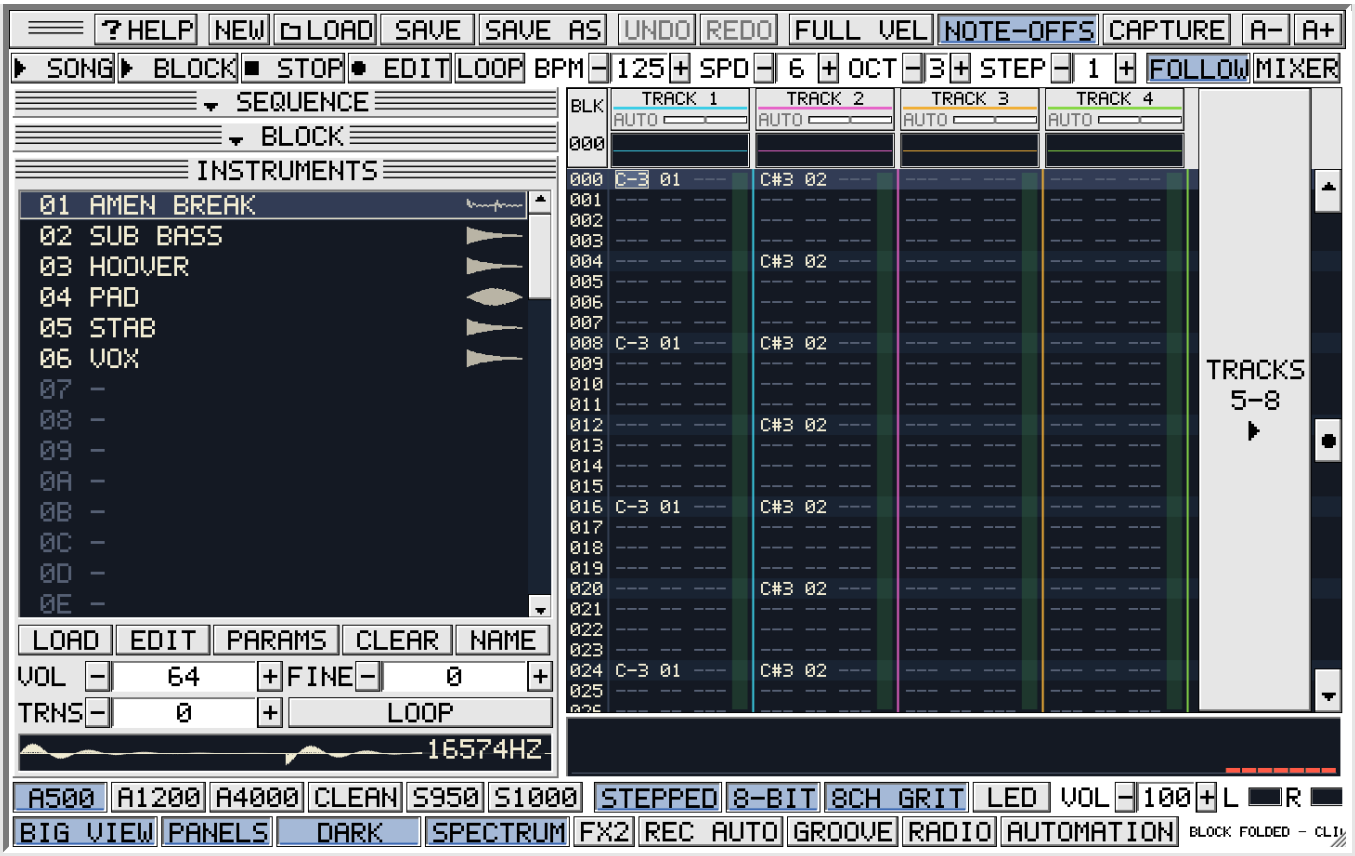
SHOW ALL: THE WHOLE SONG IN COLUMNS

FOLDING THE SIDEBAR

Click the title of a sidebar panel – **SEQUENCE**, **BLOCK** or **INSTRUMENTS** – to fold it down to just its title; its room goes to the others. Fold SEQUENCE and BLOCK and the instrument list fills the side. Click a folded title (marked with ▼) to open it again. Right-click a folded title for **TABS**, one panel open at a time (click a title to switch), or **OPEN ALL PANELS**. The sidebar is kept as you left it.

PICTURES

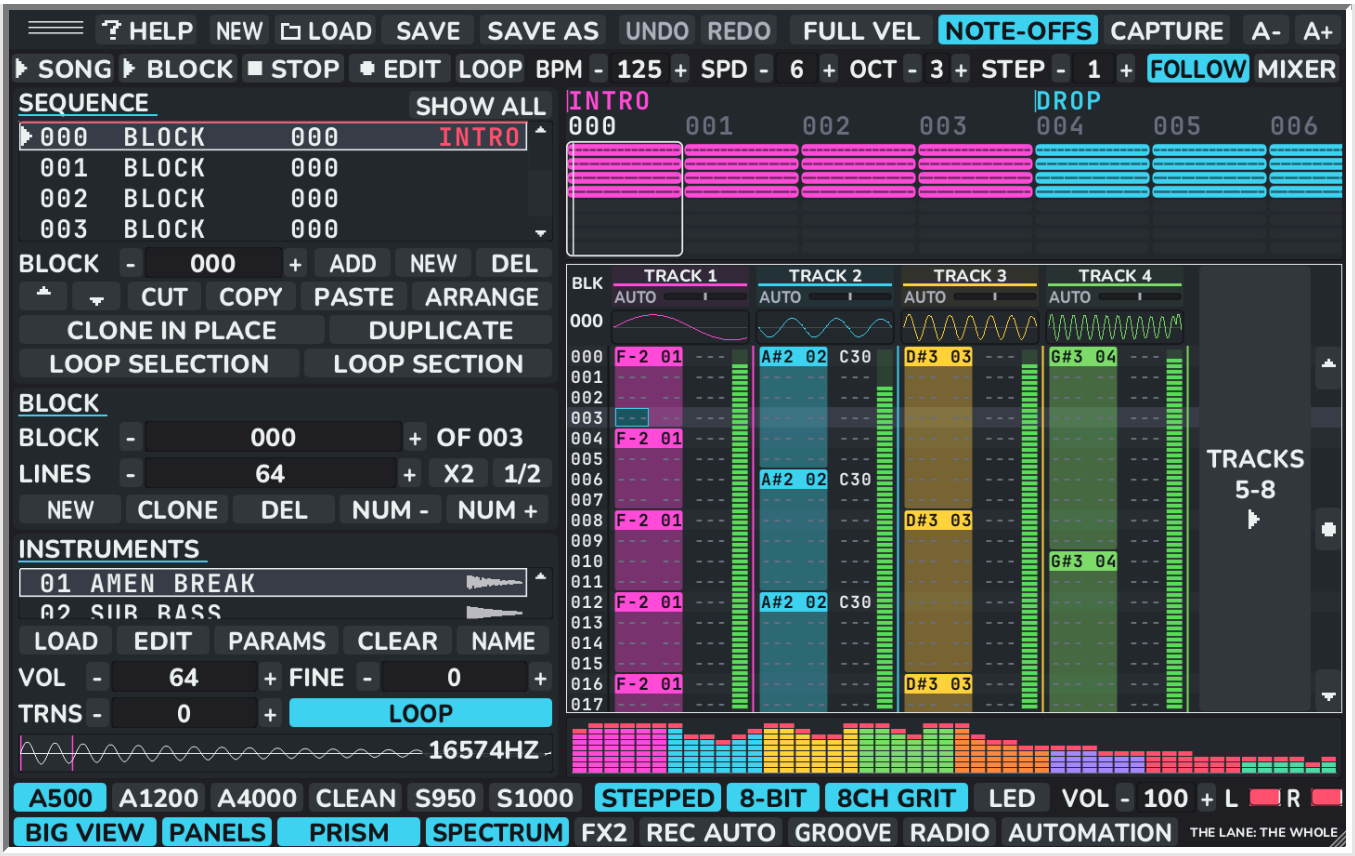
The instrument list shows each sample's waveform at the right of its row, when the name leaves room. On the mixer, a track whose EQ is on has its EQ curve drawn behind its EQ button. HELP, LOAD and other buttons in the top row get a small icon beside the label when the window has room for both – never instead of the label, and never making it smaller. The arrange page has a minimap of the whole song (see Blocks & the song).



SEQUENCE AND BLOCK FOLDED: THE INSTRUMENTS, WITH THEIR WAVEFORMS, GET THE ROOM

THE LANE

The whole song as a strip over the pattern: one column per song position, and in it a **clip** for each track, in its section's colour (a track's own colour before the first section), with the notes drawn small inside - across is when, up is how high. Section names run along the top, over the position numbers. The position you're editing is ringed; while the song plays, a playhead runs through the playing position and the lane scrolls along with it. **Click** a column to go to that position; the mouse wheel scrolls the lane. Turn it on from the sidebar menu (right-click a folded title), VIEW > THE LANE, or CMD+K and **LANE**. It's kept for next time.

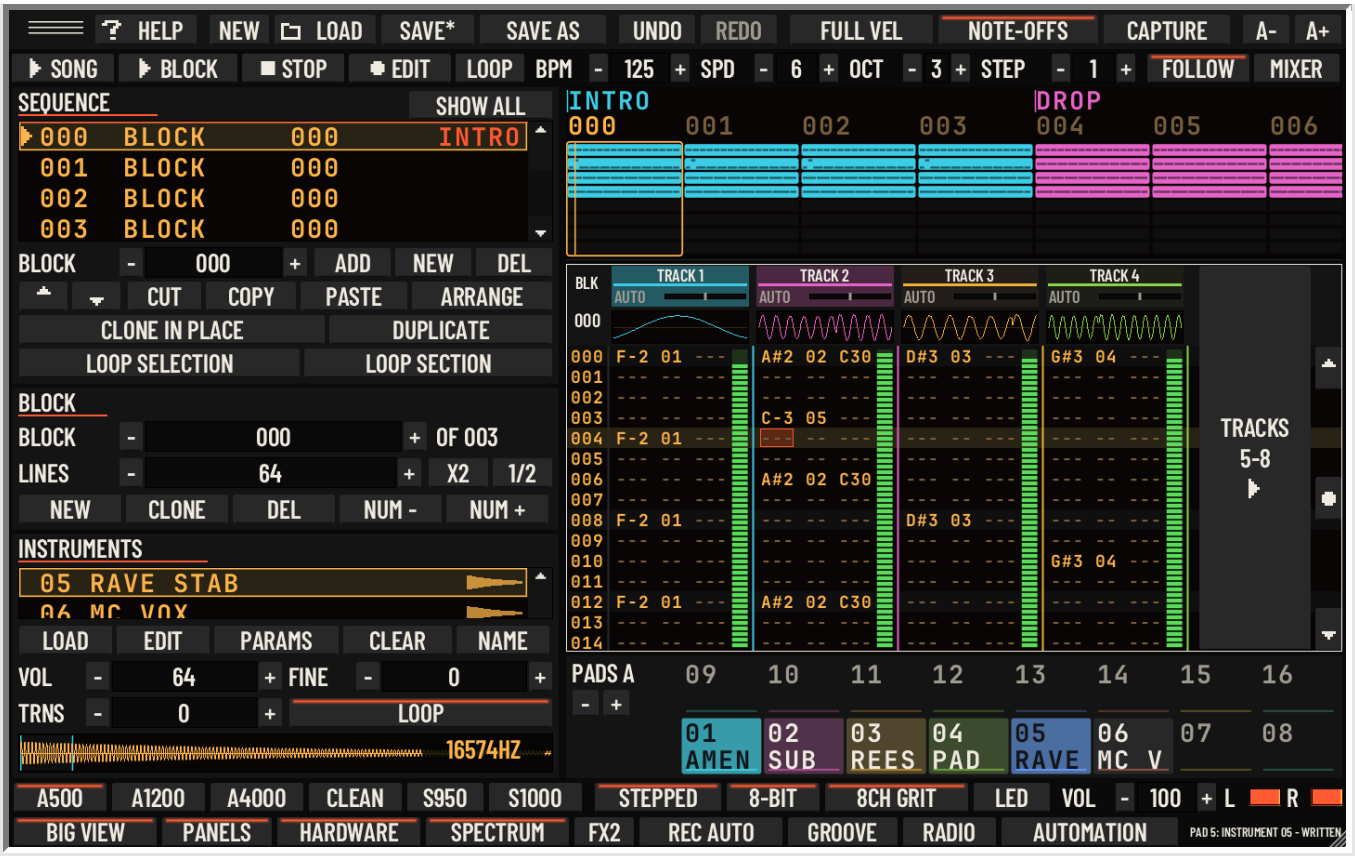


THE LANE: THE SONG AS CLIPS, COLOURED BY SECTION

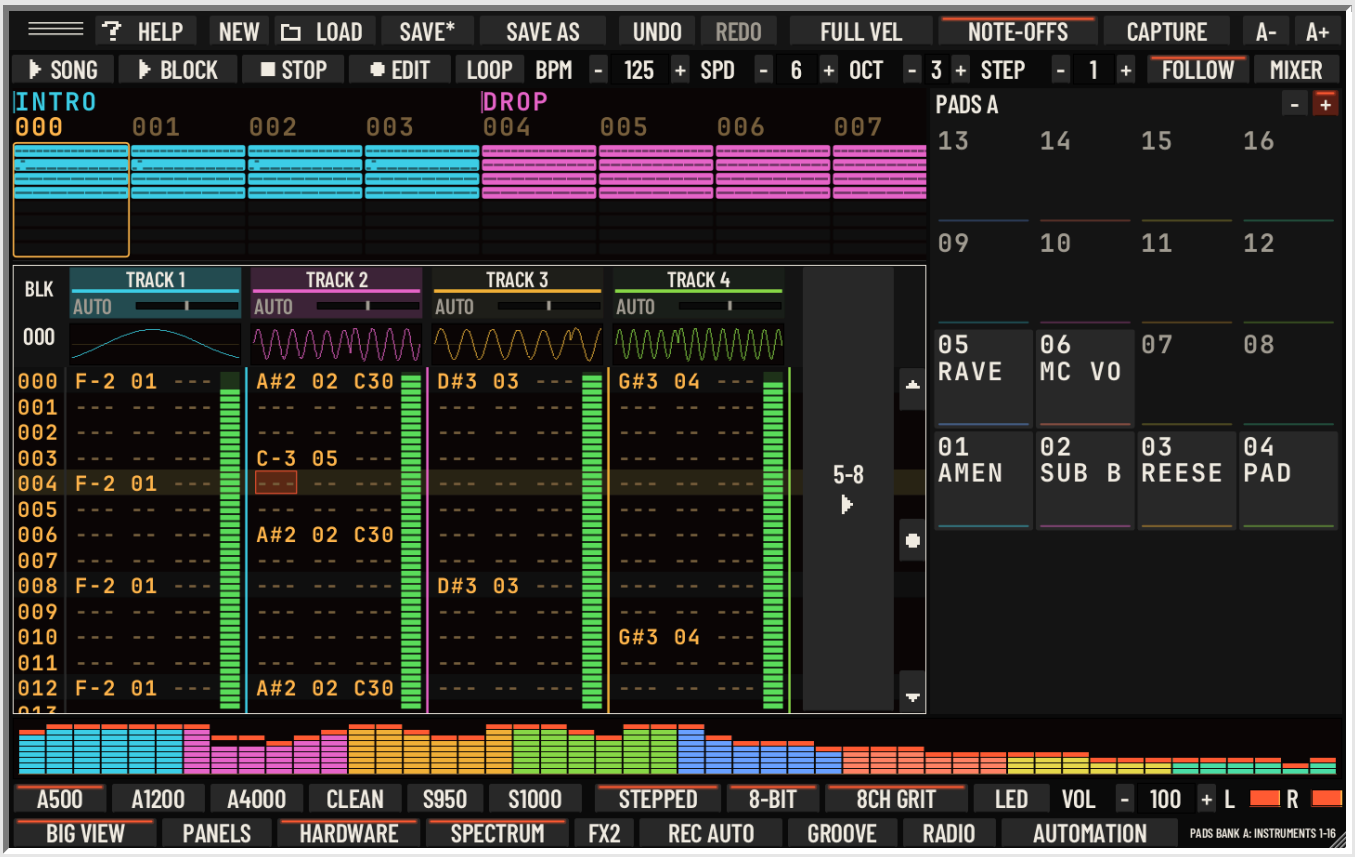
PADS

Sixteen pads, one instrument each, like a drum machine's. Press a pad to choose its instrument and play it at C of the octave (**OCT**); with **EDIT** on it's written at the cursor, just as a key would be, and the cursor steps on. Pad 1 is at the bottom left. A pad lights whenever its instrument plays, on any track. `-` and `+` change the **bank**: A is instruments 1-16, B 17-32, C 33-48, D 49-63.

With the sidebar showing, the pads are two rows of eight under the pattern, in the spectrum's place. With the sidebar off (`PANELS`) they're four by four beside the pattern. Either way the pattern keeps its size - its text never shrinks to make room. Turn them on from the sidebar menu, `VIEW > PADS`, or `CMD+K` and **PADS**.



PADS UNDER THE PATTERN (WITH THE SIDEBAR), LIT AS THEIR INSTRUMENTS PLAY



FOUR BY FOUR BESIDE THE PATTERN (SIDEBAR OFF)

WRITING PATTERNS

Each track has a note, an instrument number and an effect command per line: `C-2 01 956` means note C-2, instrument 01, command `9` with value `56`. A second effect column appears when you switch `FX2` on.

NOTES AND THE CURSOR

- With `• EDIT` on, the bottom rows of keys play and enter notes from the octave in `OCT`; the `Q` row is the octave above. `1` enters a note-off (`OFF`).
- In the instrument and command columns, type hex digits `0`–`9` `A`–`F`, or a letter `G`–`Z` for the extra commands.
- `SHIFT` `3` in a command column types `#`, the chord command (see Chords).
- `DELETE` clears the field under the cursor. `SHIFT` `DELETE` removes the line and pulls the track up; `ALT` `DELETE` inserts a blank line.
- A MIDI keyboard plays the current instrument and enters notes when EDIT is on. MIDI 60 = `C-3`.

RECORDING AND NOTE-OFFS

With EDIT on (`ESC` or `\`), playing while the song or block plays records the notes where you play them. On a plug-in or MIDI-out instrument, letting go of a key writes an `OFF` on the line you let go, so the notes are as long as you held them. When stopped (step entry), note-offs go in only if you ask: right-click `EDIT` and turn on `STEP ENTRY NOTE-OFFS`.

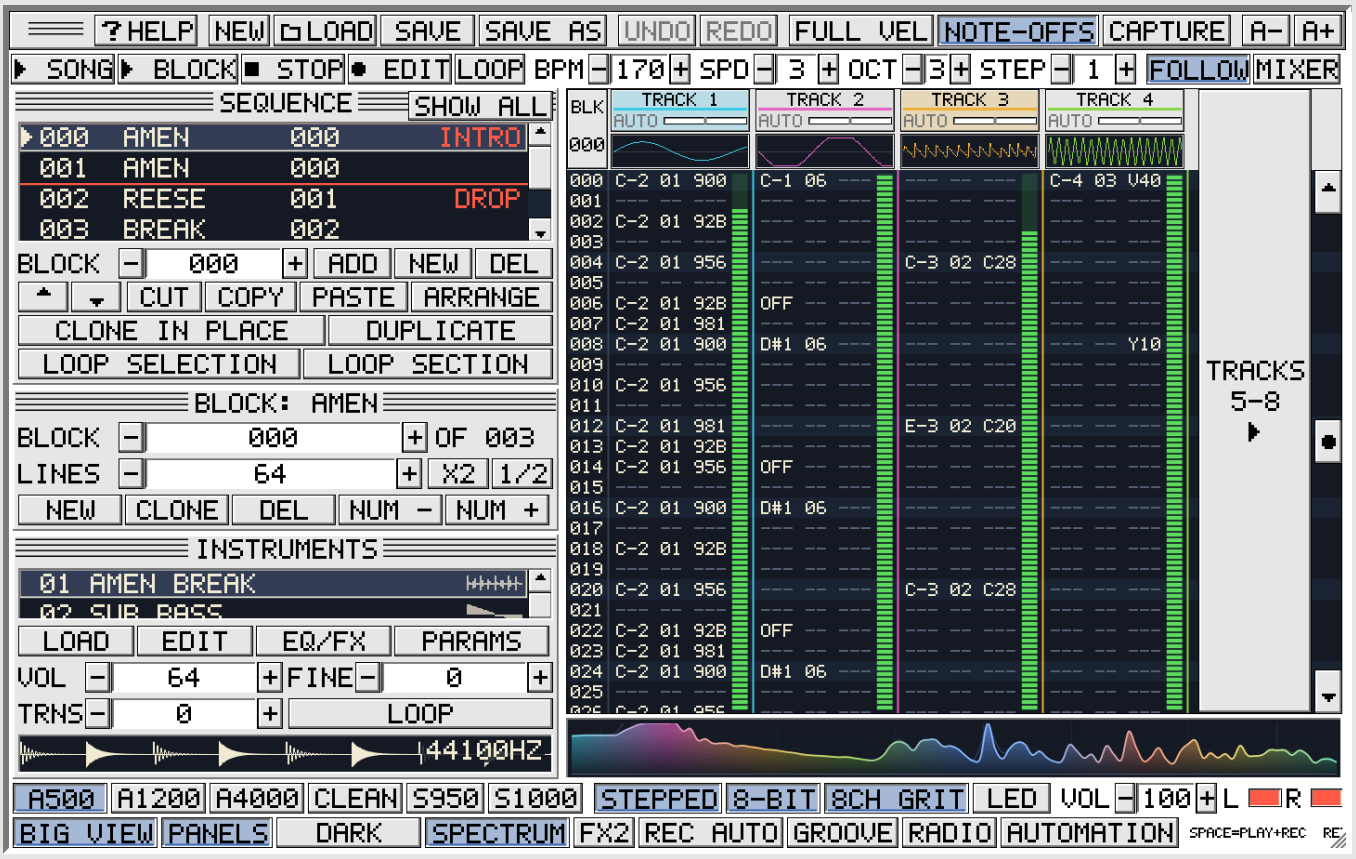
- **Legato:** let go of a key as (or just after) the next one goes down and no `OFF` is written – the next note ends the old one by itself. The OFF never lands on the next note, and the next note isn't pushed onto another track. CAPTURE does the same.
- `NOTE-OFFS` in the title bar (also right-click EDIT, and **Record Note-Offs** in the View menu) switches note-off recording off altogether: live, step entry and CAPTURE then write notes only.

- FULL VEL in the title bar (also right-click EDIT, and **Full Velocity** in the View menu) ignores how hard you hit a MIDI keyboard: every note plays and records at full velocity (127).
- **FOLLOW INSTRUMENT** (on to start with; right-click EDIT, or the View menu): clicking a note in the pattern - its note, instrument or effect columns - picks its instrument in the INSTRUMENTS list. A cell without an instrument number uses the last one above it on that track; an empty cell changes nothing.

These switches are remembered.

FOLLOW AND THE ARROWS

With FOLLOW on, the cursor moves with playback but the view doesn't scroll by itself: the play line runs down the screen and can go off the bottom. Scroll with the wheel or the arrows at the right of the pattern: ▲ / ▼ move a page (hold to keep going), • jumps to the play line. A new block starts at its top. When stopped, the arrows move the cursor a page.



PLAYING ON PAST THE BOTTOM: THE VIEW STAYS WHERE YOU LEFT IT

CAPTURE: KEEP WHAT YOU JUST PLAYED

The app is always listening to the MIDI keyboard and the laptop keys, even with EDIT off - the last two minutes. Noodle away; if you like what you played, press CAPTURE in the title bar or CMD R (or CAPTURE WHAT I PLAYED in the View menu, or right-click EDIT) and it is written into the pattern, one undo step.

- Played while the song or block was playing: the notes go where they were played, on the nearest line, with an Mxx for notes more than a third of a tick early or late.
- **One take is one time round.** Each time playback goes back to the start (the end of the song, a looping block, the loop lines) a new take begins, so loops never pile up. CAPTURE takes the latest time round that had notes in it: play on the 2nd time round, let it go round 5 more times, and it still writes the 2nd. A note held over the wrap gets its OFF where you let go, early in the block, so it rings over the loop as played.
- Played while stopped: the notes since the last pause of more than 4 seconds go in from the cursor, at the song's tempo.
- Notes start on the cursor's track; held notes spread over the tracks to the right. Chords that match the chord table go in as one note + #xy (see Playing chords in). Plug-in and MIDI-out instruments get note-offs (not between legato notes, and none with NOTE-OFFS off).

HIGHLIGHTING WITH THE MOUSE

Drag across the pattern to highlight notes and effects; it snaps to whole fields and can cover several tracks. SHIFT-click highlights from the cursor to the click, and CMD A highlights the whole block.

- CMD C / CMD X copy or cut the highlight; CMD V pastes at the cursor, each part into the same kind of field. Without a highlight they work on the whole track (add SHIFT for the block).
- CMD ↑ / CMD ↓ transpose the track a semitone, with SHIFT an octave.

- Right-click inside a highlight to bounce it to a new sample.

LOOPING SOME LINES

Drag down the line numbers to mark lines; they turn red and playback goes round them. CMD [and CMD] set the start and end at the cursor. LOOP on the transport row turns the loop off and on again.

PLAYING A BLOCK

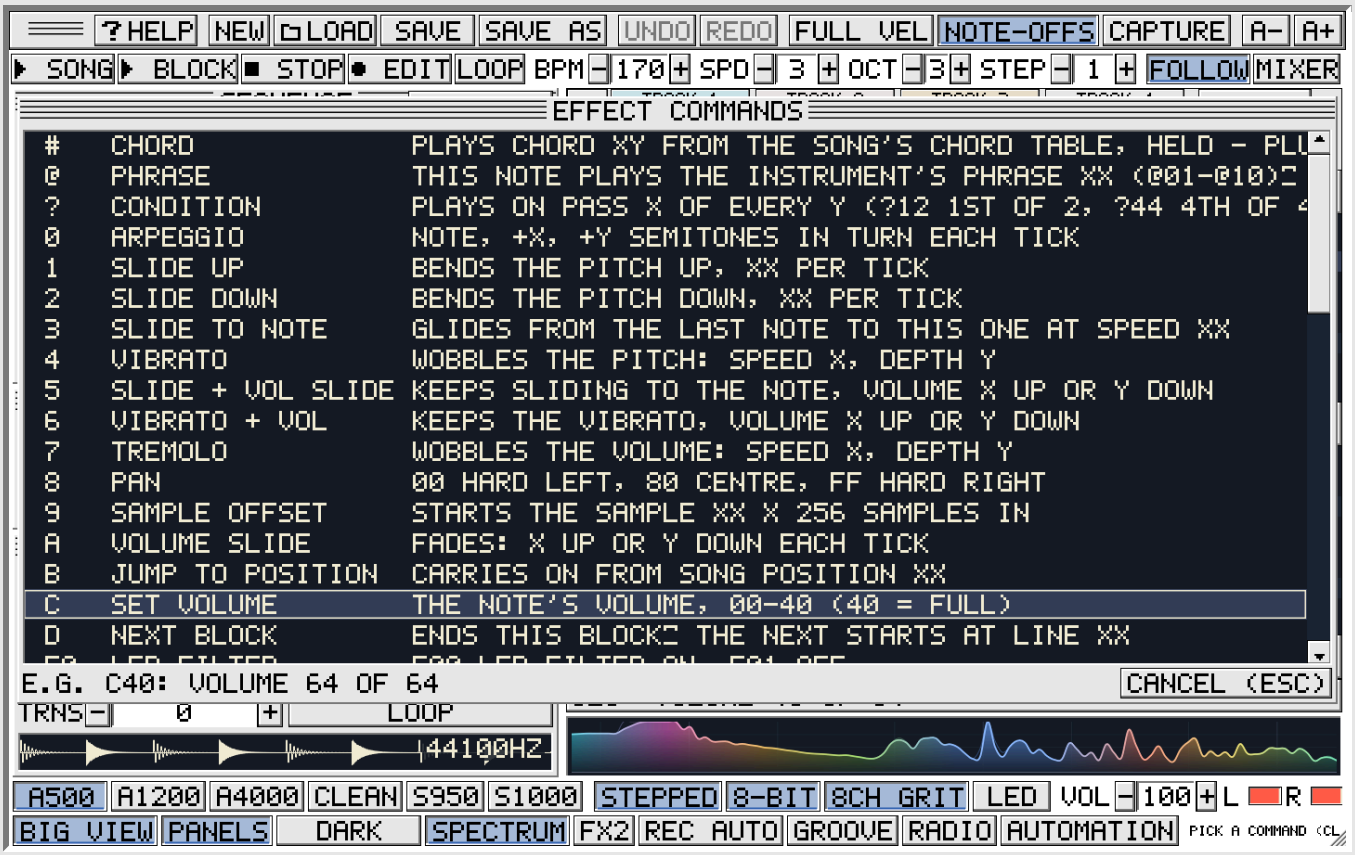
While a block plays, picking another block (in the sequence, the BLOCK box or the BLK corner) carries playback over to it on the same line, without stopping, so you can try blocks against each other in time.

UNDO

One history covers everything: notes, blocks, the sequence, instruments, samples, the mixer and song settings. CMD Z and CMD SHIFT Z work on every page, and the status line says what was undone. It keeps the last 200 steps.

EFFECT COMMANDS

Right-click an effect cell for a list of every command with what it does. Click one, or type its letter, and it goes into the cell with a useful starting value, the cursor ready on the value. Put the cursor on any command to see it explained in the help strip under the pattern.



RIGHT-CLICK AN EFFECT CELL: THE COMMAND LIST

THE CHORD COMMAND (#)

`#xy` plays chord slot `xy` from the song's chord table, held, on a plug-in or MIDI-out instrument. It is at the top of the command list; clicking it opens the chord list. See Chords.

CLASSIC COMMANDS (0–F)

The same numbering as ProTracker, so MOD files load straight in.

CMD		EFFECT
`0xy`		Arpeggio (+x, +y semitones), as in ProTracker
`1xx`	`2xx`	Slide pitch up / down

CMD	EFFECT
3xx	Slide to the note
4xy	Vibrato (speed x, depth y)
5xy 6xy	Slide to note / vibrato, plus volume slide
7xy	Tremolo
8xx	Pan: 00 left, 80 centre, FF right
9xx	Sample offset (xx × 256)
Axy	Volume slide (x up, y down)
Bxx	Jump to song position xx
Cxx	Volume (00 - 40); velocity on plug-in and MIDI-out instruments
Dxx	Next block, starting at line xx
E0x	LED filter: E00 on, E01 off
E1x E2x	Fine slide up / down
E4x E7x	Vibrato / tremolo waveform
E5x	Finetune
E6x	Loop: E60 marks the start, E6x repeats x times
E8x	Coarse pan
E9x	Retrigger every x ticks
EAx EBx	Fine volume up / down
ECx	Cut the note at tick x
EDx	Delay the note by x ticks
EEx	Delay the line by x lines
Fxx	01 - 1F speed, 20 - FF BPM, 00 stop

EXTRA COMMANDS (G–Z)

Type the letter in an effect column, then two hex digits.

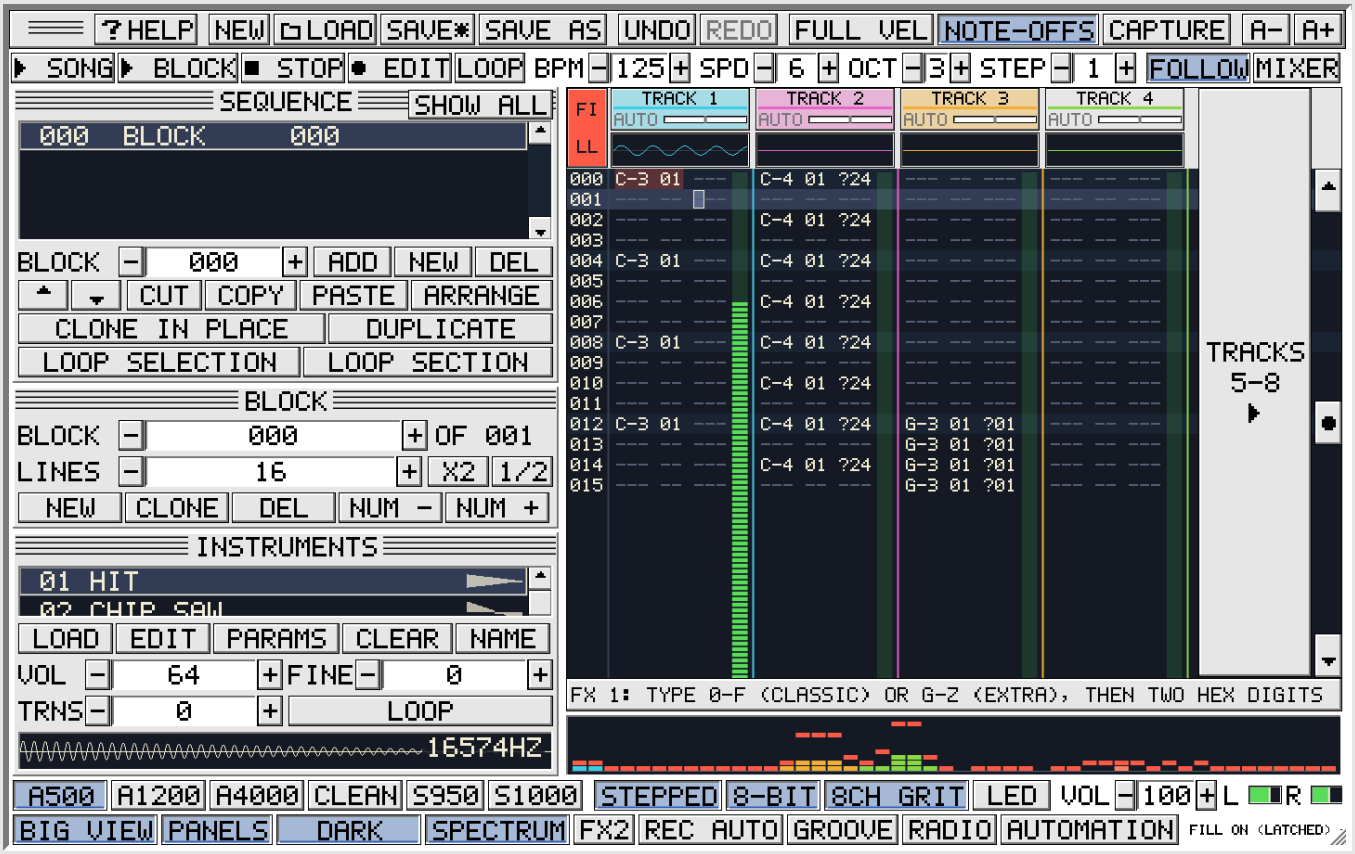
CMD	EFFECT
Gxx	Glide to the note over xx ticks

CMD	EFFECT
Hxx Lxx Wxx	High-pass / low-pass / band-pass filter, cutoff xx (00 20 Hz - FF 20 kHz). On a plug-in or MIDI-out instrument, its learned knobs instead.
Qxx	Filter resonance (or a learned knob)
Ixx	Transpose just this note: 80 none, 8C +12, 74 -12
Kxx	Chance: the step plays xx/FF of the time
?xy	Condition: plays only on pass x of every y (?01 FILL only, ?02 not in a FILL, ?03 first time only, ?04 not the first) - see below
@xx	Phrase: the note plays the instrument's phrase xx (@01-@10); @00 plays the sample - see Phrases
!xy	Dub throw: this track into the dub echo on SEND B, x how much, y lines (0 = to the next note); !0y rides the echo's feedback - see Mixer
Mxx	Micro-move: 01-7F late by xx/128 of a line, 81-FF early
Nxx	Note length: release after xx ticks
Oxx	Overdrive (on plug-in / MIDI-out instruments: the mod wheel)
Pxx	Start at xx/256 of the sample
Rxy	Roll: retrigger every x ticks, y 1-7 quieter or 9-F louder each time
Sxx	Play slice xx+1
Txx	Micro-tune: 80 = 0, each step 1 cent
U01 U00	Play backwards / forwards
Xxy	Randomise: x 0 note ±y, 1 volume, 2 pan, 3 start
Zxx	Bit crush to xx bits
Vxx Jxx	The track's automation knobs AUTO A and AUTO B (see Automation)
Yxx	The next knob change on this track glides over xx ticks

CONDITIONS AND FILL

A short block can play differently each time round, like the conditional steps on Elektron boxes.

- `?xy` in an effect column: the step plays only on pass **x** of every **y** - `?12` the first time of every two, `?22` the second, `?14` to `?44` one time in four, `?48` the fourth of every eight. Passes count from when play starts: each time the block starts again (looping, the song coming back to it, another position playing it) is the next pass.
- `?01` plays only in a **FILL**, `?02` only when not in one, `?03` only the first time the block plays, `?04` every time but the first.
- **FILL**: hold F2 while it plays - the pattern's corner turns red - or SHIFT F2 to latch it on and off. The BREAK page has a FILL button too. Put a snare roll on `?01` and the kick it replaces on `?02`, and F2 drops the fill in.
- **Highlight notes and right-click → CONDITION (WHICH PASSES PLAY)...** for the usual ones. It goes in a free effect column, or replaces a condition already there; NONE takes it off. Type `?` in an effect column to write one by hand.
- `Kxx` works with it: a step with both plays on its pass, xx/FF of the time.



?24 ON TRACK 2, A ?01 ROLL ON TRACK 3, FILL HELD

BLOCKS & THE SONG

A song is a list of positions, each playing a block. The same block can play at many positions; change it once and every position hears the change.

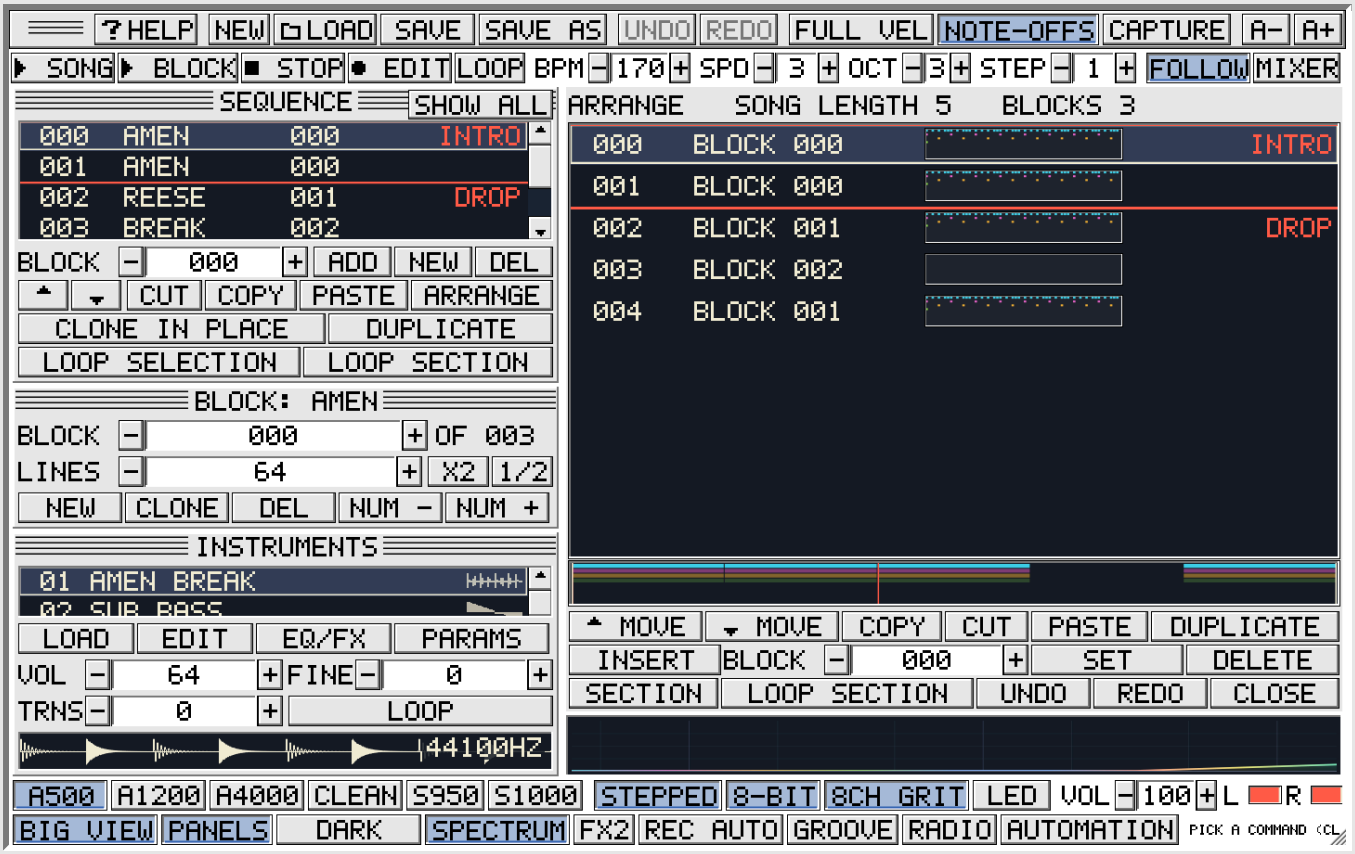
THE SEQUENCE PANEL

- ADD puts a new position after the selected one, playing the same block; change it with BLOCK - +. NEW makes a new empty block in a new position.
- Drag over the names to highlight positions, then CUT COPY PASTE DUPLICATE DEL. To move them, drag by the **position numbers** on the left (a hand shows there), or use ▲ ▼.
- **COMBINE INTO ONE BLOCK** (right-click a highlight): the highlighted blocks, one after another, become one new block in their place (up to 256 lines).
- LOOP SELECTION (or right-click) loops the highlighted positions while the song plays. The list follows the position playing.
- CLONE IN PLACE gives a position its own copy of its block, in the next free block, so you can change it without touching the other places it plays.
- Click the label at the right of a row (or + SECTION) to start a named **section** there (INTRO, DROP…). LOOP SECTION loops the section you're in; pick a position in another section and playback goes there when the current block ends.
- **Block names:** double-click a position (or right-click → BLOCK NAME...) to name its block (AMEN, REESE…). The name takes the place of the word BLOCK - 000 AMEN 003 DROP - with a gap before the block number (which never changes) and before the section name. A name too long for its space scrolls along on the playing, highlighted or pointed-at row. The BLOCK panel's title shows it too.

THE ARRANGE PAGE

ARRANGE shows the song order full size, with a picture of each block's notes. It shares the highlight and clipboard with the sequence panel. INSERT adds the BLOCK number after the cursor, SET makes the highlighted positions play it.

Under the list is the **minimap**: the whole song in one strip, a column for each position with its notes in the tracks' colours (brighter = busier), the sections marked in red, the playing position as a line and a frame round the positions the list is showing. Click or drag in it to go there.



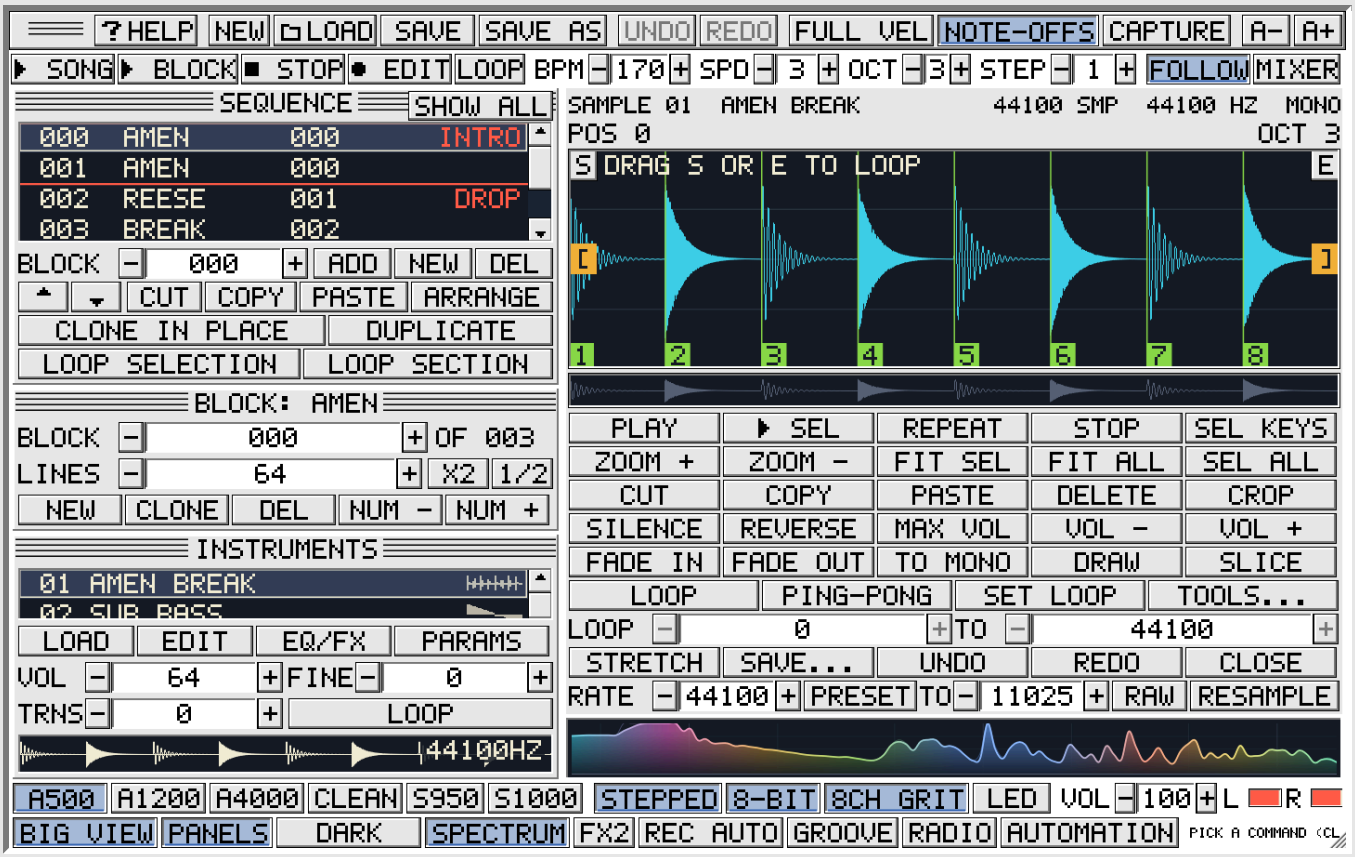
THE ARRANGE PAGE, WITH TWO SECTIONS

THE SAMPLE BROWSER

LOAD opens the browser in place of the pattern, with four tabs along the top for what the instrument plays: SAMPLES, VST / AU (your instrument plug-ins), MIDI OUT (hardware) and CV OUT (a modular; it asks first). Right-clicking an instrument offers the same. Every sample you highlight plays straight away, and the note keys or a MIDI keyboard play it at other pitches. Shortcut buttons go to Home, Music, Desktop, Downloads, Documents and Drives, and four FAV buttons hold your own folders (click an empty one to set it; SHIFT-click changes it, ALT-click clears it). ENTER or a double-click loads the sample.

THE SAMPLE EDITOR

EDIT (or a double-click on the little waveform) opens the editor.



THE SAMPLE EDITOR, WITH SLICES

- Drag to select; grab either end of a selection to move just that end. The note keys and MIDI play the sample through the full instrument sound; SEL KEYS plays just the selection.

- CUT | COPY | PASTE | DELETE | CROP | SILENCE | REVERSE | MAX VOL | VOL - | VOL + | FADE IN | FADE OUT | TO MONO, and DRAW to draw the wave.
- TOOLS...: GAIN, NORMALISE, REMOVE DC, TRIM SILENCE, INSERT SILENCE, CROSSFADE LOOP (a smooth jump back), PITCH (semitones, resampled), SWAP LEFT / RIGHT, START / END from the selection, SAVE SAMPLE. SAVE... writes the sample to a WAV wherever you choose.
- **START / END:** drag the green [and] handles at the middle of the wave: only the part between them plays, the rest is shaded but kept. They're separate from the loop (which is cut at END); 9xx counts from START.
- SPACE stops whatever plays, the song too; when nothing does, it plays the sample.
- **Loop:** drag the S and E handles at the top of the wave; you hear the loop while you drag. PING-PONG plays it forwards then backwards.

RATES

Each instrument has a RATE: the speed it plays at on C-3. Samples load at their own rate, so they sound as recorded. RATE changes it without touching the data (lower rate, lower and slower); RESAMPLE converts the data to a new rate and keeps the pitch, with RAW for the rough, unfiltered old-sampler way.

SLICING

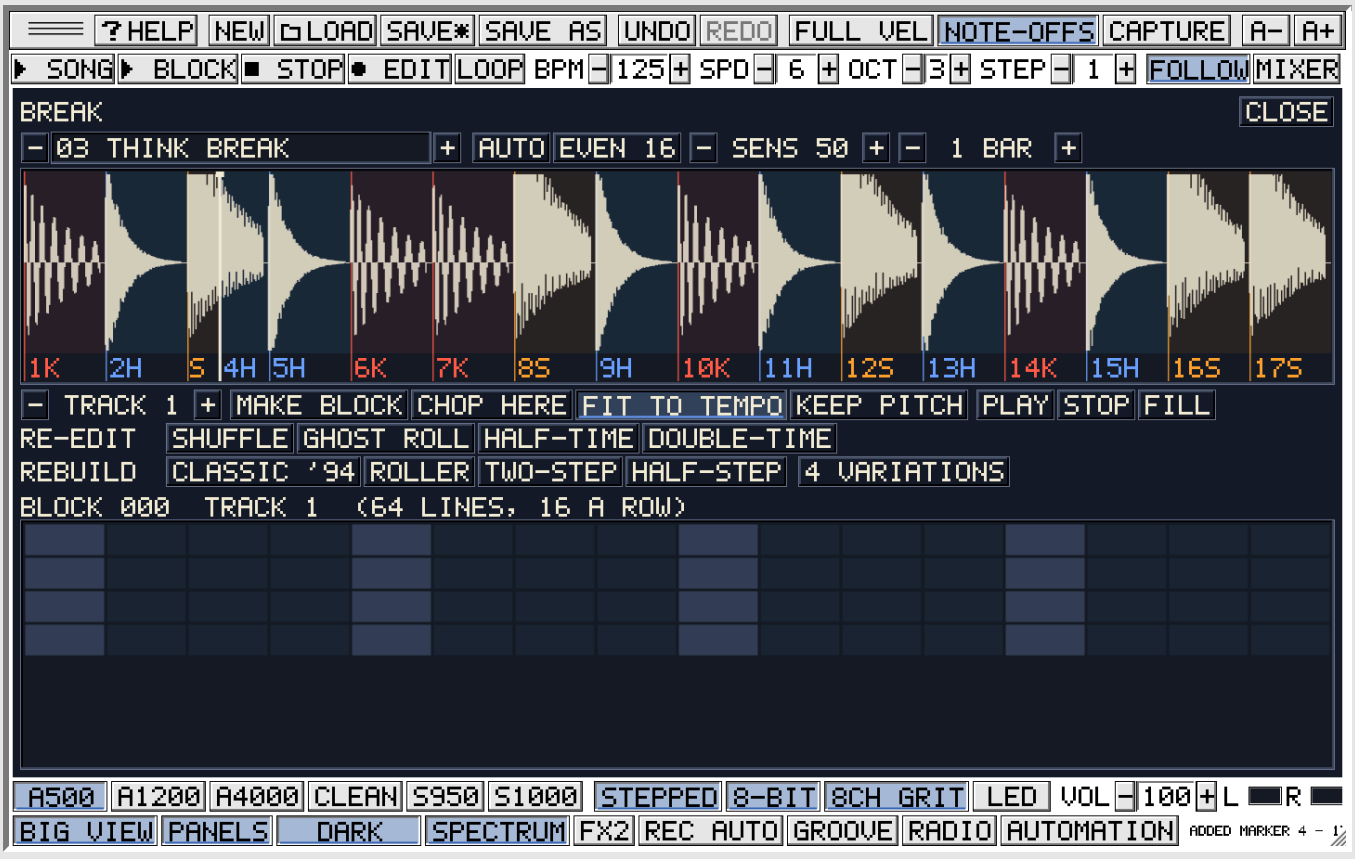
SLICE opens the slicer. AUTO finds the hits (SENS sets how quiet), EVEN cuts equal slices. Drag a tab to move a slice point, double-click the wave to add one, ALT-click a tab to remove it. With SLICE TO KEYS on, C-3 plays slice 1, C#3 slice 2 and so on; without it, Sxx picks the slice.

THE BREAK PAGE

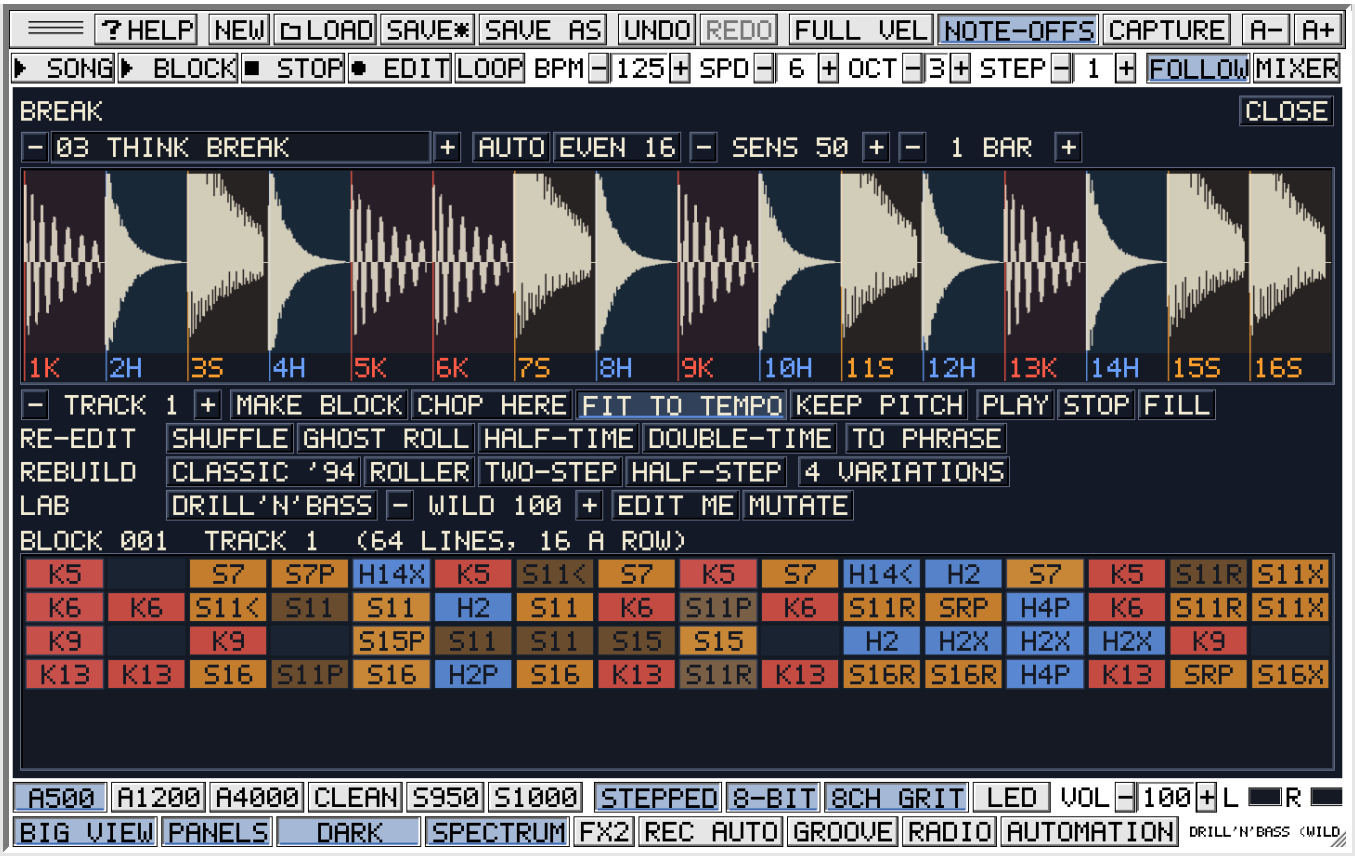
CMD 6 - or right-click an instrument and choose **BREAK PAGE...** - opens one page for a break, start to finish. CMD 6 again, CLOSE or ESC goes back.

- **The waveform** shows the break's slices, each tinted and tagged with what it is: **K** kick (red), **S** snare (orange), **H** hat (blue), **O** anything else. Paula works it out from each hit's first 60 ms - how much low end, how noisy, how loud. Click a slice to hear it; right-click it to put it right when it guessed wrong.
- **The markers** - where each slice starts - are yours to fix. **Drag** one left or right to move it; it can't pass its neighbours, and the arrow keys nudge the one you picked (Shift for bigger steps). **Double-click** the wave to **add** a marker there: it lands on the nearest zero crossing, so the slice starts without a click. **Double-click** a marker, **Alt-click** it, or pick it and press DELETE to take it away. Right-click has **ADD A MARKER HERE** and **DELETE MARKER** too. The first marker always stays at the start. The kinds are worked out again for the new slices, and every change can be undone. Slices are numbered left to right, so adding or deleting one changes which slice the notes after it play.
- AUTO finds the hits (**SENS** - / + for quieter ones); EVEN 16 cuts it into 16ths. **BARS** (1, 2 or 4) is how long the break is.
- MAKE BLOCK adds a new block after the current position with the break chopped onto **TRACK**, each slice where it falls, repeated to the block's end. CHOP HERE does it on the current block. FIT TO TEMPO (on to start with) sets BEAT SYNC so the break plays in time at any BPM; with KEEP PITCH as well, it's stretched to fit rather than repitched. PLAY / STOP (or SPACE) play the block.
- **RE-EDIT:** SHUFFLE (the rhythm stays, which slice plays changes - again for another), GHOST ROLL (the last quarter becomes a snare roll, quiet to loud), HALF-TIME, DOUBLE-TIME. TO PHRASE: the slices into a new phrase (see CHOP TO PHRASE).
- **REBUILD** it from its own kicks, snares and hats, 16 steps a bar: CLASSIC '94 (the cut-up amen - kick pushed, a ghost snare before the second half, a snare on the last 16th), ROLLER, TWO-STEP, HALF-STEP. Ghost snares go in quiet, as C14.
- 4 VARIATIONS makes four new blocks after the current one, each a different rebuild with different hits, the last ending in a ghost roll.

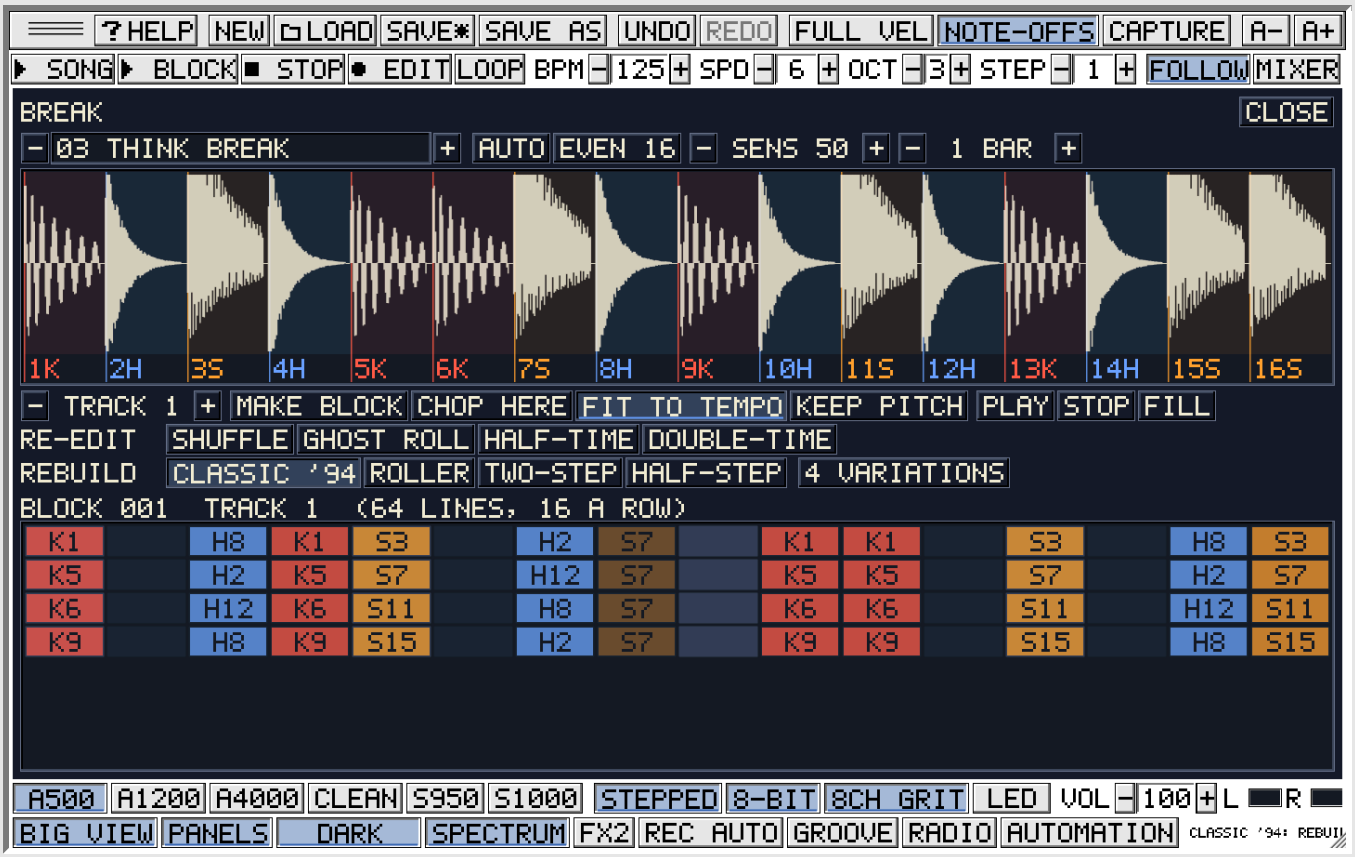
- **BREAK LAB** (the **LAB** row) writes jungle edits for you on the break's track. **STYLE** (click for the next): '94 ROLLERS (busy, rolling, ghost snares, hats a touch late), '96 TECHSTEP (tight and sparse, the kick skipping, hats cut short), AMEN CHOPS (reversed snares, pitched-down hits, stutters), DRILL'N'BASS (buzzing rolls, stutters, pitch everywhere). **WILD** - / + sets how many edits (0 = just the groove). EDIT ME writes a whole new edit - a groove from the style, then ghost snares, rolls (Rxy), reversed hits (U01), pitch drops (Ixx), stutters cut short (ECx), a gap before the drop, now and then a half-time bar, with every second bar a fill. Press it again for another. MUTATE changes a few things and keeps the rest, so an edit can evolve. In the grid an edited hit shows its edits: **R** roll, **<** reversed, **P** pitched, **X** cut short. It's all ordinary notes and commands, every press one undo step.
- The grid at the bottom is the block's track, a bar (16 lines) a row, each note tinted by kind with its slice number; it lights the line that's playing. Every change is one undo step.



A MARKER ADDED WITH A DOUBLE-CLICK (THE PICKED MARKER IS THE THICK WHITE LINE)



BREAK LAB: DRILL'N'BASS AT WILD 100

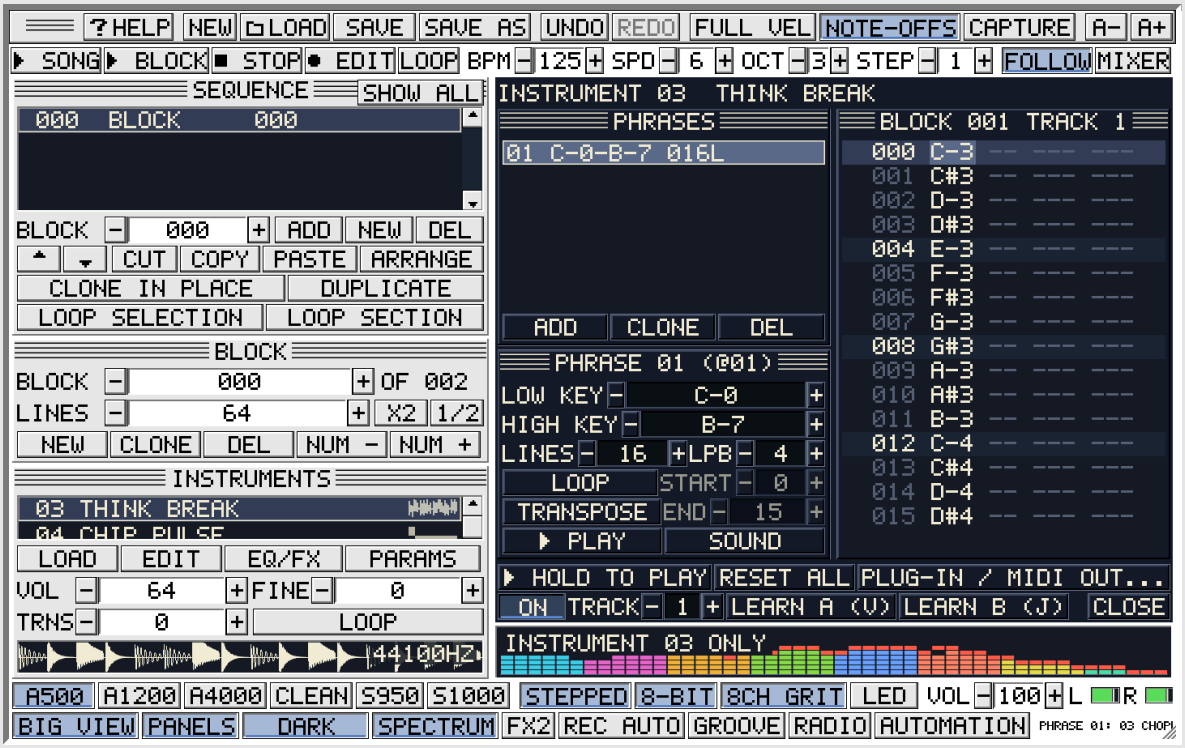


THE BREAK PAGE, AFTER CLASSIC '94

GROOVE AND BREAK TOOLS

- **CHOP TO PHRASE...** (right-click an instrument): the break cut up - BY ITS HITS, ITS OWN SLICES, or 4, 8, 16, 32 or 64 EQUAL SLICES -

into a new phrase, every hit a note (`C-3` = hit 1), each where it falls, at enough lines a beat that every hit has a line. TRANSPOSE is off, so any note plays the chopped break. The PHRASES page opens on it. The BREAK page's `TO PHRASE` does the same with its slices.



A BREAK CHOPPED INTO A PHRASE

- **CHOP TO PATTERN** (right-click an instrument): the break is sliced by its hits (if it has no slices yet) and laid out as notes on the cursor's track, each slice on the line where it falls, with SLICE TO KEYS on.
- **Right-click a highlight:** `REVERSE THE LINES`, `STUTTER` (the first quarter over and over), `RATCHET` (rolls that speed up across it, `R40` to `R10`) and `RE-SEQUENCE THE SLICES` (the rhythm stays, the slices shuffle; again for another order).
- `GROOVE` (bottom row): SWING 16THS, SWING 8THS, LAZY BREAK, PUSHED HATS or HUMANISE, at 25–100%. Some lines' notes play a little late as the song plays; the pattern isn't changed. Notes with their own `Mxx`, `EDx` or a slide keep their timing.

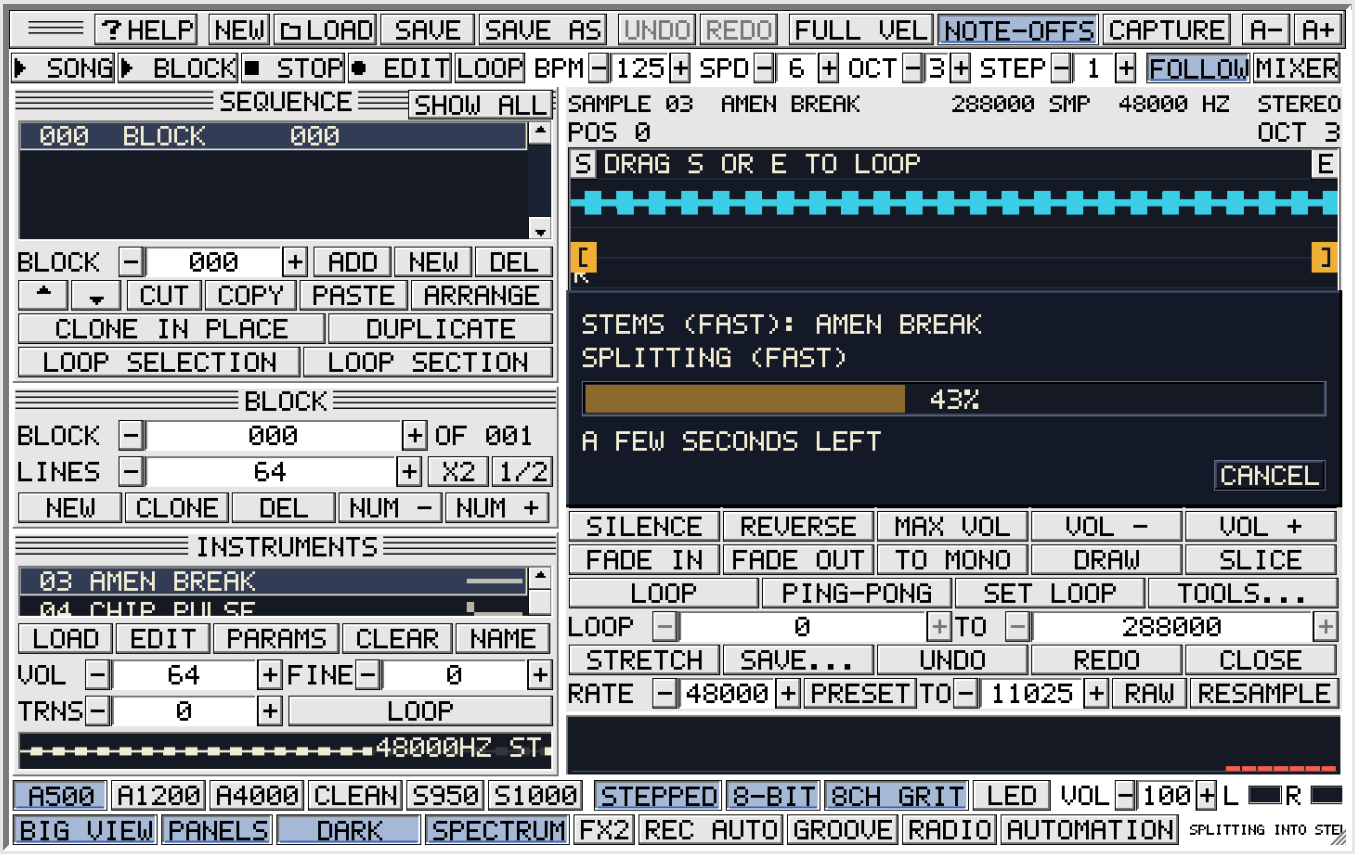
TIME STRETCH

`STRETCH` opens the old-sampler stretch: **CLASSIC** (the Akai sound, matched to recordings of the real thing), **REVISED** (cleaner joins),

HARD (no blending) and **AKAIZER** (Akaizer v1.2's granular method, rebuilt). Set TIME %, SEMI and CENTS and the CYCLE length; FIT TO LINES makes the selection last a number of lines at the song's tempo. PREVIEW, then APPLY.

STEMS: SPLITTING A SAMPLE

TOOLS... → **SPLIT INTO STEMS...** in the sample editor splits the sample into **DRUMS**, **BASS**, **OTHER** and **VOCALS**: four new instruments in the first empty slots, stereo at 44.1 kHz. It runs **Demucs v4** (Meta AI Research's hybrid transformer) on your computer - nothing is uploaded.



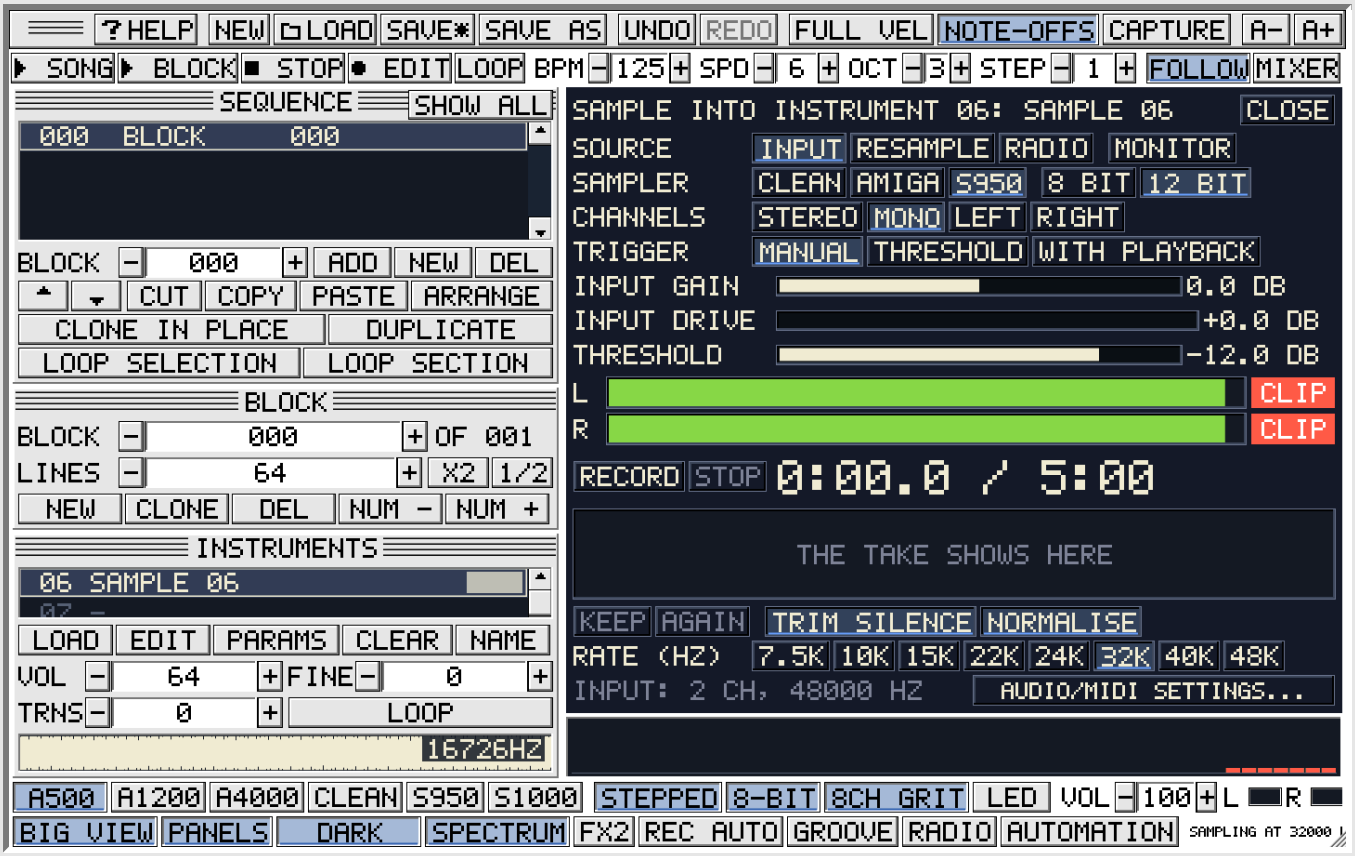
SPLITTING A BREAK: THE MUSIC AND THE OTHER PAGES CARRY ON

- **BEST**: four fine-tuned models, each the specialist for its stem - the cleanest split, about four times slower. **FAST**: one model, for a quick listen.
- The models download once, the first time (FAST about 175 MB, BEST about 700 MB), checked against their SHA-256.
- As a guide, FAST takes about as long as the sample lasts and BEST about four times as long (more on an older computer). A box shows

the progress and the time left; CANCEL stops it. The four stems add back up to the original, and each is one undo step.

SAMPLING: RECORDING AUDIO

Record straight into the selected instrument, from an audio interface, the mic or line-in, or from the app's own output. Open the SAMPLE page with CMD 4, **Sample Audio…** in the View menu, right-click an instrument → **SAMPLE (RECORD AUDIO)…**, or click an empty instrument's **NO SAMPLE - CLICK TO RECORD**.



SAMPLING THROUGH THE S950, ITS INPUT DRIVEN INTO THE CLIP LIGHTS

CONTROL	WHAT IT DOES
INPUT RESAMPLE RADIO	Record the audio input (chosen in **AUDIO/MIDI SETTINGS…**, bottom right), or everything the app plays: play the song or a block, or jam, and record what comes out - or the internet radio as it comes in (see below)
MONITOR	Hear the input through the app. Off to start with: with a mic, use headphones or it feeds back
STEREO MONO LEFT RIGHT	What goes into the instrument (MONO mixes left and right)
MANUAL	Records from the moment you press RECORD

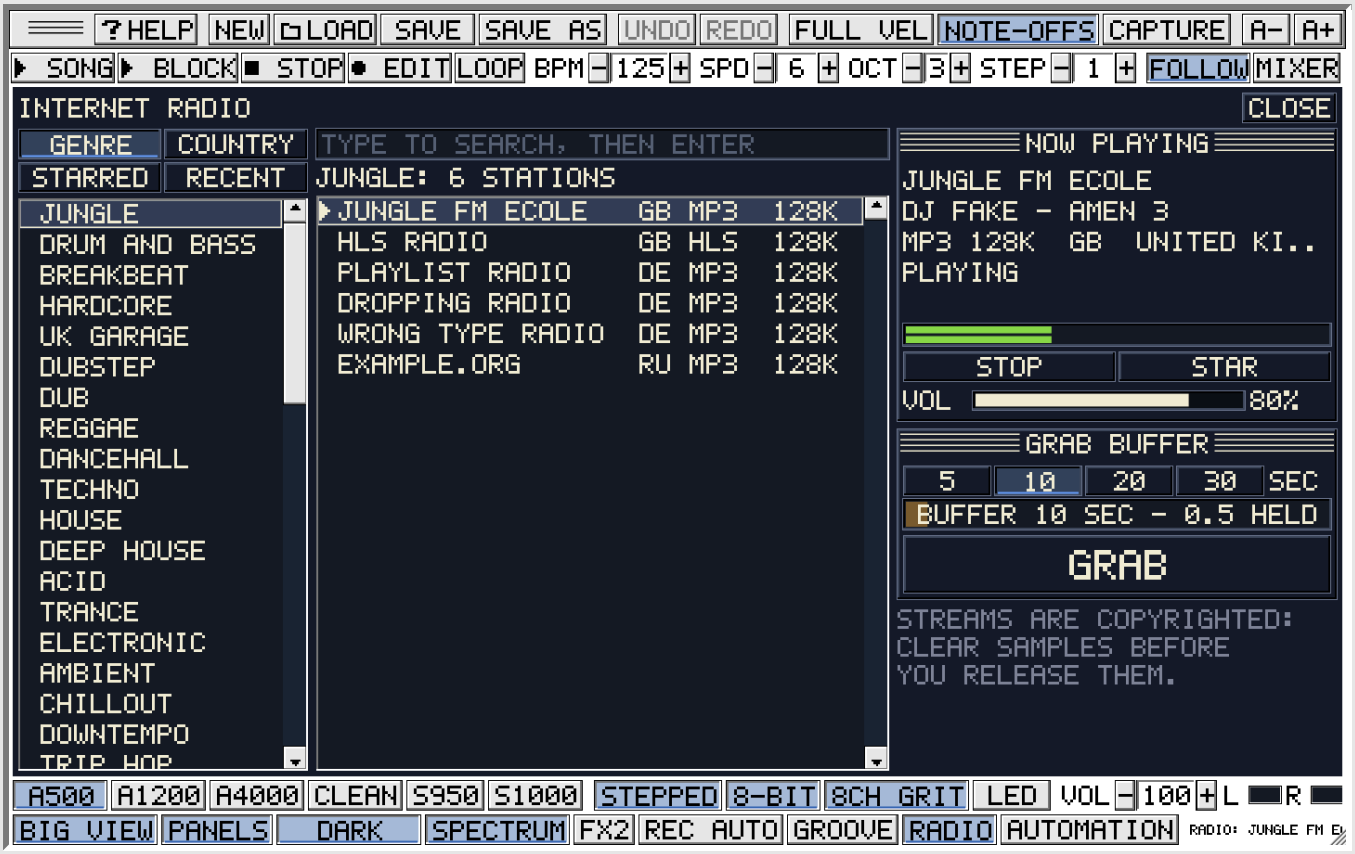
CONTROL	WHAT IT DOES
THRESHOLD	RECORD arms it; it starts when the sound goes over the THRESHOLD level (the red line on the meter), keeping the 20 ms before so the attack isn't cut
WITH PLAYBACK	RECORD arms it; it records from when the song or block starts to when it stops: one block, exactly
INPUT GAIN	-24 to +24 dB. The meter shows the level, a peak hold and a CLIP light
RECORD STOP	Start (or arm) and end it (R). Up to 5 minutes. It carries on if you go to another page
KEEP AGAIN	The take into the instrument, one undo step (ENTER), or thrown away and recorded again (BACKSPACE)
TRIM SILENCE NORMALISE	Cut quiet at both ends; bring the loudest point to full level. Both on to start with
CLEAN AMIGA S950	How the sound gets in. CLEAN (the default): the interface as it is. AMIGA: 8-bit at 8363 / 16726 / 22050 / 28867 Hz with no anti-alias filter (the aliasing grit). S950: 12-bit at 7.5K-48K through its filter, which follows the rate. You hear it as you sample, with MONITOR on
8 BIT 12 BIT	The bits, for either sampler (an S950 at 8 bits, an Amiga at 12)
INPUT DRIVE	0 to +24 dB into the sampler's input: the S950 goes soft and fat then hits the ceiling (the CLIP lights), like the classic overloaded records; an Amiga sampler just clips
AMIGA RATE	CLEAN only: NONE keeps the recording's rate; 8363, 16726, 22050 or 28867 Hz resamples it as you keep it, at the same pitch

The first time on a Mac, macOS asks whether Paula's Revenge may use the microphone: say OK, or the input is silent (System Settings → Privacy & Security → Microphone changes it later). After KEEP it's a sample like any other: EDIT to slice or loop it, and it's saved in the song.

INTERNET RADIO

RADIO (bottom row, or CMD 5) opens stations from all over the world to listen to and sample from - straight into the sampler, no

effects. The stations come from Radio Browser, a free community directory; it needs an internet connection.



THE RADIO: A GENRE, A STATION PLAYING, GRAB READY

- GENRE (jungle, drum and bass, breakbeat, dub, reggae, techno, house, jazz, funk, soul, hip hop, classical, world…), COUNTRY (the busiest first), or **type** to search by name or genre and press ENTER. STARRED and RECENT are your starred stations and the last 30 played.
- Stations that worked at the directory's last check are listed, the most played first, with their format: **MP3**, **AAC**, **AAC+** (HE-AAC) and **HLS** – the playlists of short pieces big broadcasters such as the BBC use. AAC is decoded by the Mac's or PC's own decoder. Ogg, Opus, FLAC and encrypted streams aren't played yet. The last lists are saved, so the page opens straight away, even offline.
- **Double-click** a station (or ENTER, or PLAY). It gathers 3 seconds before it starts, so a slow connection doesn't stutter; a dropped stream is tried again by itself (RECONNECTING 2/5), waiting longer each time. NOW PLAYING shows the track the station says is on.

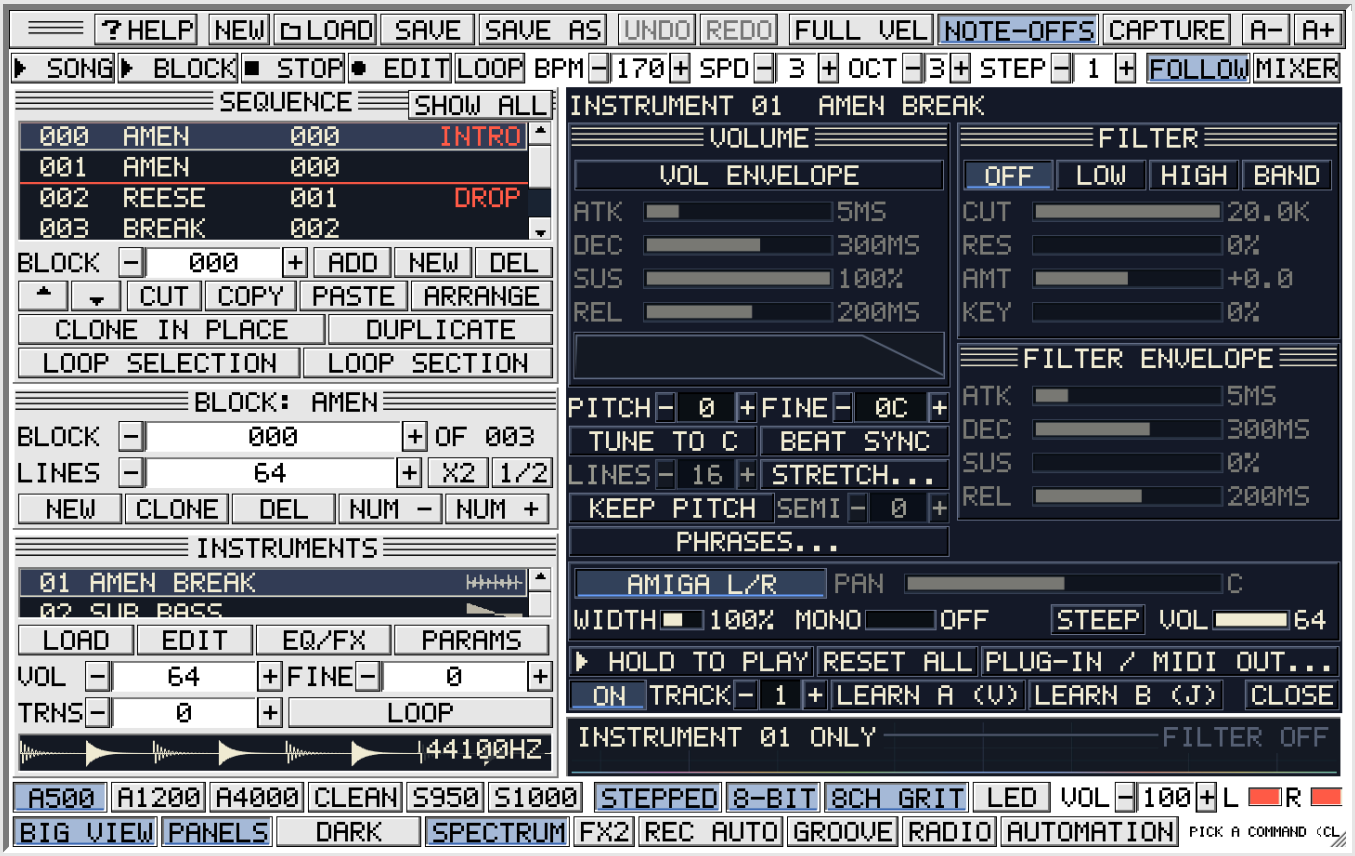
- VOL is its own. The radio goes **straight to the output**: not through the mixer, effects or the Amiga / Akai sound, and never into renders or bounces. It plays on while you work on other pages (RADIO stays lit).
- **GRAB BUFFER** 5 | 10 | 20 | 30 seconds: the radio always keeps that much of what it just played. GRAB puts it in the **first empty instrument**, named after the station, exactly as it came off the stream (stereo, not trimmed or normalised), stops the radio, and opens it in the sample editor. One undo step.
- Streams are copyrighted: sample for your own sketches, and clear samples before releasing them.

INSTRUMENT VOL AND SWITCHING OFF

VOL is a level for the whole instrument on top of each note's Cxx. Any instrument can be switched off with its ON button (on its PARAMS page): its notes are silent and the list shows it dimmed with OFF.

SOUND SHAPING

PARAMS opens the current instrument's page. Everything starts off or neutral, so MODs and older songs sound exactly as they did.



PARAMS: ENVELOPE, FILTER, PAN AND UTILITY, AND THE ON BUTTON

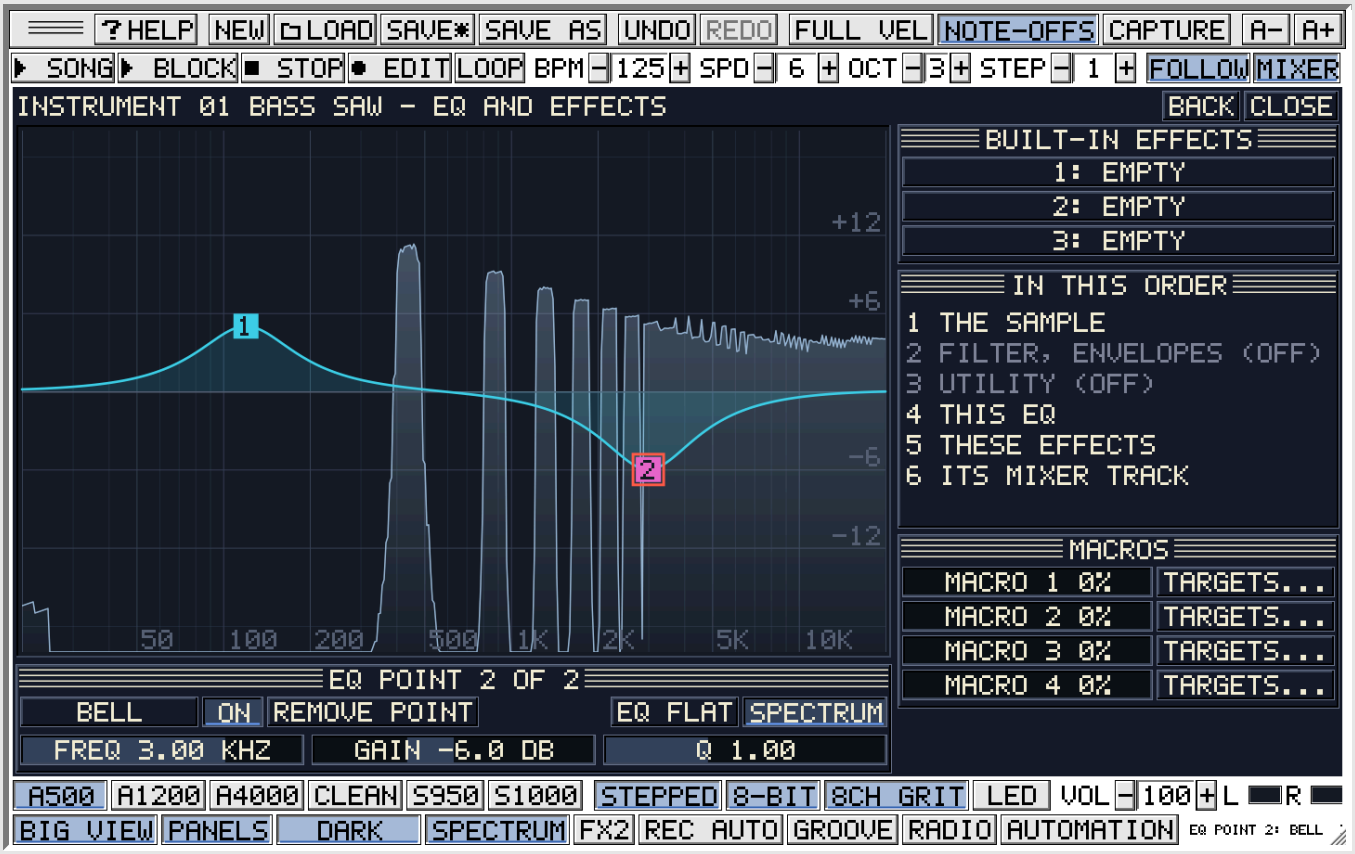
- **Volume envelope:** VOL ENVELOPE on, then ATK, DEC, SUS, REL (1 ms – 10 s). With it on, a note-off starts the release instead of cutting the note.
- **Filter:** OFF / LOW / HIGH / BAND, CUT, RES, AMT (how far the filter envelope moves it) and KEY (how far the cutoff follows the note). The filter envelope has its own ATK, DEC, SUS, REL. The spectrum strip under the page shows the instrument's sound with the **filter's curve** over it and a line at the cutoff.
- **Pan:** AMIGA L/R follows the Amiga track layout; switch it off to give the instrument its own PAN.
- **UTILITY** (the row under PAN, and on a plug-in instrument's page): WIDTH 0% (mono) to 200%; MONO makes everything below a frequency mono (OFF, or <20HZ to <500HZ, a smooth 24 dB/octave crossover, or 48 with STEEP) – mono the bass; VOL is the

instrument's VOL itself, so moving either one moves the other. Before the instrument's own effects; not on MIDI out or CV out. At 100% and OFF it does nothing.

- **Spectrum:** every spectrum (the strip, the filter, the EQ pages) is drawn as an analyser - a smooth line at 1/12 octave, the fill fading under it, a faint peak hold above and a light grid. While PARAMS (or a plug-in instrument's page) is up, the spectrum strip under it shows just that instrument's sound, marked `INSTRUMENT 01 ONLY`.
- **Tuning:** `PITCH` (±48 semitones) and `FINE` (±100 cents). `TUNE TO C` listens to the sample and sets them so `C-3` plays a C.
- **BEAT SYNC** (like Renoise): the sample lasts `LINES` lines at `C-3` at any tempo, repitched live to fit. `STRETCH...` opens the time stretch set to those lines, to fit it at its own pitch instead.
- **KEEP PITCH**: BEAT SYNC stretches the sample instead of repitching it, with the old-sampler stretch - the break stays at its own pitch and keeps the grainy 90s sound. The stretch is made again whenever BPM, SPD or LINES change, and the original is kept: switch KEEP PITCH off and it's back exactly as it was (songs save the original, and make the stretch again when they load). Right-click KEEP PITCH for **CLASSIC** (the Akai S-series sound) or **AKAIZER** (smoother grains). **SEMI** (-24 to +24) moves the pitch without changing the length - a break pitched up and still in time, no chipmunk. Tempo changes while playing (`Fxx`) are followed by repitching a little until the next stretch. Editing the stretched sample keeps the stretch as the sample (KEEP PITCH goes off).
- `▶ HOLD TO PLAY` or the keys to hear it; `RESET ALL` puts everything back; double-click any slider for its default.
- `ON` switches the instrument off and on. To switch it from the pattern, pick a track, press `LEARN A (V)` or `LEARN B (J)`, then click ON: that track's `V00` switches it off and `V01`-`VFF` on.
- `PLUG-IN / MIDI OUT...` plays the instrument with a VST3 / AU instrument or with your hardware instead (see Plug-ins and MIDI out).

INSTRUMENT EQ AND EFFECTS

Every instrument – a sample or a VST3 / AU instrument – has ITS OWN EQ and THREE INSERT SLOTS OF ITS OWN, like a mixer strip's: the built-in filter, drive, compressor, delay, reverb and chorus, or a VST3 / AU plug-in. EQ/FX on the instruments panel, next to LOAD and EDIT, opens them (a plug-in instrument's page has them too).



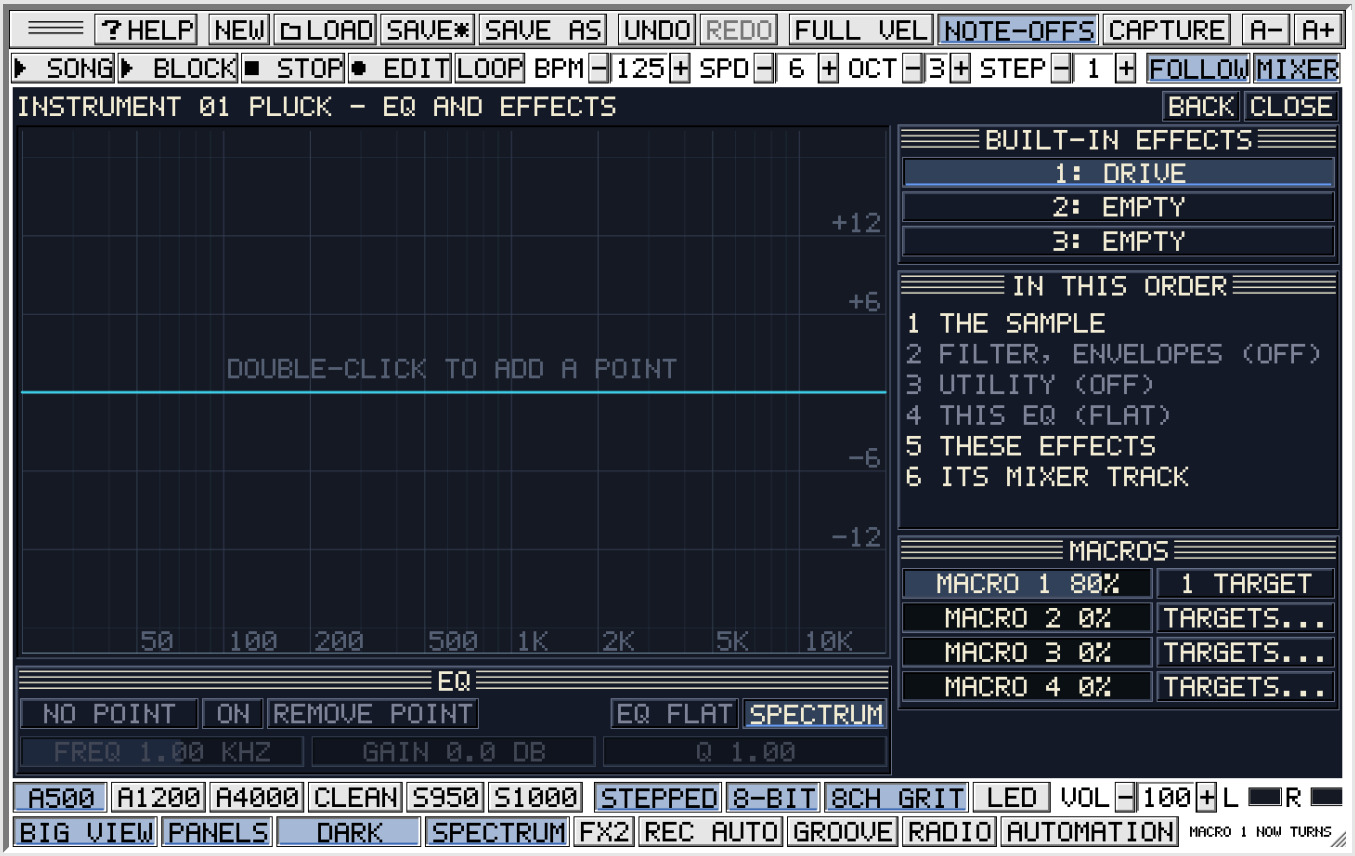
EQ OPENS THE INSTRUMENT'S OWN EQ AND EFFECTS PAGE

- EQ/FX opens the **EQ AND EFFECTS** page: the mixer's EQ graph for this instrument alone – up to 8 points, double-click to add one, drag, the wheel for Q, right-click for its type, EQ FLAT for none – with the instrument's own spectrum behind it. Its 3 effects are on the right, and **IN THIS ORDER** shows the way the sound goes: the sample or plug-in, filter and envelopes, UTILITY, this EQ, these effects, then its mixer track. BACK returns to the instrument.
- The EQ is saved in the song, its changes are undo steps of the instrument, and renders and bounces include it. Not on MIDI out or CV out.

- Click a slot to open it on the mixer's effect page; BACK returns to the instrument. Right-click for OPEN, BYPASS, COPY, PASTE and CLEAR. A lit slot is on.
- The sound goes through the instrument's effects first, then into the strip of the track it plays on. Every note goes through the one chain, so a delay or reverb carries on across notes; its output goes to the track of the instrument's **latest** note.
- The effects belong to the instrument: saved in the song, undo steps, and copied with it (COPY / PASTE / DUPLICATE). RENDER WAV and bouncing include them. A plug-in's own delay on an instrument slot isn't compensated.

MACROS

The EQ AND EFFECTS page has **MACROS**: four knobs, each turning up to four controls of the instrument's own effects together. TARGETS... adds one (the insert, then the control); click a target there to turn it the other way or remove it. Envelopes and V / J can turn them too.



MACRO 1 TURNING DRIVE UP

PHRASES

A phrase is a little pattern stored with an instrument: its note plays the phrase instead of the sample, as in Renoise. An instrument can have up to **16**. PHRASES... at the bottom left of the PARAMS page turns it over to them; SOUND turns it back.

- **The list**: each phrase's number, keys, length and LOOP. ADD makes an empty 16-line one, CLONE copies the one picked, DEL deletes it. Right-click one for EDIT IT IN THE PATTERN, MOVE UP / DOWN, ALL KEYS, its TRACK, ANOTHER BLOCK... and DELETE.
- **The phrase**, on the right, is written in the pattern editor itself, showing just its track: everything the pattern does works - note keys, 1 for OFF, the command list, STEP, SHIFT-click or drag to highlight, DELETE, copy / cut / paste, transpose, undo. EDIT is always on; the octave and STEP are the pattern's. Write it from C-3; the note played moves it. A note with no instrument plays this one's sample.
- LOW KEY / HIGH KEY: the notes that play it. A note plays the **first** phrase that holds it; a note none holds plays the sample.
- LINES: its length. LPB: lines a beat - 4 is the song's speed, 2 half, 8 double.
- LOOP with START and END: round from START after END until the next note or OFF.
- TRANSPOSE: on, its notes follow the note played; off, they play as written.
- PLAY hears it from C-3; the line playing lights up, in the song too.

@xx in an effect column picks the phrase: @01 the first… @10 the sixteenth; @00 plays the sample. Without it, the keys decide. A phrase stops at its end (unless it loops), on the next note, or on OFF.



TWO PHRASES: A RIFF ON THE LOW KEYS, AN ARPEGGIO ON THE HIGH ONES

RESAMPLING IN ONE STEP

SHIFT CMD B (or RESAMPLE in the bounce menu) records the highlighted notes - or the whole block - with all their effects into a new instrument: cut to its length, a slice a beat, notes playing the slices, BEAT SYNC'd to stay in time at any tempo.

FOLLOWING A MIDI CLOCK

View → Follow MIDI Clock: another app or a drum machine leads. The tempo follows its clock; Start, Continue (from the last song position) and Stop work too.

PAULA & THE AKAIS

The output strip picks the machine you hear. None of these change your samples; switch between them any time.

CONTROL	WHAT IT DOES
A500	The A500 (and A600, A2000): a 4.42 kHz output filter, warm and dull
A1200	The A1200: much brighter, about -3 dB around 14 kHz, as measured
A4000	The A4000: about 4.6 kHz, as measured
CLEAN	Paula with no output filtering
S950	Akai S950: 12-bit, stepped variable-rate playback, its 6-pole filter
S1000	Akai S1000: 16-bit, interpolated playback, clean 44.1 kHz-style output
STEPPED	Paula's stepped playback, exactly as the chip makes it (off = smooth)
8-BIT	Samples play at 8-bit resolution
8CH GRIT	OctaMED's 8-channel sound on all 8 tracks. The Amiga has 4 channels, so 8-channel trackers mixed two tracks into each (1+5, 2+6, 3+7, 4+8) at half level and squashed them back to 8 bits; this does the squashing, then brings the level back up. On for the Amiga models. Every track plays at full level either way, and a safety limiter stops eight loud tracks clipping
LED	The Amiga's "LED" low-pass filter (12 dB/oct at 3.2 kHz) at the start of the song (`E00` / `E01` switch it while playing)
SEP	Right-click a machine button (or View → Stereo Separation...): how far tracks lean to the Amiga sides (1, 4, 5, 8 left; 2, 3, 6, 7 right). 20% to start; 0% = all centred, 100% = the full Amiga split
VOL	The master volume after everything, 0–200% (100 = as it is), beside the meter

HOW THE PAULA MODES WORK

- THE STAIRCASE, WITHOUT FALSE NOTES. Paula holds each sample until the next, so its output is a staircase whose copies of the sound go far above hearing. Played straight, those copies fold back as out-of-tune tones. With STEPPED on, every change of level

is a band-limited step instead: everything a real Paula puts out below 20 kHz is there, its own staircase images included, and nothing folds back. It sounds the same at 44.1, 48 or 96 kHz.

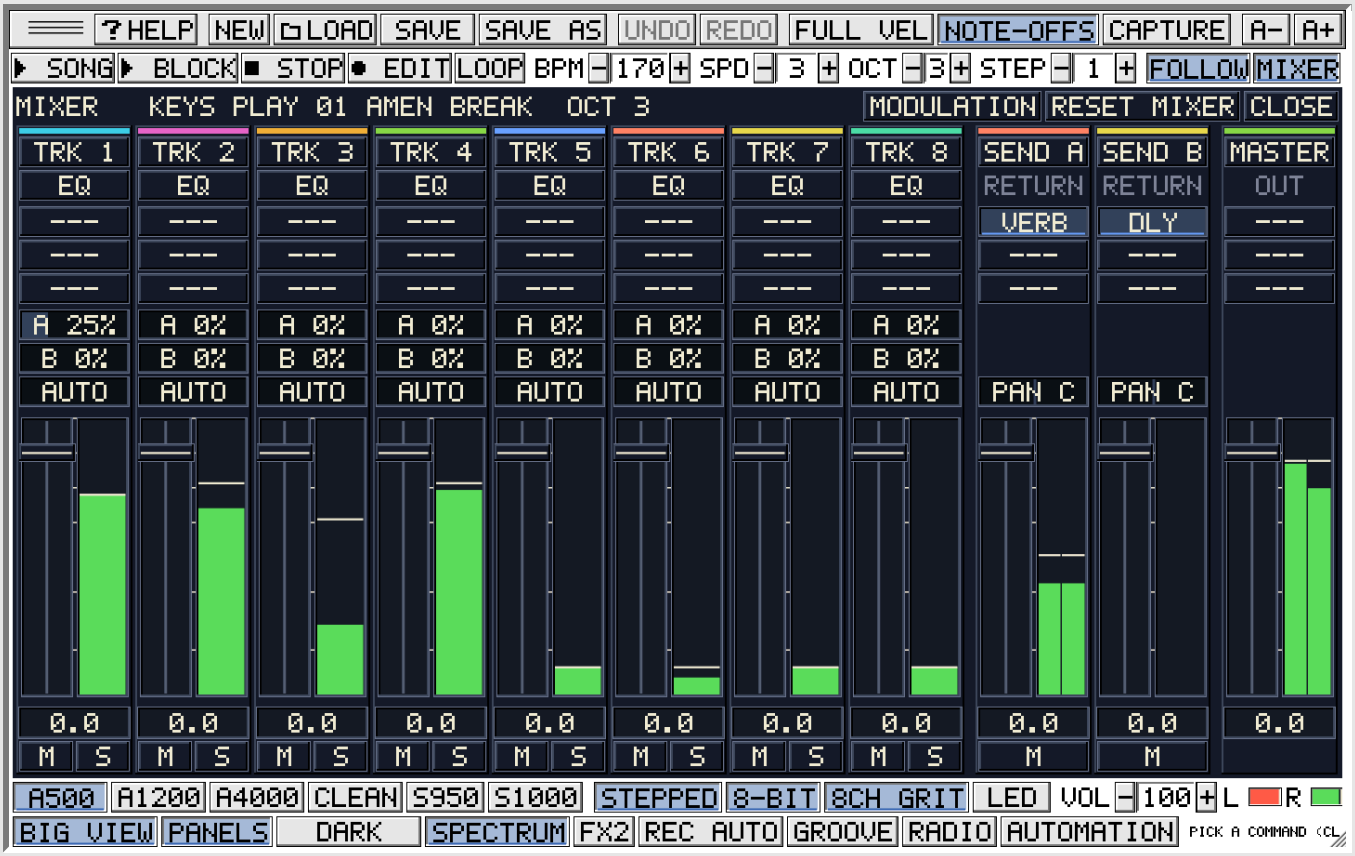
- **VOLUME BELOW 64.** At 64 a real Paula changes level exactly when the next sample is due; below 64 it works on a fixed 55.4 kHz grid, so each change waits for the next grid point. That adds a faint grit to everything not at full volume. A500, A1200 and A4000 do this; CLEAN doesn't.
- **THE FILTERS** are the real ones: the A500's 4.42 kHz RC filter (100 nF, 360 Ω), the A1200's and A4000's measured responses, the LED filter as a Butterworth. (From Henryk Richter's *Amiga Paula vs. System Theory*; the band-limited steps follow OpenMPT and Antti S. Lankila.)

PANNING, IN ORDER

1. `8xx` in the pattern (an absolute position, not scaled by SEP).
2. The track's pan, the bar under its name (AUTO = none).
3. The instrument's PAN, when its AMIGA L/R is off.
4. Otherwise Amiga style, by SEP. Stereo samples play centred with their own stereo image.

THE MIXER

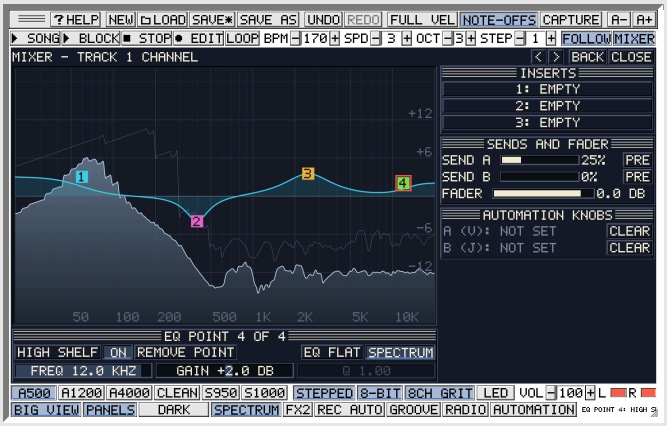
MIXER on the transport row (CMD 2) opens the mixer across the window; ESC or CMD 1 goes back. The keys still play the current instrument while it's open.



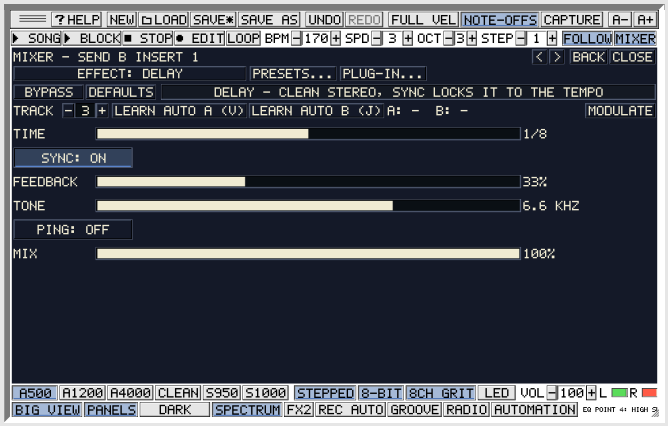
THE STRIPS: TRACKS, SEND A, SEND B, MASTER

STRIP PART	WHAT IT DOES
Name / EQ	Opens the track's channel page. EQ is lit when a band is on; EQ AB when automation knobs are set.
3 insert slots	Click for the effect page; SHIFT-click bypasses (a bypassed slot is dimmed).
A, B	How much goes to SEND A / B. AP / BP = pre-fader. Double-click = off.
PAN	The track's pan. Double-click = AUTO.
Fader	-INF to +6 dB with the meter beside it. SHIFT = fine, double-click = 0 dB.
M / S	Mute and solo.

EQ → INSERTS → PRE SENDS → FADER → POST SENDS → MASTER → PAULA / AKAI OUTPUT



CHANNEL PAGE: EQ POINTS OVER THE SPECTRUM, INSERTS, SENDS



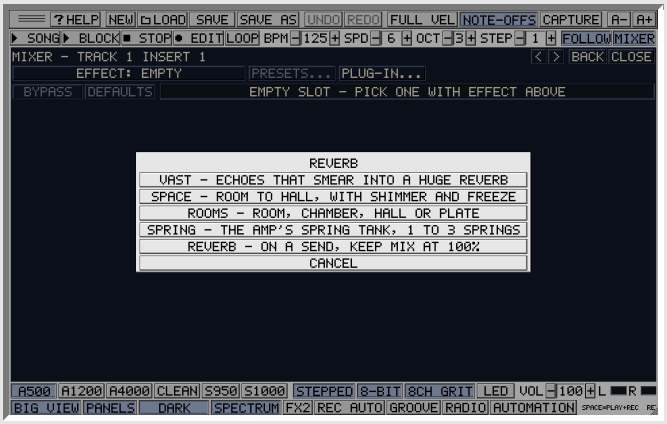
EFFECT PAGE: SEND B'S DELAY

- **Channel page – the EQ:** up to 8 points on the curve, over the track's live spectrum, smoothed with a faint peak-hold line (SPECTRUM switches it off; it stays put when you stop). Right-click SPECTRUM or an empty part of the graph for its colour: ICE, CYAN, GREEN, LIME, AMBER, ORANGE, MAGENTA or VIOLET. < > step through the tracks.
  - **Double-click** the graph to add a point there: a LOW SHELF below 80 Hz, a HIGH SHELF above 10 kHz, a BELL in between.
  - **Drag** a point: across is frequency, up and down is gain, with a big readout as you go. The **wheel** sets a bell's Q.
  - **Right-click** a point for its type (BELL, LOW / HIGH SHELF, HIGH- / LOW-PASS), SWITCH OFF, RESET GAIN and REMOVE POINT; DELETE removes the selected one.
  - Under the graph: the selected point's type, ON, REMOVE POINT, FREQ, GAIN and Q. EQ FLAT removes them all. Songs from before v0.25 keep their bands that were on.
- **Effect page:** EFFECT: … opens the menu of effects in groups – REVERB, DELAY, MODULATION, DRIVE & CRUSH, FILTER & DYNAMICS, PLUG-IN (VST3 / AU), EMPTY – then the effect: 23 built in. PRESETS... has the factory presets and yours. Every page has BYPASS and DEFAULTS; more than 6 controls go in two columns.

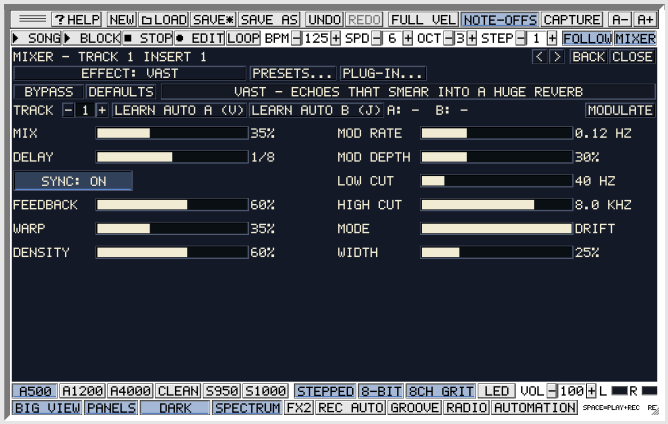
- **Returns:** SEND A starts with a reverb and SEND B with a delay, fully wet, silent until a track sends to them.
- **Dub throws:** `!xy` on a note's line throws that track into the dub echo on SEND B - `x` how much (1-F), `y` for how many lines (0 = just that hit, until the track's next note). `!0y` rides the echo's feedback up (`!0F` nearly endless), `!00` puts it back. Type `!` with Shift+1. The first `!` in a song makes SEND B a dub echo - DUB DELAY into SPRING - if SEND B is still as a new song has it (an undo step of its own). The throw fades in and out over a few milliseconds, so it never clicks, and the echoes ring on after it.
- **Delay compensation:** plug-ins that give their sound back late (lookahead limiters, linear-phase EQs) are lined up, so every track stays in time, sends included. The strips page title shows the total delay.

A mixer with nothing changed is skipped, so the song sounds exactly as it would without one. Everything in it is saved in the song, undoable and in renders.

THE STUDIO EFFECTS



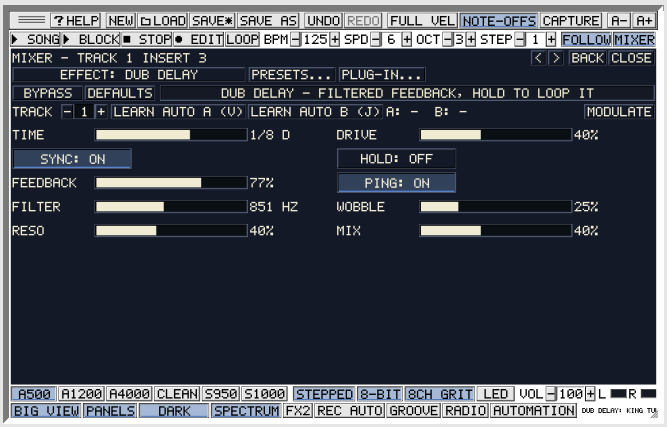
EFFECT: A GROUP, THEN THE EFFECT



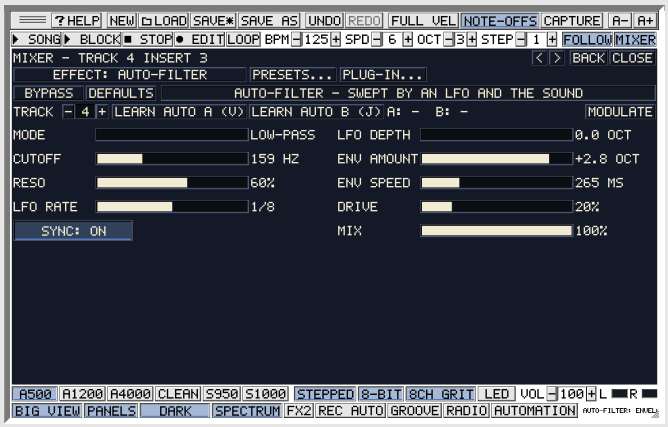
VAST: TWELVE CONTROLS IN TWO COLUMNS

EFFECT	WHAT IT IS
VAST	Echoes that turn into a huge reverb: 8 delay lines feeding each other. **DENSITY** at 0 gives clean repeats on the beat, turned up a cloud; **WARP** smears each repeat more; **FEEDBACK** at 100% goes on for ever. **DELAY** in ms or, with **SYNC**, a note length. **MODE**: NEBULA, ABYSS (longer, darker), BLOOM (swells in), DRIFT (wandering lines).
SPACE	A room, plate or hall: **SIZE**, **DECAY** 0.3–40 s, **PRE-DELAY** with **PRE ECHO**, **DAMP**, **LOW CUT**, **MOD**, **CHARACTER** (GROUNDED, AIRY, COSMIC), **SHIMMER** (an octave up in the tail), **FREEZE** (holds the tail) and **WIDTH**.
TAPE ECHO	A tape loop with three heads at 1, 2 and 3 times **TIME** (**HEADS** picks them). **FEEDBACK** past 100% for runaway dub (the tape saturates, it doesn't explode), **WOW**, **FLUTTER**, **SATURATE**, **TONE**, **WEAR** (hiss, dropouts), **SPREAD**.
ENSEMBLE	The polysynth bucket-brigade chorus: **MODE** I, II, I+II or STRINGS (three voices: the string-machine sound), **RATE**, **DEPTH**, **WIDTH**, **TONE**, **NOISE**.
ROOMS	A room, chamber, hall or plate (**TYPE**) with early reflections: **SIZE**, **DECAY**, **PRE-DELAY**, **EARLY** (reflections against the tail), **DAMP**, **LOW CUT**, **HIGH CUT**, **MOD**, **WIDTH**.
SPRING	An amp's spring tank: 1–3 **SPRINGS**, **DECAY**, **TONE**, **DRIP** (the splash on hits), **DRIVE**.
DUB DELAY	Repeats through a band-pass **FILTER** with **RESO** and **DRIVE**, feedback past 100%. **HOLD** loops what's in it and lets nothing new in. **PING**, **WOBBLE**.
BBD DELAY	An analog bucket-brigade echo up to 1 s: darker the longer it is. **MOD RATE**, **MOD DEPTH**, **AGE** (hiss, duller), **DRIVE**.
MULTI-TAP	Up to 8 **TAPS** in a **PATTERN** (STRAIGHT, DOTTED, TRIPLET, SWING, SPEED UP, SLOW DOWN, SCATTER), each quieter by **DECAY**, bounced by **SPREAD**.
GRAIN DELAY	**MODE** REVERSE plays each stretch backwards; GRAINS sprays grains from the past. **PITCH** -12 to +12, **SPRAY**, **DENSITY**, **FEEDBACK**.
DIMENSION	4 modes of wide, clean stereo that sums back to mono cleanly.
FLANGER	**RATE**, **DEPTH**, **MANUAL**, **FEEDBACK** (±), **STEREO**; **THRU-ZERO** sweeps through the dry sound for the tape-flanging cancel.

EFFECT	WHAT IT IS
PHASER	4, 8 or 12 **STAGES**, **RATE**, **DEPTH**, **CENTRE**, **FEEDBACK** (±), **STEREO**.
ROTARY	A spinning organ speaker, horn and drum: **FAST** switches speed and they ramp like the real thing (**ACCEL**). **HORN** balance, **DRIVE**, **WIDTH**.
TREMOLO	TREMOLO, AUTO-PAN or HARMONIC; **RATE** or **SYNC**; **SHAPE** sine to square; **STEREO**.
FREQ SHIFTER	SHIFT moves every frequency by the same Hz (spirals with **FEEDBACK**); RING MOD for metallic, robotic tones.
AUTO-FILTER	Low-, band- or high-pass moved by an LFO (synced or free) and by the sound's own level (**ENV AMOUNT** ±, **ENV SPEED**): envelope funk, wah, sweeps.
The originals	FILTER, DRIVE, COMP, DELAY, REVERB and CHORUS are still there, in their groups.



DUB DELAY: HOLD LOOPS IT



AUTO-FILTER: AN LFO AND THE SOUND'S OWN LEVEL MOVE IT

- **Presets:** PRESETS... lists the factory presets, then yours as **MY _**. SAVE THESE SETTINGS AS MY PRESET... keeps them under a name on this computer, for any slot in any song; DELETE ONE OF MY PRESETS... removes one. Choosing a preset is one undo step.
- Every control glides to a new value (no zipper noise), so they all automate smoothly with V and J. The saturation runs at twice the sample rate, and everything is true stereo.
- Songs that use VAST, SPACE, TAPE ECHO or ENSEMBLE need v0.34 or later; the others, v0.35.

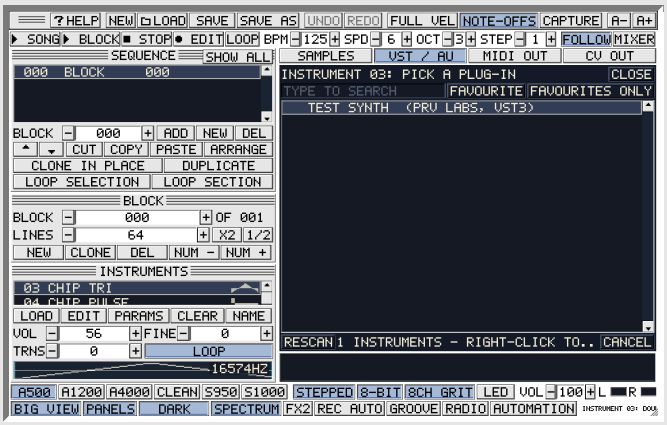
PLUG-INS

EFFECTS IN THE MIXER

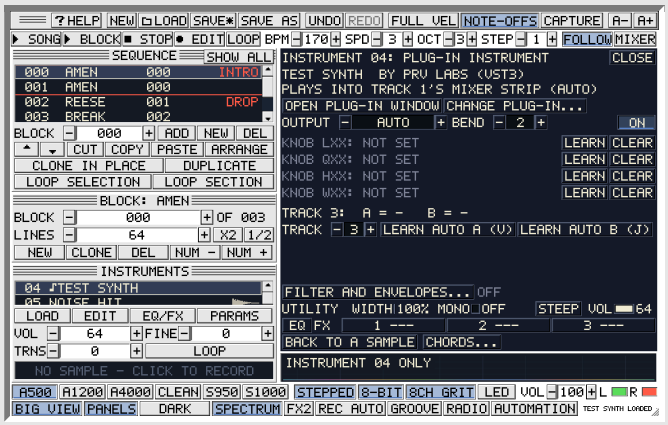
On any insert slot's effect page press PLUG-IN... . The first time, it looks for the plug-ins on your Mac, each one in a separate hidden copy of the app, so one that crashes can't take Paula's Revenge down. Double-click one to load it; OPEN PLUG-IN WINDOW shows its own controls. **To find one, just type:** the list narrows to every word in the name, maker or format (VALHALLA , U-HE ZEB); Esc clears it. **Favourites:** right-click a plug-in (or FAVOURITE) to star it - stars come first, and FAVOURITES ONLY shows just them. The instrument list works the same way. Its settings are saved in the song. If a song uses a plug-in this Mac doesn't have, the slot shows !NAME , passes the audio through and keeps the settings.

INSTRUMENTS

Pick an instrument, press LOAD and the VST / AU tab (or right-click the instrument → **LOAD A VST / AU SYNTH...**), then double-click a plug-in. (The SYNTHS tab has the built-in SUB, REESE and GRANULAR synths, below.) Type to search. PARAMS → PLUG-IN / MIDI OUT... gets there too.



LOAD → VST / AU



A PLUG-IN INSTRUMENT'S PAGE



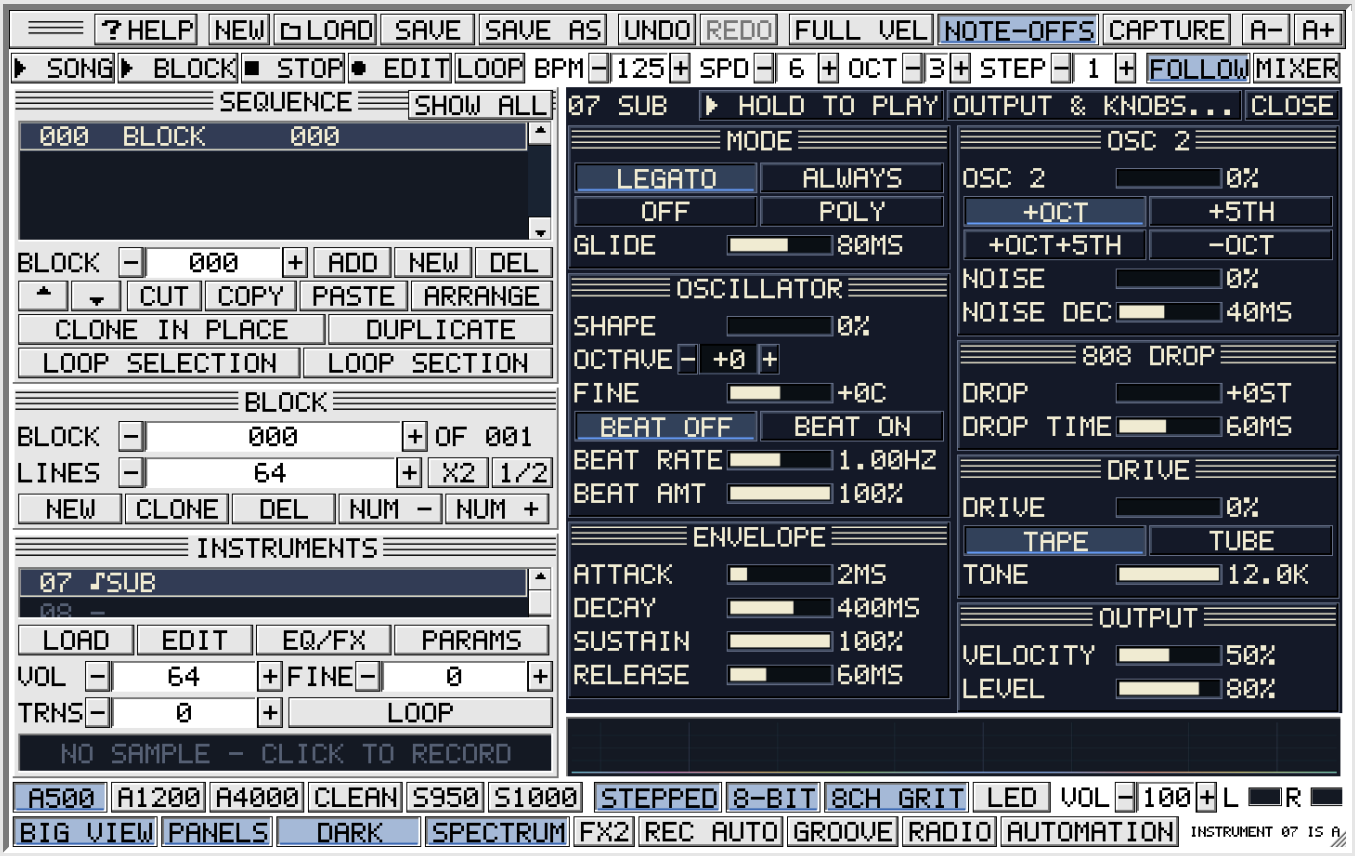
FILTER AND ENVELOPES... ON A VST3 / AU INSTRUMENT

- `EDIT` opens the plug-in's own window. The laptop keys keep playing it while that window is in front.
- `FILTER AND ENVELOPES...` opens the PARAMS page for the plug-in: the same `VOL ENVELOPE`, `FILTER` and `FILTER ENVELOPE` as a sample, on the plug-in's sound. They start with each note played on it (from where they are, no click) and release when its last note ends - a slow fade-in for a pad, a pluck from any synth. `BACK TO THE PLUG-IN` returns; the button is lit, with what's on written next to it.
- `EQ` and `FX` on its bottom rows: its own EQ and effects (see Instrument EQ and effects).
- `OUTPUT` picks the mixer strip it plays into; AUTO is the track that played its latest note (the same in the song, a render and a bounce). `BEND` should match the plug-in's own pitch bend range, so slides and vibrato land in tune.
- Notes play one at a time per track. `Cxx` is velocity, pitch commands become pitch bend, `Oxx` is the mod wheel, and `ECx` `EDx` `Nxx` `Kxx` `Mxx` `E9x` `Rxy` work. Sample-only commands are ignored.

- Chords: `#xy` on a note's line plays chord slot `xy` from the song's chord table, held. See Chords.
- Letting go of a key while recording writes a note-off (see Recording and note-offs).
- `LEARN` next to `L` `Q` `H` `W`, then move a knob in the plug-in's window: those commands then move that knob on this instrument's notes.

THE BUILT-IN SUB SYNTH

SUB is a sub bass synth built into Paula's Revenge. Pick an instrument, press `LOAD` → the `SYNTHS` tab (or right-click the instrument → **LOAD A BUILT-IN SYNTH (SUB, REESE, GRANULAR)...**) and pick SUB. Its page opens; `EDIT` opens it again later.



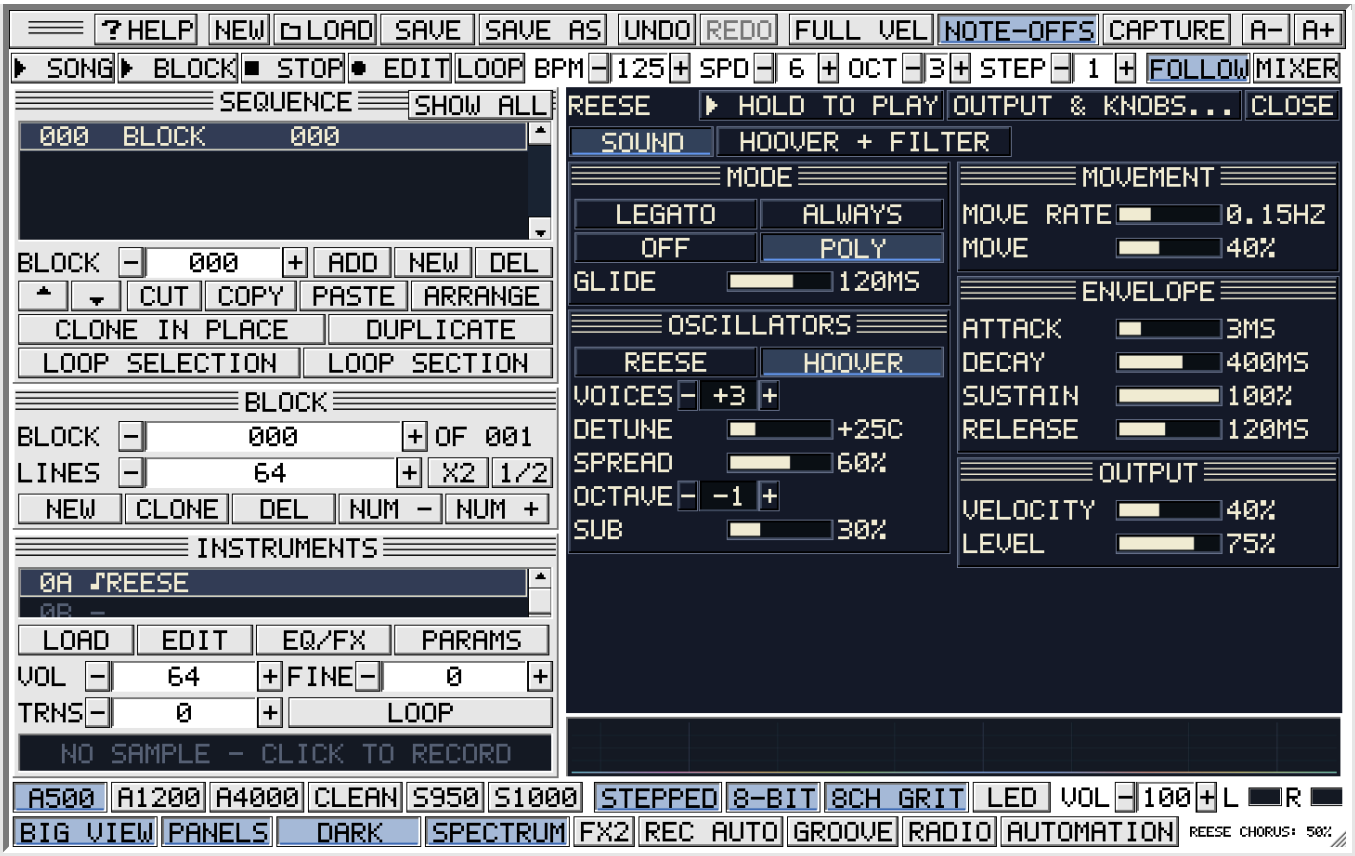
THE SUB PAGE

It never clicks when notes overlap or follow each other. Most mono synths click there because they restart the oscillator or drop the envelope to zero. SUB never restarts its oscillator (a new note only moves its pitch), its envelope carries on from where it is, and anything that has to start afresh starts where the wave crosses zero or fades in.

- **MODE:** LEGATO - one note at a time; a note played while another is held slides to it without a new attack. ALWAYS - always slides. OFF - every note gets a new attack (still no click). POLY - up to 8 notes. **GLIDE** sets how long a slide takes.
- **OSCILLATOR:** SHAPE from a pure sine to a rounder, brighter wave; OCTAVE and FINE. **OSC 2:** a second sine (an octave up, a fifth up, both, or an octave down) and NOISE, a short burst at the start of each note (NOISE DEC).
- **808 DROP:** each note starts DROP semitones higher and falls to its pitch over DROP TIME.
- **DRIVE:** TAPE (soft) or TUBE (warmer) saturation; TONE is a low-pass after it.
- **ENVELOPE** (ATTACK, DECAY, SUSTAIN, RELEASE) and **OUTPUT** (VELOCITY: how much Cxx changes the level; LEVEL).
- **BEAT** (in the OSCILLATOR box): BEAT ON adds a copy of the oscillator slightly higher, so the sub swells and dips on a single note - the beating of two notes a little out of tune. BEAT RATE is how many times a second (0.1–10 Hz, the same on every note), BEAT AMT how deep the dips go. Switching it on or off never clicks, even mid-note; right-click the BEAT buttons to put them on a knob and switch BEAT from the pattern.
- **Right-click** any control to make it a knob (L Q H W) or the track's AUTO A / B, so the pattern can move it. Hold HOLD TO PLAY or play the keys to hear it. OUTPUT & KNOBS... opens the plug-in instrument page above - OUTPUT, BEND, ON, FILTER AND ENVELOPES, EQ / FX and chords all work for SUB.

THE BUILT-IN REESE SYNTH

REESE is the reese and hoover bass: detuned saws drifting against each other so the phasing never sits still, or the hoover's hollow pulse-saws swooping up into each note. It's click-free like SUB, so it pairs with SUB for a mid bass over a clean sub. LOAD → SYNTHS → REESE. Its controls are on two tabs, SOUND and HOOVER + FILTER.

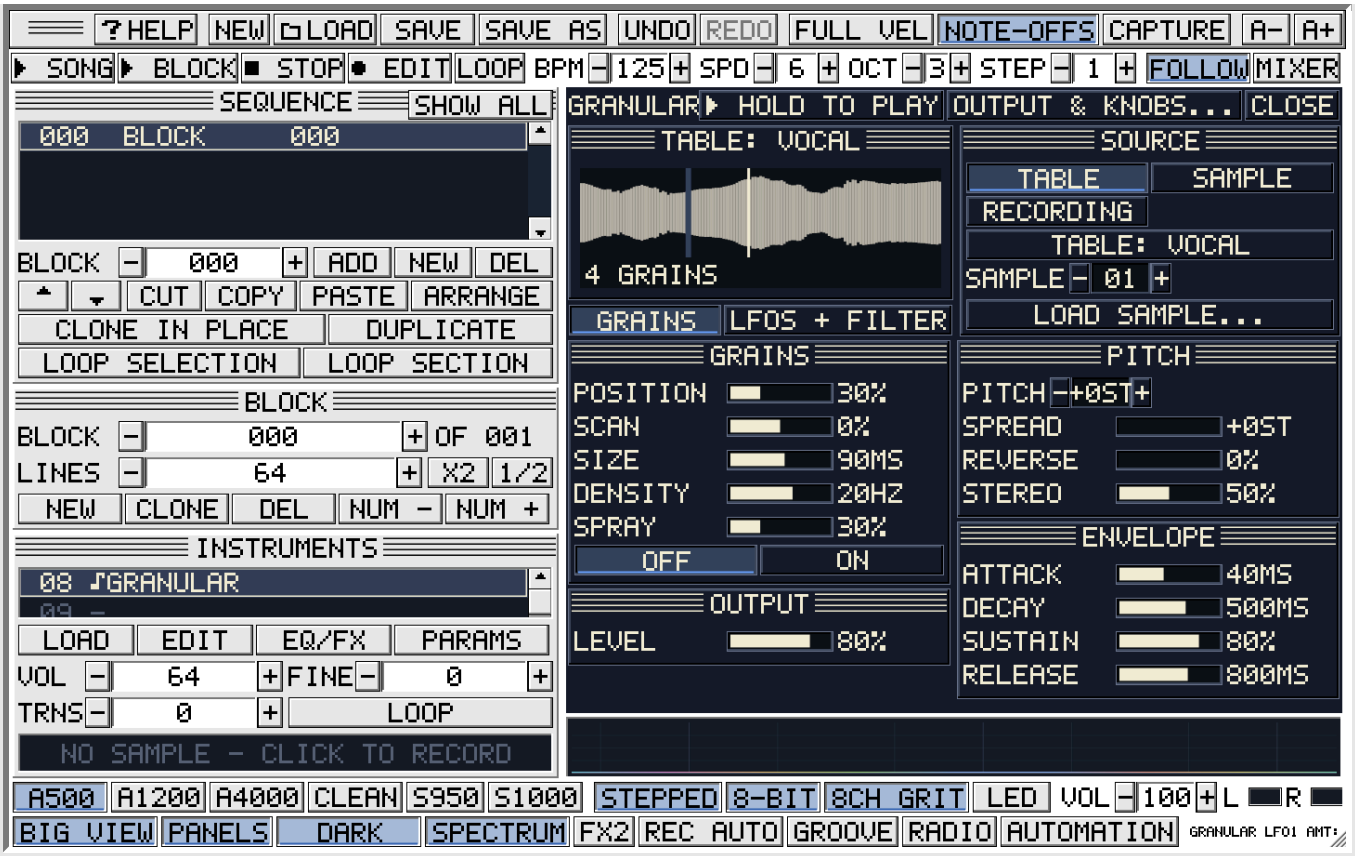


THE REESE PAGE

- **MODE** and **GLIDE** as on SUB (POLY plays up to 6 notes).
- **OSCILLATORS:** TYPE `REESE` (saws) or `HOOVER` (pulse-saws, a square sub); VOICES (1–7), DETUNE (wider and faster phasing), SPREAD (left and right), OCTAVE, SUB (an octave below). **MOVEMENT:** MOVE breathes the detune wider and narrower at MOVE RATE.
- **HOOVER:** BEND – each note swoops up from that many semitones below over BEND TIME; PWM and PWM RATE move the pulse width.
- **FILTER:** a 24 dB low-pass with CUTOFF, RES, ENV AMT (how far each note opens it) and ENV DECAY; **WOBBLE** puts an LFO on the cutoff. **DRIVE** and **CHORUS** (the hoover's wide swirl).
- **ENVELOPE** and **OUTPUT**. Right-click, HOLD TO PLAY, OUTPUT & KNOBS..., saving and undo work as on SUB.

THE BUILT-IN GRANULAR SYNTH

GRANULAR plays a sound as a cloud of tiny overlapping pieces – grains – so it can hold one moment of a sound for ever, smear it into a pad, stretch it or scatter it into glitter. LOAD → `SYNTHS` → GRANULAR. Notes play it at their pitch (`C-3` = the sound's own pitch), up to 6 at once.



THE GRANULAR PAGE: CLICK THE SOUND TO SET WHERE THE GRAINS COME FROM

- **SOURCE:** `TABLE` – 8 built-in sounds that change from start to end (SINE, SAW, SQUARE, VOCAL through A E I O U, BELL, GLASS, NOISE, CHOIR). `SAMPLE` – press `LOAD SAMPLE...` under the SOURCE box and pick a sample in the browser, just as for any instrument: it goes into GRANULAR (which stays GRANULAR) and is saved with the song. Or set SAMPLE to another instrument's number to use its sample. `RECORDING` – the audio input, live: the last 8 seconds of what comes in.
- Across the top, beside the SOURCE box, is the sound itself, with a line at POSITION and a thin line where the grains are reading. **Click or drag on it** to set POSITION. Two tabs under it show the rest: `GRAINS` and `LFOS + FILTER`.
- **GRAINS:** POSITION; SCAN (moves the position on as the note plays – 100% plays through at the sound's own speed, less stretches it, minus goes backwards); SIZE; DENSITY (grains a second); SPRAY (scatters where and when grains start); **FREEZE** (holds the position – and on RECORDING stops taking in sound, so you can play what was caught).

- **PITCH:** PITCH, SPREAD (each grain randomly out of tune), REVERSE (how many play backwards), STEREO (how wide they spread).
- **ENVELOPE**; **LFO 1** (sine) and **LFO 2** (triangle) with RATE, AMT and TO; **ENV MOD** – each can move POSITION, SIZE, DENSITY, SPRAY, PITCH, STEREO, CUTOFF or LEVEL.
- **FILTER:** CUTOFF and RES (a resonant low-pass), DIFFUSE (smears the grains into a wash). Then LEVEL.
- On a TABLE, a note's grains line up with each other, so with SPRAY and SPREAD at 0 it is a clean tone; turn them up to break it apart. Right-click, HOLD TO PLAY, OUTPUT & KNOBS..., saving and undo work as on SUB.

AUTOMATION

Every track has two automation knobs, **AUTO A** (`Vxx`) and **AUTO B** (`Jxx`). Point one at any effect control and the pattern moves it.

1. Open an insert's effect page in the mixer. Pick the track in the `TRACK` box.
2. Press `LEARN AUTO A (V)`, then move the control: drag a built-in effect's slider, or turn a knob in the plug-in's window.
3. In the pattern, `V00` is the control's lowest and `VFF` its highest. The value holds until the next one.
4. Put `Yxx` before or beside the `V` to glide there over xx ticks instead of jumping.

- **On and off:** press LEARN, then click the effect's `BYPASS` (or an instrument's `ON`) instead. `V00` then switches it off and `V01`–`VFF` on.
- The channel page lists what each knob controls under AUTOMATION KNOBS, with CLEAR buttons. The help strip and the command list name it too.

RECORDING BY HAND: REC AUTO

Switch on `REC AUTO` on the bottom row, turn EDIT on and play. Moving a learned control writes its command on the line that's playing: `V` or `J`, or `L` `Q` `H` `W` for an instrument's own knobs on the track playing it.

- It goes in the first effect column if that's free, otherwise in FX2. It never overwrites other commands; that line is skipped and the status line says so.
- One value per line per knob, and only when it changes. What the pattern plays back isn't recorded again.
- A whole pass, until you stop, is one undo step.

ENVELOPES: DRAWING AUTOMATION

`AUTOMATION` on the bottom row opens the envelope panel under the pattern, where the spectrum was. Drag its top edge to resize it. It

shows the block and track you're on.



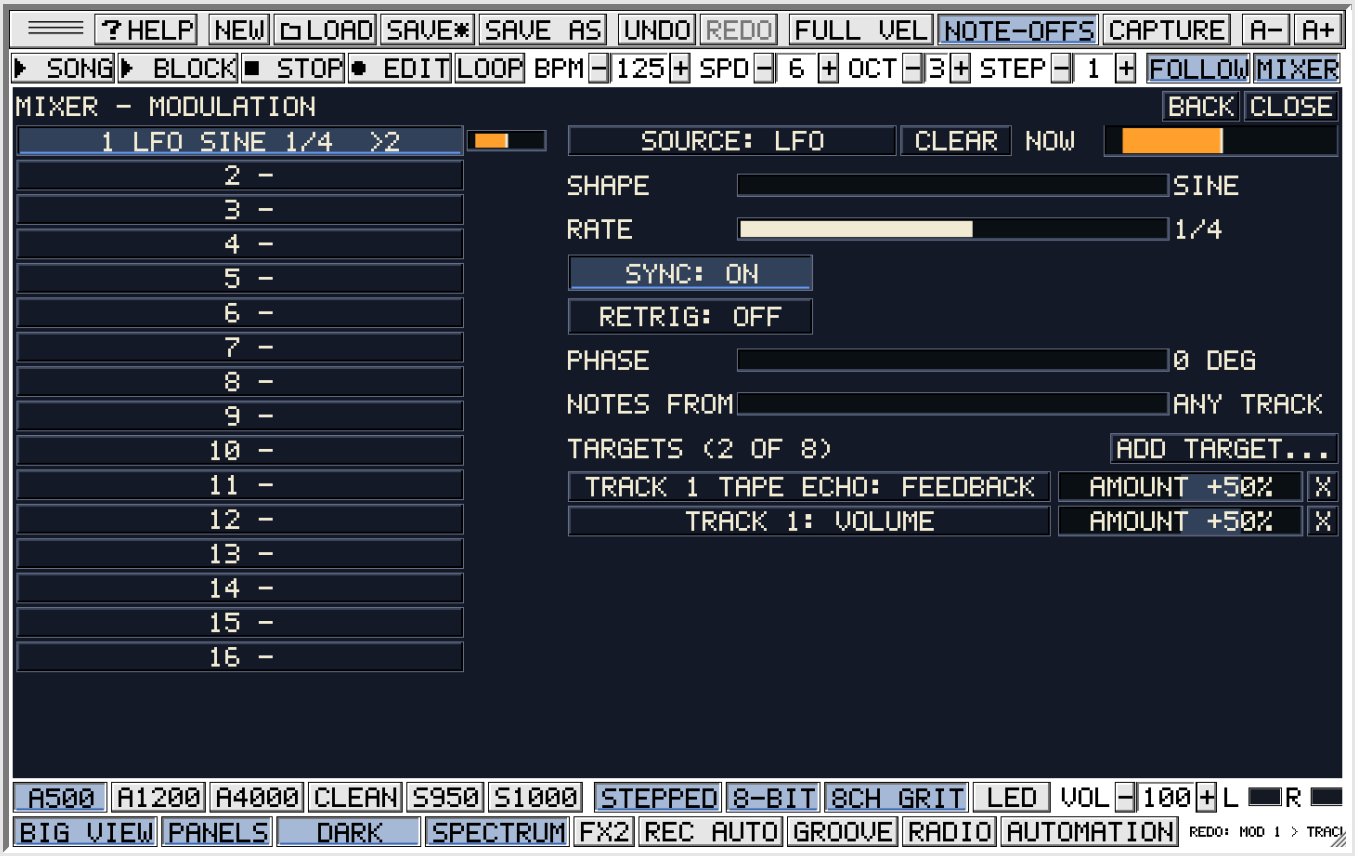
AN ENVELOPE ON TAPE ECHO'S FEEDBACK, BENT WITH CURVES, WHILE THE BLOCK PLAYS

DO	WHAT HAPPENS
Pick a control (left)	VOLUME, SENDS, the track's inserts, the instrument's own inserts and parameters, MACROs. `*` = has an envelope here. Type to search; AUTOMATED ONLY.
Double-click	Adds a point (snapped); on a point, removes it. DELETE removes the selected one.
Drag a point	Moves it, with a readout of the line and the real value. SHIFT leaves the snap.
POINTS / LINES / CURVES	Steps, straight lines, or bends: drag a line up or down to bend it.
DRAW / LINE	Paint a point on every line as you drag / drag a straight ramp.
SNAP, COPY, PASTE, CLEAR	Lines, beats or off; copy an envelope to another control or block; remove it.

- Envelopes belong to their block, and DOUBLE / HALVE take them along. When the block stops the control goes back to its own setting.
- Saved in the song (v0.37 or later), undoable, and in renders.

MODULATION

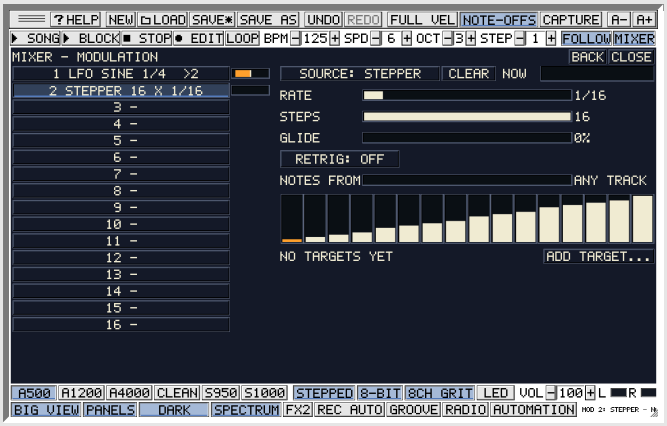
MODULATION on the mixer's strips page opens 16 modulators. Each has a **SOURCE** and up to 8 **TARGETS** - any built-in effect control, a plug-in's parameter, or a track's VOLUME, SEND A or SEND B - each with its own **AMOUNT** (-100% to +100%).



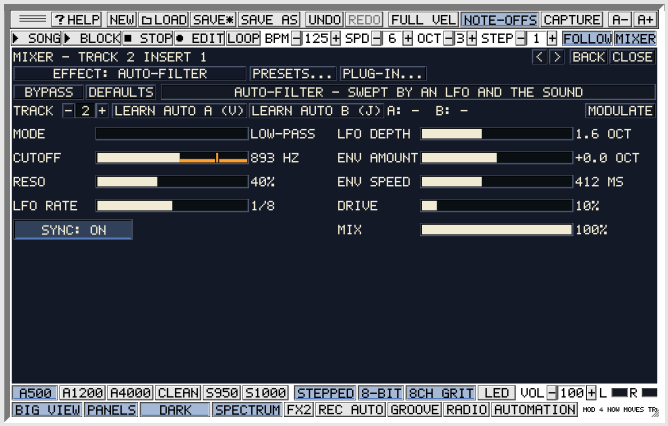
AN LFO IN TIME WITH THE SONG, MOVING AN ECHO'S FEEDBACK AND A TRACK'S VOLUME

SOURCE	WHAT MOVES IT
LFO	SINE, TRIANGLE, SAW UP / DOWN, SQUARE or RANDOM S&H; RATE in Hz or SYNC'd from 1/32 to 4 BARS (locked to the song position); RETRIG on each note; PHASE.
FOLLOWER	A track's level (ATTACK, RELEASE, GAIN). Aim it at another track's VOLUME for sidechain ducking (it starts at -100%).
STEPPER	16 steps you draw with the mouse, at a RATE in time with the song, with GLIDE.

SOURCE	WHAT MOVES IT
RANDOM	A new value each RATE, stepped or SMOOTH.
KEY / VELOCITY	How high / how loud the notes are, from a track or ANY TRACK (your playing too).
MACRO	One knob for many controls, each its own amount. AUTO A (V) / AUTO B (J) hand it to a track's automation knob.



A STEPPER: DRAW THE STEPS

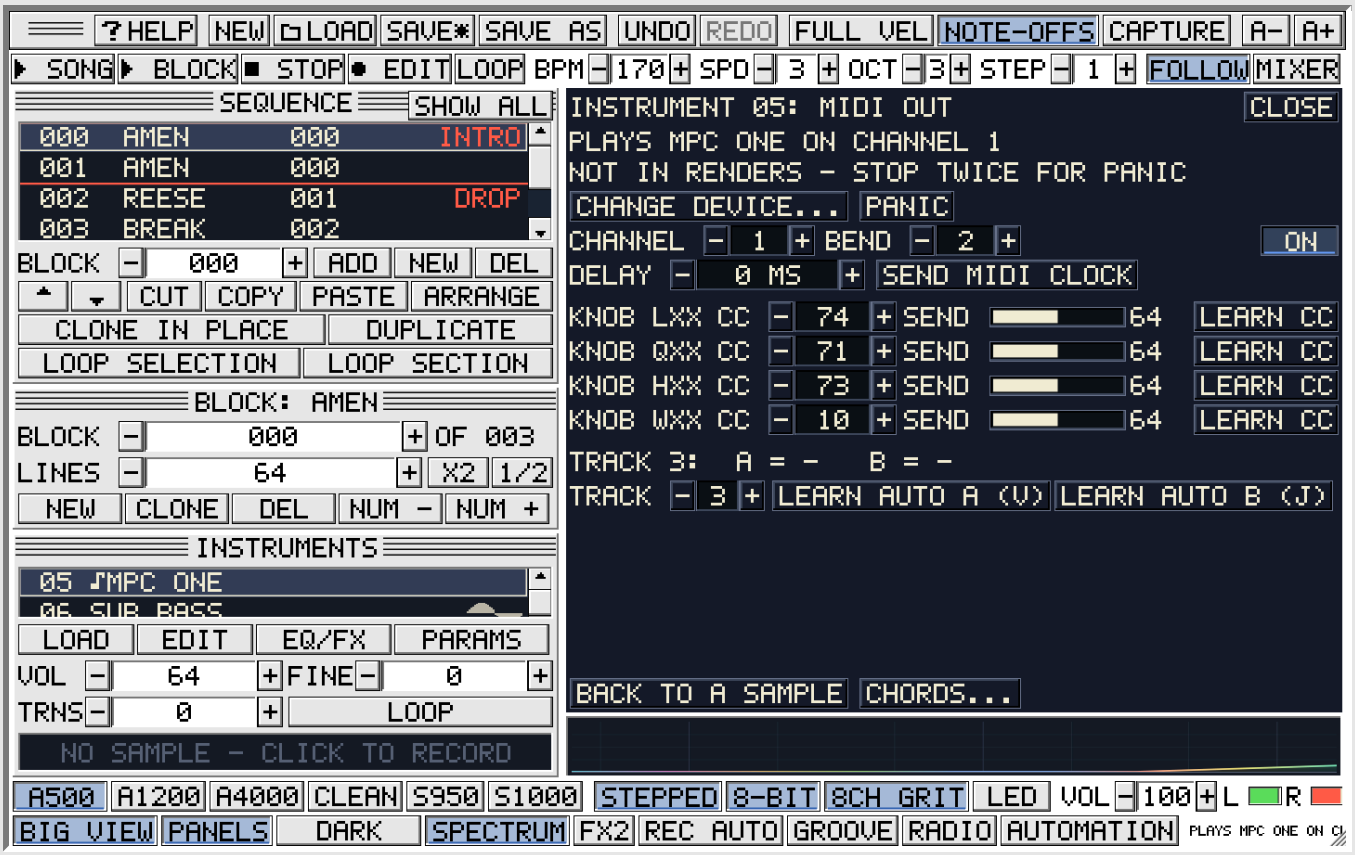


MODULATE, AND THE RING ON A MODULATED CONTROL

- **Targets:** ADD TARGET... lists the tracks' VOLUME and SENDS and every effect in the song. Or press MODULATE on an effect page and move a control: the selected modulator gets it. Click a target for TURN IT THE OTHER WAY, GO TO THE EFFECT or REMOVE.
- A modulated slider shows an orange band for its range and a line where it is now. Your setting stays put; modulation moves the sound around it.
- Saved in the song (v0.36 or later) and undoable. VOLUME and SENDS glide; switches flip past the middle.

MIDI OUT

An instrument can play your hardware (an MPC, a microKORG, an NTS-1) over MIDI. Press LOAD and the MIDI OUT tab (or right-click the instrument → MIDI OUT TO HARDWARE...), then double-click a MIDI output. (For a modular, see CV out.)



A MIDI-OUT INSTRUMENT'S PAGE

CONTROL	WHAT IT DOES
CHANNEL	MIDI channel 1–16
BEND	The hardware's pitch bend range, so pitch commands land in tune
ON	As for plug-in instruments; #xy plays chords here too
DELAY	-50 to +200 ms to line the hardware up with the audio. Positive: the MIDI goes later. Negative: all the audio goes later instead.
SEND MIDI CLOCK	24-a-beat clock with start, stop, continue and song position (4 lines = a beat), so the hardware follows the song

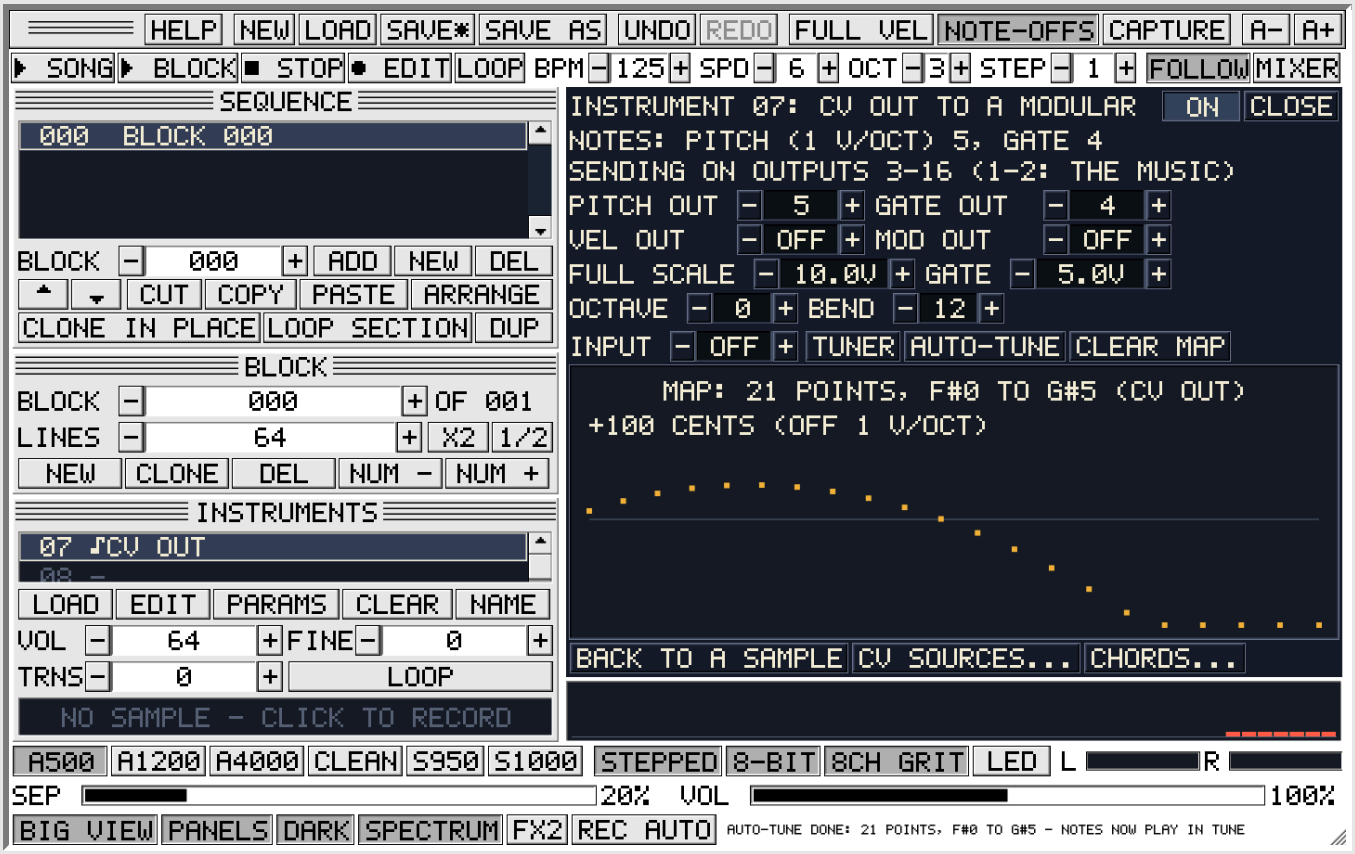
CONTROL	WHAT IT DOES
KNOB L Q H W	The CC each sends (74, 71, 73, 10 to start). LEARN CC takes the number from the next knob you turn on a controller; the SEND bar sends it now (and REC AUTO records it).
PANIC	All notes off on every MIDI output. STOP pressed while already stopped does the same.

Notes, velocity, pitch bend, the mod wheel and the timing commands work as for plug-in instruments. If the device isn't connected, the instrument shows !NAME, sends nothing and keeps its settings; CHANGE DEVICE... and the same device again reconnects it. MIDI-out instruments aren't in renders or bounces (they're hardware).

CV OUT TO A MODULAR

With a **DC-coupled** interface (an Expert Sleepers **ES-9**, or anything that can put out steady voltages) Paula's Revenge plays and clocks a Eurorack modular. Ordinary interfaces block DC, so CV won't work through them.

1. Choose the interface in **Audio/MIDI Settings…**. With more than 2 outputs the app switches them all on (up to 16), in order: output 5 in the app is the interface's output 5. Outputs **1–2 are the music**; CV goes on **3–16**.
2. Press LOAD and the CV OUT tab, or right-click the instrument → **CV OUT TO A MODULAR...**.
3. Patch the PITCH and GATE outputs to an oscillator and an envelope, and play.



A CV OUT INSTRUMENT AFTER AUTO-TUNE: THE MAP'S POINTS, IN CENTS FROM 1 V/OCTAVE

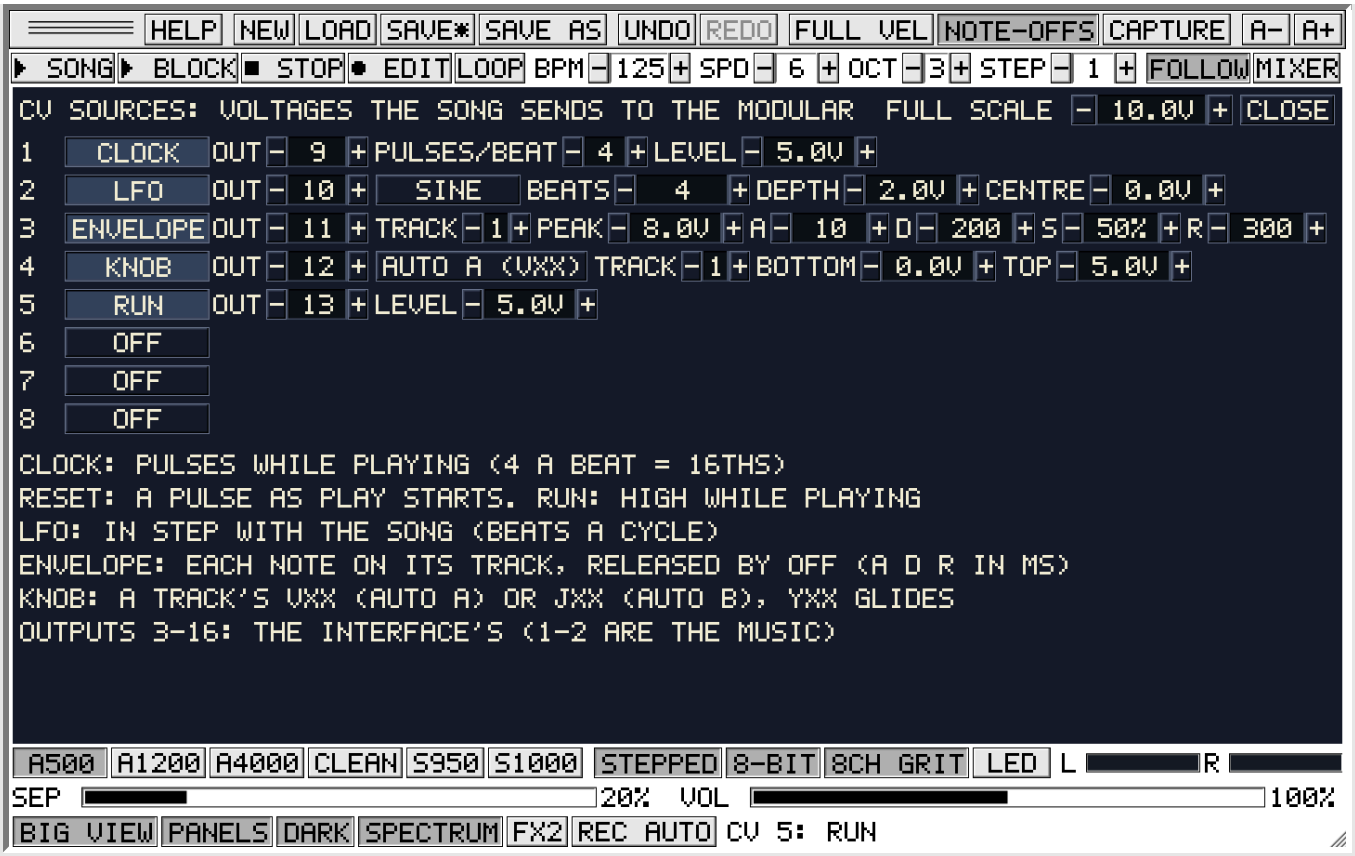
CONTROL	WHAT IT DOES
PITCH OUT	1 V / octave, C-3 = 0 V
GATE OUT	High while a note plays, at the GATE level

CONTROL	WHAT IT DOES
VEL OUT	The note's velocity (or Cxx) as 0 V to the gate level
MOD OUT	The Lxx knob as a voltage
FULL SCALE	The volts the interface puts out at full level: 10 V for the ES-9
OCTAVE BEND	Shift the pitch by octaves; the range pitch commands bend over
INPUT TUNER	INPUT is where the oscillator comes back in. TUNER holds C-3's voltage with the gate open and shows the pitch it hears: tune the oscillator to a C
AUTO-TUNE	Steps the pitch CV from -2.5 to +2.5 V, measures each step (about 10 seconds; leave the oscillator alone) and makes a **map**: every note is then sent at the voltage that plays it in tune, even on an oscillator that isn't exactly 1 V / octave. The map is saved with the instrument, and kept on this computer under the instrument's name (name it after the synth first); MAP... loads one of those, or CLEAR MAP goes back to plain 1 V / octave

It's mono: the newest note wins, a released key goes back to one still held, and a new note while the gate is open closes it for 2 ms so envelopes start again. OFF closes the gate; slides, vibrato and pitch bends move the pitch CV. It plays in step with the song, sample by sample. RENDER WAV writes the voltages too: one 32-bit float WAV per output used (song CV OUT 3.wav …), the same length as the audio. It isn't in bounces.

CV SOURCES

CV SOURCES... (on the CV OUT page) gives the song up to 8 voltages of its own, each on an output, saved in the song.



CV SOURCES: CLOCK, LFO, ENVELOPE, KNOB AND RUN

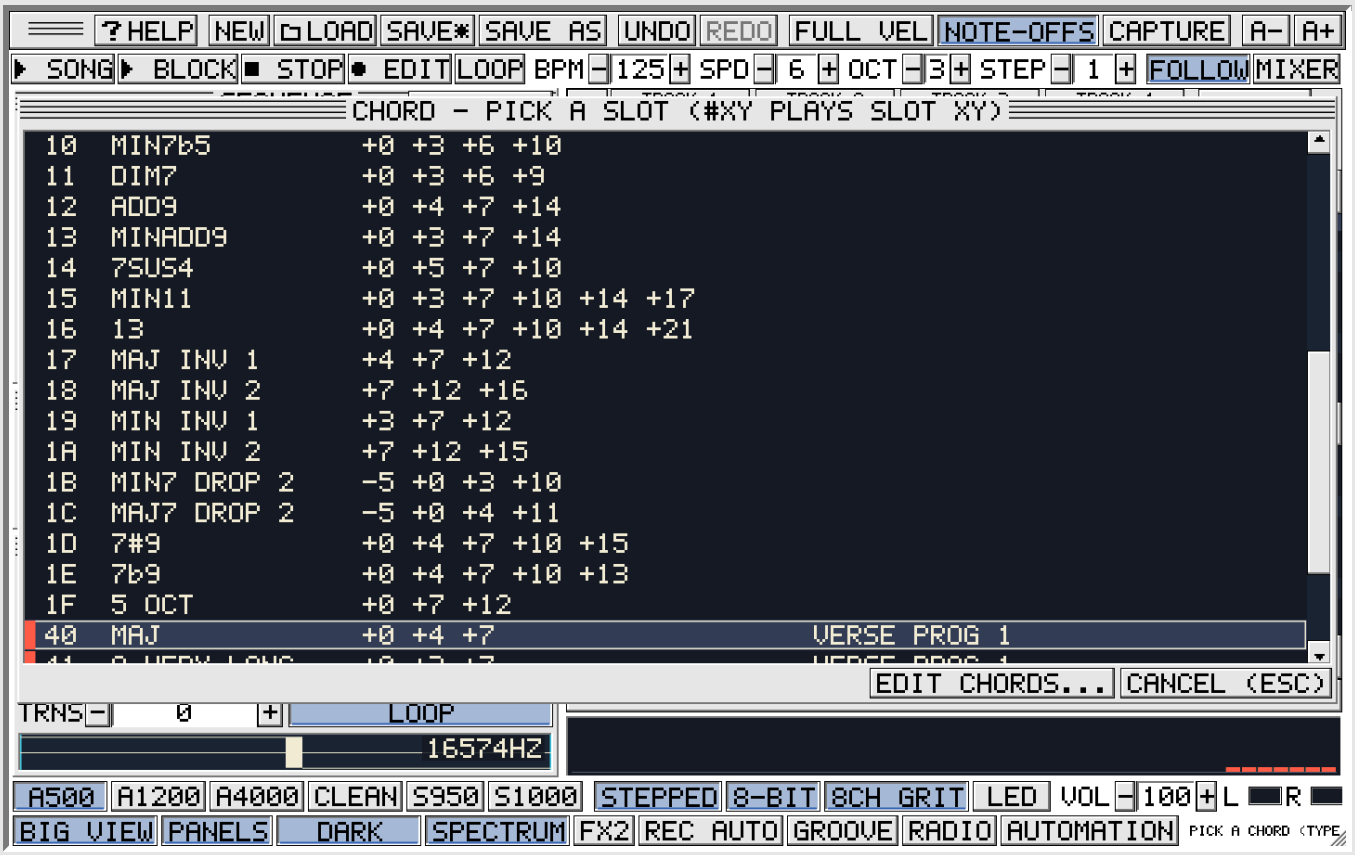
SOURCE	WHAT IT SENDS
CLOCK	Pulses while playing: PULSES/BEAT 4 = 16ths, 24 = 24 ppqn
RESET RUN	A pulse as play starts; high while playing
LFO	Sine, triangle, saw, square or random, BEATS a cycle in step with the song, with DEPTH and CENTRE
ENVELOPE	Attack, decay, sustain, release (ms), started by each note on its TRACK and released by OFF, up to PEAK
KNOB	A track's Vxx (AUTO A) or Jxx (AUTO B) as a voltage between BOTTOM and TOP (Yxx glides it): automation for the modular

CV DELAY (top of CV SOURCES, -100 to +100 ms) lines the modular up with the music: positive sends the voltages later; negative sends the music on 1-2 later, for a modular heard back through the inputs.

Safety. Voltages are clamped to the interface's range. STOP closes every gate and drops RUN; STOP twice (panic) also lets go of a tuner or AUTO-TUNE voltage.

CHORDS

A plug-in or MIDI-out instrument plays a whole chord from one note. Put `#xy` on the note's line and it plays `CHORD SLOT XY` from the song's chord table, held. The note you type is the chord's root: `C-3` with `#01` (slot 01, MIN) plays C, D#, G.



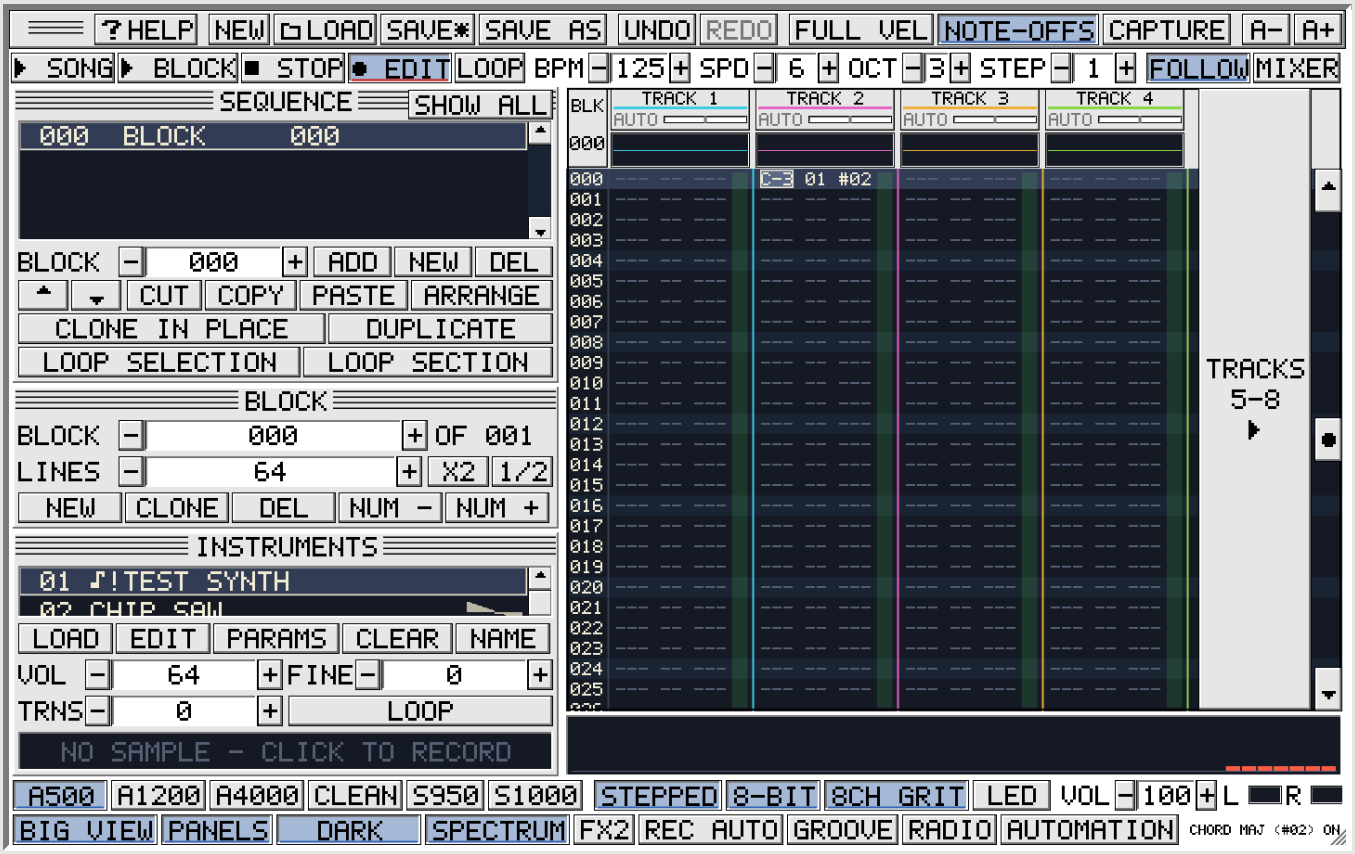
RIGHT-CLICK A `#` CELL: THE CHORD LIST

- Three ways in: type `SHIFT` `3` (`#`) in the command column, then the slot number; or right-click an effect cell and click `# CHORD` at the top of the list, then pick the chord; or right-click a cell that already has a `#` to change its chord.
- In the chord list, click a chord, or use the arrows and Enter; typing a slot number jumps to it. `EDIT CHORDS...` opens the CHORDS page on that slot, `EFFECT COMMANDS...` the usual list. Cancelling leaves `#01`.
- A slot holds up to 8 notes, below the root as well as above it. An inversion doesn't put the root at the bottom: `C-3` with MAJ INV 1 plays E, G and the C above.

- The chord ends like a note: the next note on the track, note-off, `ECx` or `Nxx`. Pitch commands bend the whole chord. On a sample instrument `#xy` plays just the note, and saving a MOD leaves it out.

PLAYING CHORDS IN

With EDIT on, or for CAPTURE, just play the chord on a MIDI keyboard or the laptop keys. On a plug-in or MIDI-out instrument, notes pressed together (within 60 ms) that match a slot in the song's chord table go in as `ONE NOTE, THE ROOT, WITH #XY` on one track. The status line says what it found, for example `CHORD MIN7 (#03) ON A-2`.



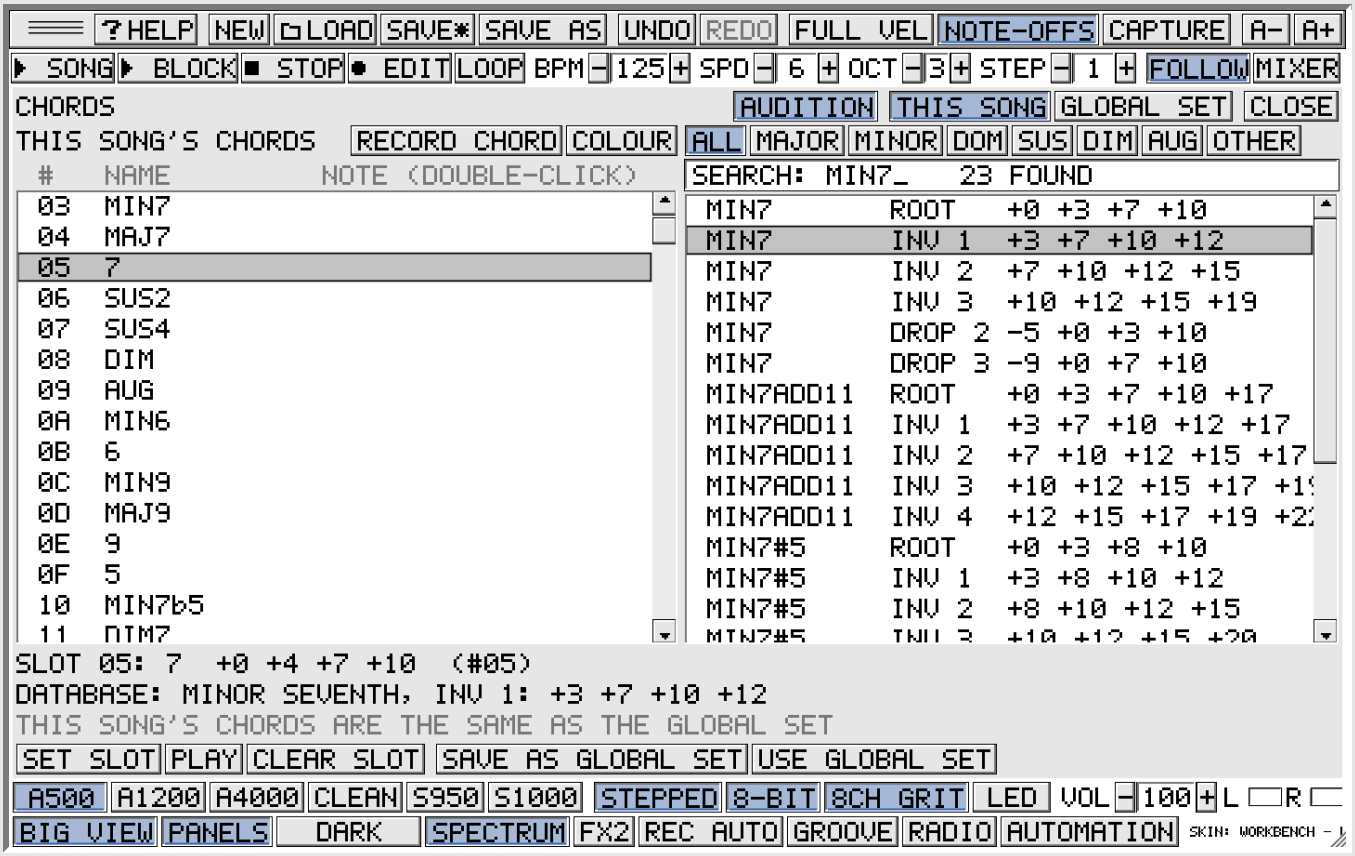
C, E AND G PLAYED TOGETHER: ONE NOTE, `#02` (MAJ)

- Inversions and spread voicings match their own slots (MAJ INV 1, MIN7 DROP 2…). If no slot has exactly the notes you played, a slot with the same note names in another octave is used, and the status line says the voicing may differ.
- When two slots have the same notes (C6 and Am7), the one whose root is your lowest note wins, then the one starting on the first key you pressed, then the lower slot.

- The notes go in as you play them; the moment they make a chord they're rewritten as one, still one undo step. In step entry the cursor moves on once for the chord. Its `OFF` goes on the line where you let go of the last key.
- `#` goes in the first effect column, or the second if the first already has a command. If both are taken, the notes go in one by one and the status line says so.
- Notes rolled slower than 60 ms stay separate notes, and sample instruments always get the notes as played. Right-click `EDIT` → `DETECT CHORDS` turns it off; it's remembered.

THE CHORDS PAGE

Press `CMD` `3`, or `CHORDS...` on the instrument's page. The slots 01–FF are on the left, the chord database on the right.

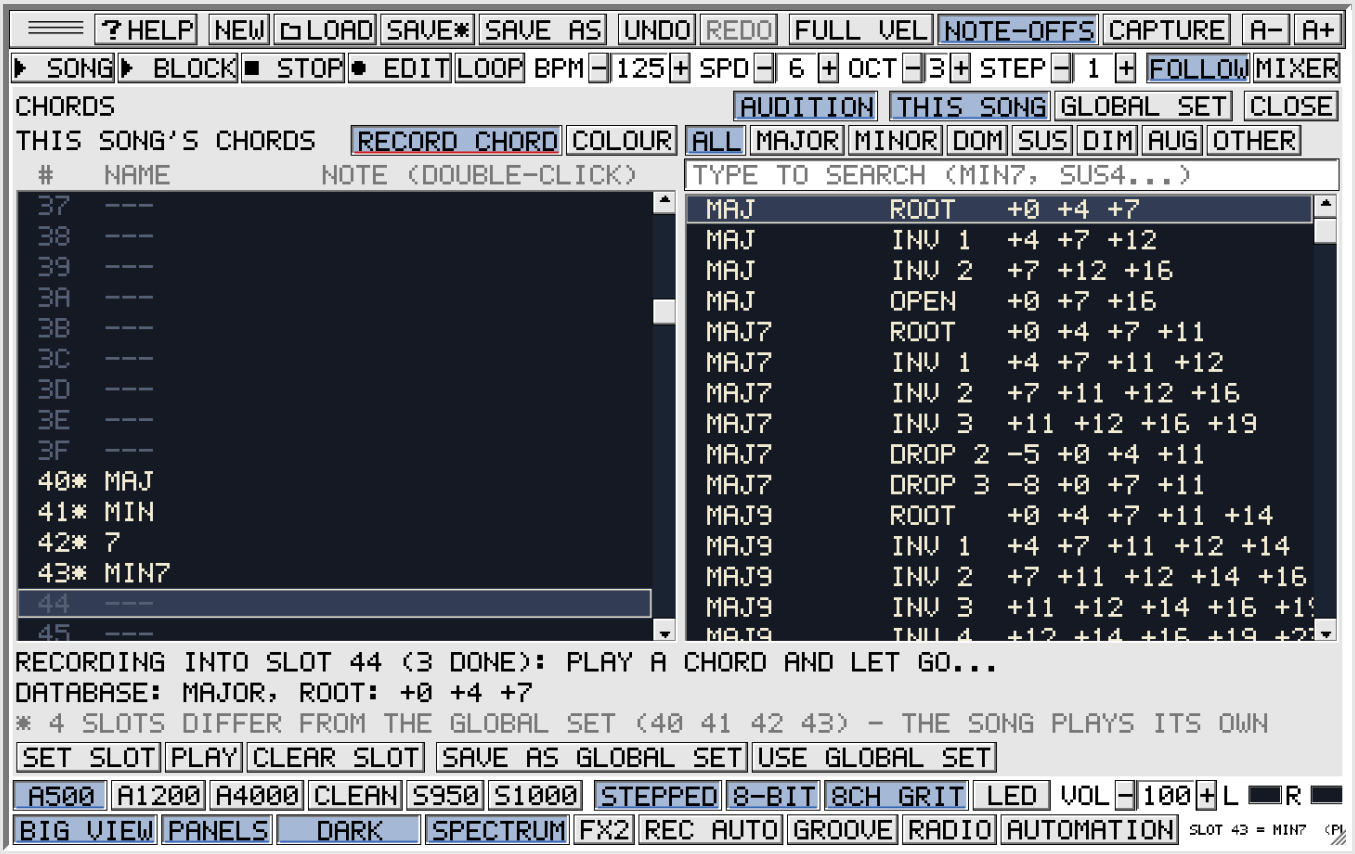


THE CHORDS PAGE, SEARCHING FOR MIN7

CONTROL	WHAT IT DOES
Slot list	Fixed columns: `#`, NAME (a long one ends in `..`) and NOTE, so the notes line up. Click a slot to fill it; drag, `SHIFT`-click or `CMD`-click to pick several. `SHIFT` `↑` `↓` moves through the slots. A `*` marks a slot that isn't the same in the global set; a coloured stripe, its group colour.
Double-click NOTE	Type the slot's note right there (Enter keeps it, Esc doesn't)
`RECORD CHORD`	Stays on: play chord after chord into the slots (see below)
`COLOUR`	A group colour for the picked slots: 8 colours or none
`AUDITION`	Clicking a chord plays it, on the last key you played
Database	Double-click a chord (or press Enter, or `SET SLOT`) to put it in the slot. `↑` `↓` move through it.
`ALL` `MAJOR` `MINOR` `DOM` `SUS` `DIM` `AUG` `OTHER`	Show one family of chords
Typing	Searches: `MIN7`, `SUS4`, `DROP 2`, `7B9` for 7♭9. Backspace and Esc clear it.
`PLAY`	Plays the chord with the current instrument, on the last key you played (Space)
`CLEAR SLOT`	Empties the slot, or all the picked ones (Delete): their `#xy` then plays just the note
`THIS SONG` `GLOBAL SET`	Which table you're editing
`SAVE AS GLOBAL SET`	The song's chords become the global set (it asks first)
`USE GLOBAL SET`	The song gets the global set (undo takes it back)

RECORDING YOUR OWN CHORDS

Click the slot to start at and press RECORD CHORD above the slot list: it stays down. Play chord after chord on a MIDI keyboard or the laptop keys, letting go between them. Each chord goes into the selected slot, then the selection moves on to the **next empty slot**, so nothing is written over; the status line counts them. Click RECORD CHORD again (or ESC) to stop.



THREE CHORDS RECORDED: 40, 41, THEN 43 (42 ALREADY HAD ONE); 44 IS NEXT

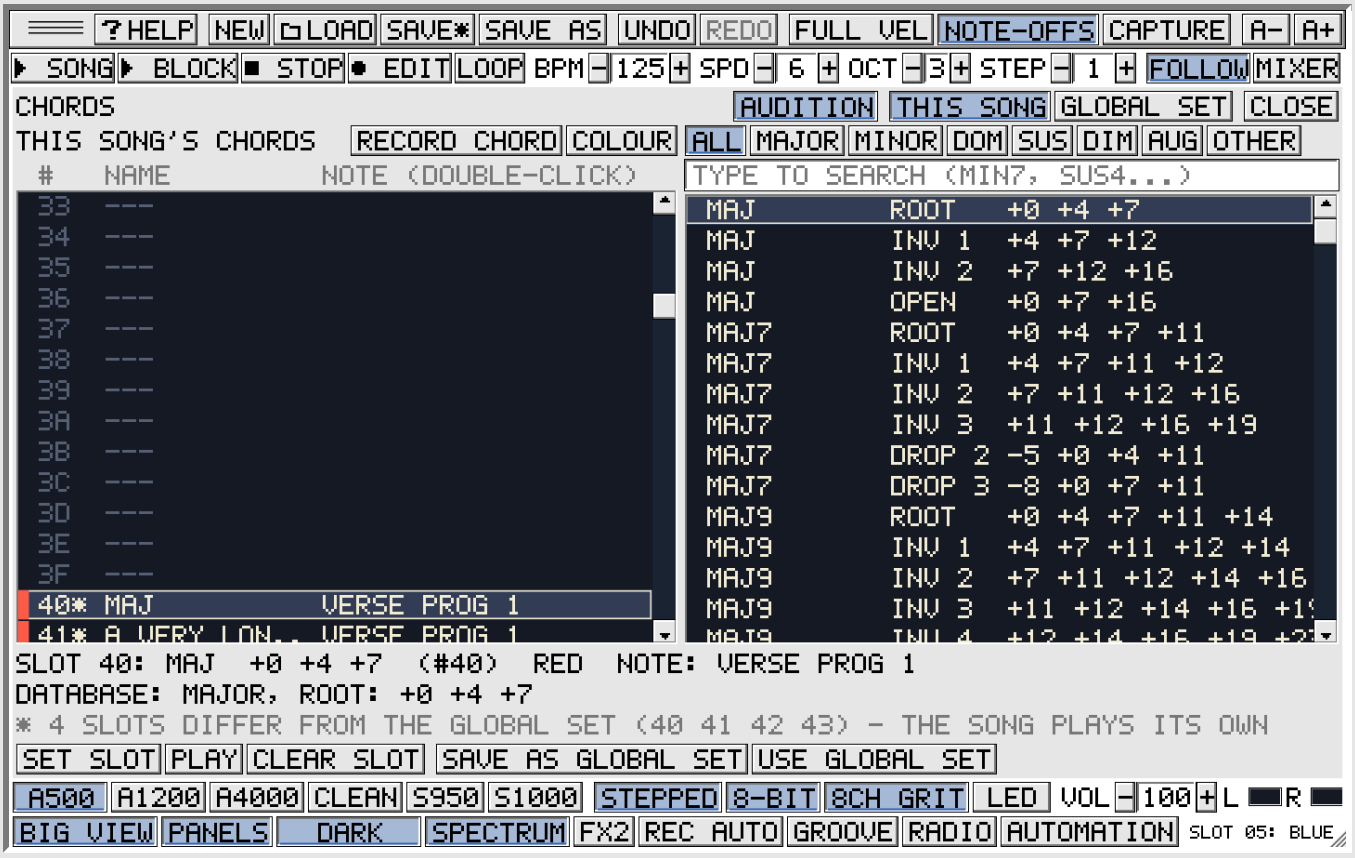
- If the notes are one of the database's chords, the slot takes that name and root: E G C becomes MAJ INV 1, so C-3 with it plays your E G C. Anything else is called USER and is measured from your lowest note.
- Up to 8 notes (with more, the lowest 8). One note on its own isn't a chord and is ignored.
- While it records, the MIDI keyboard doesn't write into the pattern and the laptop keys don't type into the search. Each chord is an undo step.

NOTES ON SLOTS

Double-click a slot's NOTE column (or right-click → NOTE...) and type a note of your own right there, such as `VERSE PROG 1`, up to 24 characters; Enter keeps it, Esc doesn't. It shows in the slot list's NOTE column, in the chord list over the pattern and in the help line for a `#xy`, and it stays with the slot when you change its chord. Notes are saved in the song and in the global set.

COLOUR GROUPS

Pick several slots (drag down the list, `SHIFT`-click a range, `CMD`-click one at a time), then `COLOUR` (or right-click → COLOUR...): red, orange, yellow, green, cyan, blue, purple, pink, or none. The colour shows as a stripe left of the slot number, in the chord list over the pattern, and on the `#xy` in the pattern itself, so a progression stands out while you write.



CHORD GROUPS IN THE PATTERN: `#40` RED, `#05` BLUE

COPY, PASTE and CLEAR work on all the picked slots. Colours are saved in the song and in the global set.

HEARING CHORDS (AUDITION)

With `AUDITION` on (top right of the CHORDS page; on to start with), clicking a slot or a chord in the database plays it with the current instrument while you hold the mouse, on **the last key you played** (C of the octave before you've played any). The arrow keys play each chord briefly as you move. In the chord list over the pattern, moving with the arrow keys or the mouse plays each chord on that cell's note with that track's instrument; clicking still picks it.

THE DATABASE

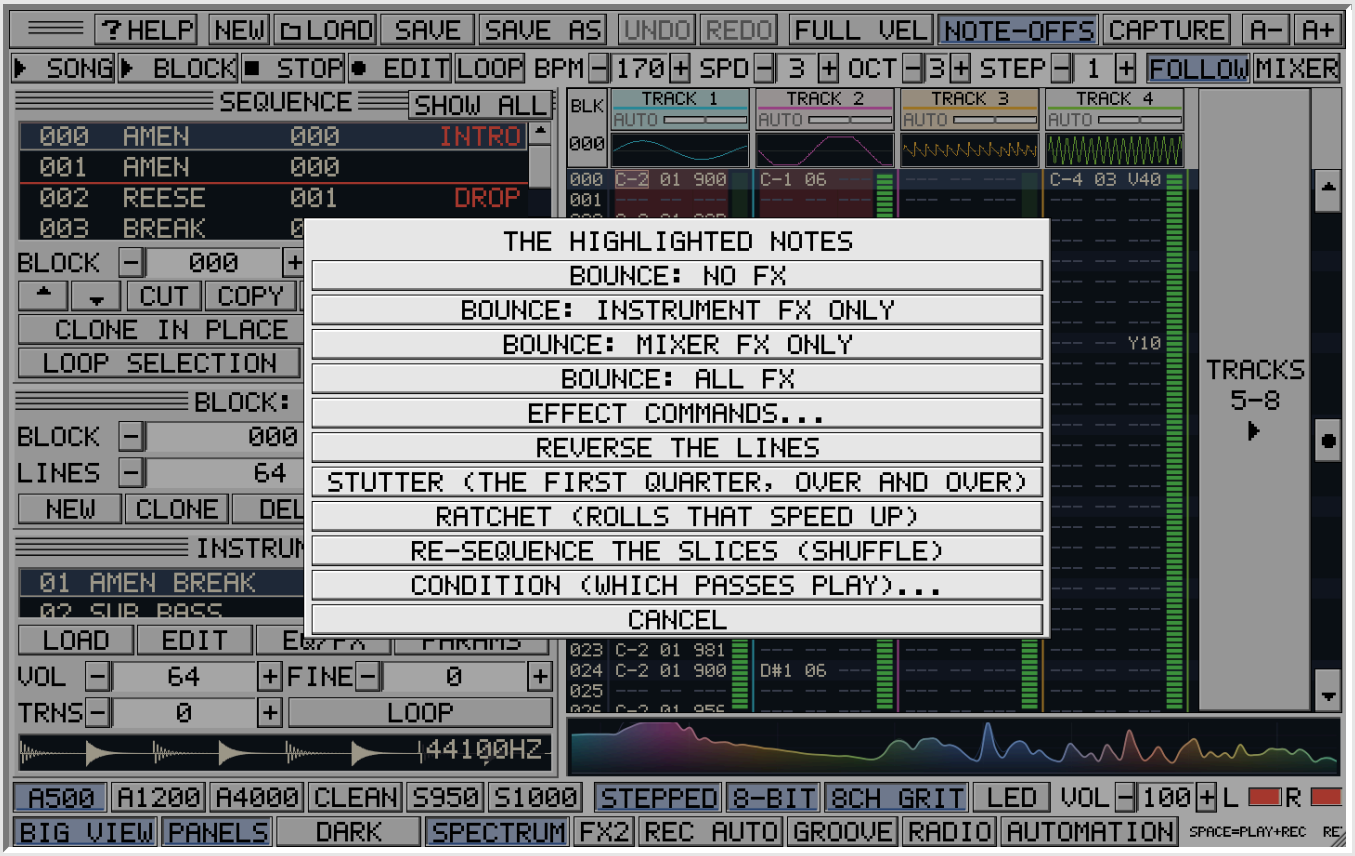
116 chord types – major, minor, dominant, suspended, diminished, augmented and more – from the tonal music library's list, which comes from Wikibooks' complete list of chord patterns. Each one is there as written (ROOT), in every inversion (INV 1, INV 2…), as DROP 2 and DROP 3 voicings for four-note chords and OPEN for triads: 638 voicings in all.

THE SONG'S CHORDS AND THE GLOBAL SET

- `EVERY SONG HAS ITS OWN TABLE`, saved in the song. The `GLOBAL SET` is kept on your Mac: every new song, and every MOD or older song without a table, starts with a copy of it.
- Changing a song's chords doesn't change the global set, and changing the global set doesn't change any song. A saved song always plays its own chords, whatever the global set has become.
- When you load a song, the status line says how its chords compare with the global set, for example `SLOTS 05 1F DIFFER FROM THE GLOBAL SET`. It plays its own either way.
- Older songs are converted when they load: v0.12–v0.14 chords (`0xy` on a CHORD instrument) become `#xy`, and v0.11 chords (`0xy` as two intervals) each get a slot with the same notes. The song sounds the same.

BOUNCING

Turn any part of the pattern into a new sample. Highlight some notes (they can cover several tracks), then right-click inside the highlight, or press CMD B.



RIGHT-CLICK INSIDE A HIGHLIGHT

CHOICE	WHAT'S IN THE SAMPLE
NO FX	The instruments' sound before any effects. Use it when the sample will play through the same effects.
INSTRUMENT FX ONLY	Through the instruments' own inserts, not the mixer.
MIXER FX ONLY	After the tracks' strips (EQ, inserts, fader) plus what they send to SEND A / B, so reverb and delay tails are in it; not the instruments' own effects.
ALL FX	Both. Never the master or the Paula / Akai output stage. The new instrument is called BOUNCE, BOUNCE+IFX, BOUNCE+MFX or BOUNCE+FX.

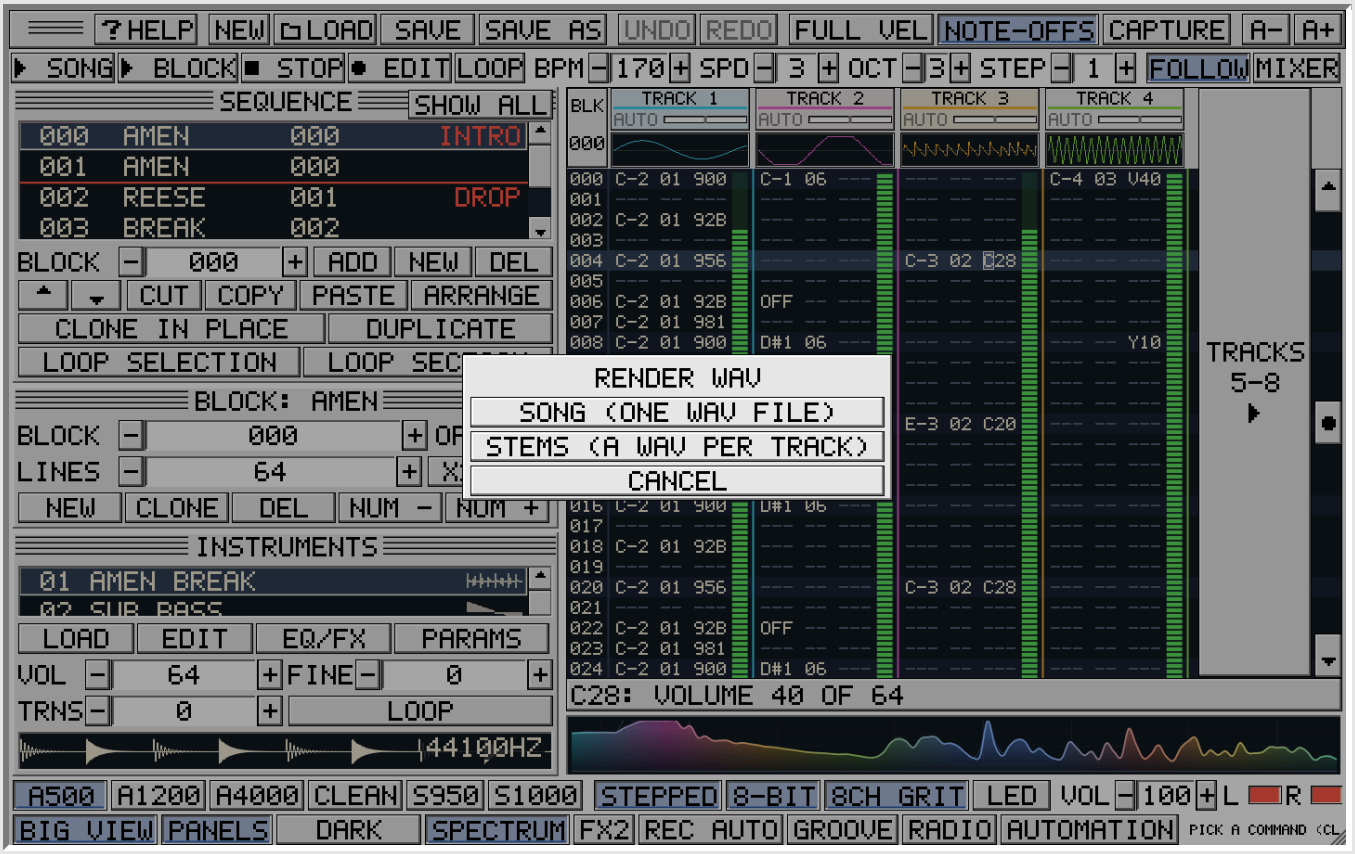
- Only the highlighted lines of the highlighted tracks play, from the first line, at the tempo and speed set there. Plug-in

instruments are included.

- It records on after the last line until the sound has died away (up to 10 seconds), so tails aren't cut.
- The sample goes into the first empty instrument slot, named like `BOUNCE T1-2 000-015`, at 44.1 kHz. `C-3` plays it exactly as recorded.
- Nothing else changes: your pattern and the instruments that played it stay as they are. Undo removes the new instrument.

RENDERING

`RENDER WAV` (File menu) asks what to render:



SONG OR STEMS

- **SONG:** the song once through as a 24-bit stereo WAV, mixer and all.
- **STEMS:** a folder next to the name you pick, `<name> STEMS`, with one 24-bit WAV per used track (after its strip), SEND A, SEND B and the master: `01 TRACK 1.wav` … `11 MASTER.wav`. They start together and are the same length, so they drop straight into Ableton.
- After the song ends it keeps going until everything is quiet for a second (up to 20 seconds), so tails aren't cut. Plug-in delay is taken off the start.
- MIDI-out instruments can't be rendered; the status line says which were left out.

These are in the **File** menu. `EXPORT SAMPLES` writes every instrument's sample to a folder as `01 NAME.wav`, `02 NAME.wav` …

SAVE MOD writes a ProTracker MOD (`M.K.` when only tracks 1–4 are used, else `8CHN`) and says what a MOD can't hold.

SAVING & FILES

- SAVE (CMD S) writes straight to the song's file; the first time it asks for a name. SAVE AS (CMD SHIFT S) always asks, for versions. The song takes the file's name. SAVE* with a star means unsaved changes. Saving happens in the background: the pattern and the music carry on. If the file was changed since you opened it (by your other computer), SAVE asks: SAVE A COPY or REPLACE IT.
- A `.prv` song holds everything: every sample (stereo too), the mixer, plug-in settings, automation, MIDI-out settings, its chord table. You can move or delete the original WAVs.
- LOAD also opens ProTracker-style `.mod` files (4, 6 and 8 channels) and MED / OctaMED `MMD0` / `MMD1` modules.
- Songs saved by older versions always load.
- **Autosave:** every two minutes, when something changed, a copy of the song goes into the app's own folder (in the background). If the app doesn't close properly – a crash, a plug-in taking it down – next time it offers RECOVER (the song as it was, still pointing at its own file, marked unsaved) or START AS NORMAL. A normal quit never asks.
- The window's title shows the song, `*` for unsaved changes, and the block you're editing.

SHARING SONGS BETWEEN YOUR MAC AND PC

- **File → Shared Folder…** lists the synced folders on this computer (Dropbox, iCloud Drive, OneDrive, Google Drive) or lets you choose one. Pick the same one on each computer; songs go in `Paula's Revenge/Songs` inside it.
- **Save to Shared Folder** and **Open from Shared Folder…** (File menu). Songs carry their samples, so nothing else needs copying.
- Plug-ins are found on the other computer by name and ID. Use the **VST3** on both (install it on the Mac too) to keep their settings; with only the AU on one side, it loads with its own defaults.
- Let the sync finish before opening a song on the other computer.

DIAGNOSTICS

The app logs anything that could cause a dropout: audio blocks that took too long, the driver's own dropout count, the window stalling, saves and loads, the radio running dry, plug-ins that wouldn't load, and the audio device's rate and buffer. **Help → Save Diagnostics Report…** puts it all in a text file on your desktop, to send in; **Help → Show the Log Folder** opens the log.

KEYS

KEY	ACTION
CMD K	The command palette: type any command, button or page
Z S X D C …	Notes, lower octave (, L . ; / carry on up)
Q 2 W 3 E …	Notes, upper octave
1	Note off
SPACE	Play and record (EDIT on) / stop (EDIT off) · with SHIFT, the song. On every page
ESC or `	Edit mode on / off
SHIFT 3	In a command column: #, the chord command
RETURN	Play / stop, EDIT left as it is · SHIFT song · CMD block from the cursor
Arrows	Move · SHIFT ↑ ↓ 4 lines · SHIFT ← → track
TAB	Next track · with SHIFT, previous
[ ]	Octave down / up, on every page (also with a plug-in's window in front)
- =	Previous / next instrument
DELETE	Clear the highlight or the field · SHIFT removes the line · ALT inserts one
CMD C X V	Copy / cut / paste the highlight, or the track (SHIFT: the block)
CMD A	Highlight the whole block
CMD B	Bounce the highlight to a new instrument
CMD ↑ ↓	Transpose the track a semitone (SHIFT: an octave)
CMD Z	Undo · with SHIFT, redo
CMD [ ]	Loop start / end at the cursor
CMD S	Save · with SHIFT, save as
CMD 2 / 1	Mixer / pattern

KEY	ACTION
CMD 3	CHORDS: the chord table
CMD 4	SAMPLE: record audio into the instrument
CMD 5	INTERNET RADIO
CMD 6	The BREAK page
F2 (hold) / SHIFT F2	FILL while held / latched on or off
CMD F	FOLLOW on / off
CMD R	CAPTURE: write what you just played into the pattern
CMD O	Load a song
ALT ↑ ↓	Move the selected song positions